

Project Data Page:





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Ammonia Scrubber

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\123-DopeWO OverfillRev 02.pvdb

Date of Analysis : Aug 13, 2026 1:40pm

PV Elite 28 U2, April 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 333.350 mm ID	
Vessel Design Length, Tangent to Tangent	16599.63 mm
Distance of Bottom Tangent above Grade	4500.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	4200.00 mm
Internal Design Temperature	100 °C
Internal Design Pressure	1.800 MPa
External Design Temperature	100 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	4 MPa
External Max. Allowable Working Pressure	2 MPa
Shop Test Pressure	5.948 MPa
Field Test Pressure	2.340 MPa
Required Minimum Design Metal Temperature	-28.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-105	...	...	K03504	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	+IT
Nozzle	SA-106 B	...	...	K03006	
Nozzle Flg	SA-105	...	...	K03504	
Basing	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
Bottom head	1.817	0.10	16.000	3.0000	No	No
Shell-1	1.815	0.10	9.000	3.0000	N/A	No
Shell-2	1.800	0.10	7.000	3.0000	N/A	No
Flange-1	1.800	0.10	5.000	3.0000	N/A	No
Flange-2	1.800	0.10	5.000	3.0000	N/A	No
Shell-3	1.800	0.10	7.000	3.0000	N/A	No
Flange-3	1.800	0.10	5.000	3.0000	N/A	No
Flange-4	1.800	0.10	5.000	3.0000	N/A	No
Shell-4	1.800	0.10	7.000	3.0000	N/A	No
Flange-5	1.800	0.10	7.000	3.0000	N/A	No
Flange-6	1.800	0.10	7.000	3.0000	N/A	No



Shell-5	1.800	0.10	7.000	3.0000	N/A	No
Flange-7	1.800	0.10	7.000	3.0000	N/A	No
Flange-8	1.800	0.10	4.000	3.0000	N/A	No

Liquid Level: 1833.34 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.920

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-1000.0	3200.0	26.0	26.0	...	...	0.70	0.70
Skirt	-0.0	1000.0	28.0	28.0	...	...	0.70	0.50
Ellipse	50.0	50.0	26.0	22.0	5.2	4.0	1.00	1.00
Cylinder	3550.0	3500.0	14.0	14.0	5.2	6.2	1.00	1.00
Cylinder	5550.0	2000.0	12.0	12.0	5.2	6.1	1.00	1.00
Body Flg	5706.4	151.9	56.0	80.0	45.4	45.4	1.00	1.00
Body Flg	5846.4	139.9	50.0	48.0	45.3	45.3	1.00	1.00
Cylinder	9046.4	3200.0	12.0	12.0	5.2	5.9	1.00	1.00
Body Flg	9190.8	139.9	50.0	48.0	45.3	45.3	1.00	1.00
Body Flg	9330.8	139.9	50.0	48.0	45.3	45.3	1.00	1.00
Cylinder	12530.8	3200.0	12.0	12.0	5.2	5.9	1.00	1.00
Body Flg	12681.2	145.9	56.0	54.0	45.3	45.3	1.00	1.00
Body Flg	12827.2	145.9	56.0	54.0	45.3	45.3	1.00	1.00
Cylinder	16397.2	3570.0	12.0	12.0	5.2	6.0	1.00	1.00
Body Flg	16547.6	145.9	56.0	54.0	45.3	45.3	1.00	1.00
Body Flg	16599.6	52.0	54.0	52.0	45.5	45.5	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N4-2"	WRC-107/537	0.134	n/a	Passed
N1-2"	WRC-297	0.249	Shell	Passed
N2-3"	WRC-297	0.633	Shell	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1600	10540.5	4913.3	113875440	76486240
20	30	3700	9294.42	4887.32	82126696	60798888
30	40	4225	8820.55	4860.12	73065528	55923196
40	50	6000	8792.87	4849.94	72625008	55680344
50	60	8750	6913.12	4631.02	45128384	39081944
60	70	9830.47	5847.22	4485.56	32336548	29941470
70	80	9976.42	5770.55	4424.89	31453546	29264234
80	90	11646.4	5699.52	4380.43	30650612	28647838
90	100	13320.9	3994.08	3861.43	15116587	15438144
100	110	13460.8	3925.08	3787.91	14562225	14902671
110	120	15130.8	3856.09	3713.06	14017523	14377584
120	130	16808.3	2150.65	2194.98	4393159	4911001
130	140	16954.2	2079.6	2076.61	4084336	4599160
140	150	18812.2	2008.13	1957.25	3785917	4304673
150	160	20674.7	105.488	421.093	11110.5	55707.5
160	170	20773.6	34.4365	252.158	895.656	6558.37

Abs Max of the all of the Stress Ratio's : 0.9582

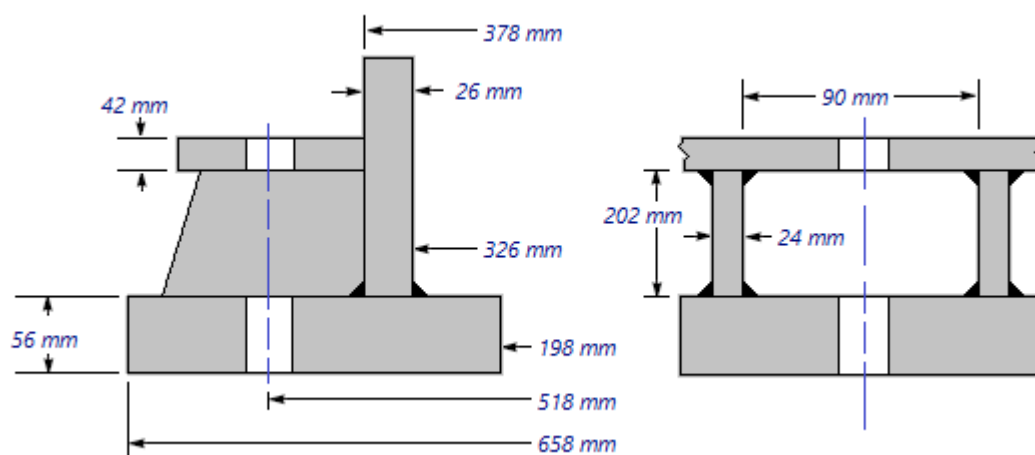
Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 56.0000 mm
Inside Diameter of Basing 198.0000 mm
Outside Diameter of Basing 658.0000 mm



Nominal Diameter of Bolts	42.0000	mm
Diameter of Bolt Circle	518.0000	mm
Number of Bolts	8	
Thickness of Gusset Plates	24.0000	mm
Average Width of Gusset Plates	130.0000	mm
Height of Gussets	202.0000	mm
Distance between Gussets	90.0000	mm
Thickness of Top Plate or Ring	42.0000	mm
Radial Width of the Top Plate	140.0000	mm
Number of Gussets	16	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	28436.	N
Total Earthquake Shear on Support	22809.	N
Wind Moment on Support	251858096.	N-mm
Earthquake Moment on Support	214468880.	N-mm
Wind Load Case Test Moment on Support	125929048.	N-mm
Test Case Wind Shear on Support	14218.	N
Maximum Initial Bolt Load	235042.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	5090.3	kgm
Shop Test - Fabricated + water (full)	6656.9	kgm
Shipping - Fab. + rem. intls.+ shipping app.	5776.3	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	5891.3	kgm
Empty - Fab. + intls. + details + wghts.	6077.8	kgm
Operating - Empty + operating liquid (no ca)	6222.7	kgm
Field Test - Empty weight + test medium (full)	7342.9	kgm



Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N4-2"	16	...	...	OK	5.19	OK	No Calc[*]
H4-6"		...	...	...		OK	No Calc[*]
K1-2"		...	...	OK	5.23	OK	No Calc[*]
K2-2"		...	...	OK	5.24	OK	No Calc[*]
N1-2"		...	...	OK	5.23	OK	No Calc[*]
H3-6"		...	...	...		OK	No Calc[*]
H2-6"		...	...	...		OK	No Calc[*]
H1-6"	7	OK	...	OK	5.23	OK	Passed
N2-3"	7	...	...	OK	5.23	OK	No Calc[*]
N3-3"	4	OK	...	OK	8.02	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4 Nozzle : N3-3"
Minimum MAWP Shells/Flanges : 4 Element : Flange-8
Minimum MAPnc Shells/Flanges : 5 Element : Flange-8

Computed Vessel M.A.W.P. : 4 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N4-2"	0.000	0.000	102.068
40	H4-6"	2870.000	0.000	310.145
40	K1-2"	1850.000	0.000	113.600
40	K2-2"	550.000	0.000	113.600
40	N1-2"	1750.000	180.000	132.800
80	H3-6"	6696.391	0.000	310.145
110	H2-6"	10180.702	0.000	310.145
140	H1-6"	13677.202	0.000	310.145
140	N2-3"	16227.201	0.000	168.862
160	N3-3"	0.000	0.000	234.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: Flange-8

Note: No Nozzle pairs found on this element.



UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N3-3"	82.2	196.03	199.4	20.55

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4-2"	2.000 in	160	WNF	2.375	8.738	...	...	173.44	600
K1-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	225.81	300
K2-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	225.81	300
N1-2"	2.000 in	Actual	WNF	2.787	10.000	...	...	273.48	600
N2-3"	3.000 in	160	WNF	3.500	11.125	...	...	280.05	600
N3-3"	2.000 in	LWN	WNF	3.787	10.000	...	...	302.05	300
H4-6"	6.000 in	80	WNF	6.625	10.973	...	...	286.80	300
H3-6"	6.000 in	80	WNF	6.625	10.973	...	...	284.80	300
H2-6"	6.000 in	80	WNF	6.625	10.973	...	...	284.80	300
H1-6"	6.000 in	80	WNF	6.625	10.973	...	...	184.80	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4-2"	SA-105	9.735	10.000	...	...	...
K1-2"	SA-105	9.735	10.000	...	...	...
K2-2"	SA-105	9.735	10.000	...	...	...
N1-2"	SA-105	9.735	10.000	...	...	...
N2-3"	SA-105	9.735	10.000	...	...	...
N3-3"	SA-105	9.735	10.000	...	...	...
H4-6"	SA-105	9.735	10.000	...	...	...
H3-6"	SA-105	9.735	10.000	...	...	...
H2-6"	SA-106 B	9.735	10.000	...	...	...
H1-6"	SA-105	9.735	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N4-2"	...	0.0	150.00	0.00	Bottom head
K1-2"	1850.000	0.0	206.38	0.00	Shell-1
K2-2"	550.000	0.0	206.38	0.00	Shell-1
N1-2"	1750.000	180.0	250.00	0.00	Shell-1
N2-3"	16227.201	0.0	250.00	0.00	Shell-5
N3-3"	...	0.0	250.00	0.00	Flange-8
H4-6"	2870.000	0.0	250.00	0.00	Shell-1
H3-6"	6696.390	0.0	250.00	0.00	Shell-3
H2-6"	10180.700	0.0	250.00	0.00	Shell-4
H1-6"	13677.200	0.0	150.00	0.00	Shell-5



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 26.0mm THK X 326.0mm ID X 3200.0mm	SA-516 70
1	SKIRT: 28.0mm THK X 324.0mm ID X 1000.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 26.0mm THK X 333.3mm ID X 50.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 333.3mm ID X 2000.0mm	SA-516 70
1	WELD NECK FLANGE: 80.0mm THK X 535.0mm OD	SA-105
3	WELD NECK FLANGE: 50.0mm THK X 535.0mm OD	SA-105
2	CYLINDER: 12.0mm THK X 333.3mm ID X 3200.0mm	SA-516 70
3	WELD NECK FLANGE: 56.0mm THK X 535.0mm OD	SA-105
1	CYLINDER: 12.0mm THK X 333.3mm ID X 3570.0mm	SA-516 70
1	BLIND FLANGE: 54.0mm THK X 535.0mm OD	SA-105
1	INSULATION: 500mm X 80mm THK	
1	INSULATION: 133mm X 80mm THK	
1	PACKING: 929.999mm	...
1	INSULATION: 3499mm X 80mm THK	
1	PACKING: 1140.000mm	...
1	INSULATION: 2000mm X 80mm THK	
1	INSULATION: 145mm X 80mm THK	
1	INSULATION: 136mm X 80mm THK	
3	PACKING: 2070.000mm	...
2	INSULATION: 3200mm X 80mm THK	
5	INSULATION: 127mm X 80mm THK	
1	INSULATION: 3569mm X 80mm THK	
1	INSULATION: 35mm X 80mm THK	
4	CLASS 300 GR 1.1, 6.0" BLIND FLANGE(S)	SA-105
1	CLASS 600 GR 1.1, 3.0" BLIND FLANGE(S)	SA-105
1	BASERING: 658.0mm OD X 198.0 ID X 56.0mm	SA-516 70
8	BASERING BOLTS: 42.0mm DIA	SA-307 A
8	BASERING BOLT NUTS: 42.0mm DIA	...
1	BASERING TOP RING: 658.0mm OD X 378.0 ID X 42.0mm	...
16	BASERING GUSSET PLATES: 130.0mm X 130.0mm X 202.0mm X 24.0mm	...
1	GASKET: 406.0mm OD X 374.0mm ID	...
80	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B7



160	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
3	GASKET: 406.4mm OD X 374.0mm ID	...
2	CYLINDRICAL SEGMENT 14.0mm THK X 416.7mm X 3500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 213.9mm X 3500.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Flange-1	[11]	A	-3	-27	-29	0.573	12.000	1.00	No
Flange-2	[11]	A	-3	-27	-29	0.573	12.000	1.00	No
Flange-3	[11]	A	-3	-27	-29	0.573	12.000	1.00	No
Flange-4	[11]	A	-3	-27	-29	0.573	12.000	1.00	No
Flange-5	[11]	A	-3	-27	-29	0.573	12.000	1.00	No
Flange-6	[11]	A	-3	-27	-29	0.573	12.000	1.00	No
Flange-7	[11]	A	-3	-27	-29	0.573	12.000	1.00	No
Flange-8	[11]	A	0	-7	-29	0.888	13.000	1.00	No
Bottom head	[10]	B	-5	-104	-29	0.259	22.000	1.00	No
Bottom head	[7]	B	0	-104		0.222	26.000	1.00	No
Shell-1	[20]	B	-18	-18	-29	1.000	14.000	1.00	No
Shell-2	[20]	B	-23	-23	-29	1.000	12.000	1.00	No
Shell-3	[20]	B	-23	-34	-29	0.808	12.000	1.00	No
Shell-4	[8]	B	-23	-48	-29	0.565	12.000	1.00	No
Shell-5	[8]	B	-23	-48	-29	0.565	12.000	1.00	No
N4-2"	[1]	B	-8	-104	-29	0.085	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
H4-6"	[1]	B	-8	-104	-29	0.165	9.601	1.00	No
Nozzle Flg	[5]	A	-18	-48					
K1-2"	[1]	B	-18	-104	-29	0.203	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
K2-2"	[1]	B	-18	-104	-29	0.204	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N1-2"	[1]	B	-8	-104	-29	0.053	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
H3-6"	[1]	B	-8	-104	-29	0.165	9.601	1.00	No
Nozzle Flg	[5]	A	-18	-48					
H2-6"	[1]	B	-8	-104	-29	0.193	9.601	1.00	No
Nozzle Flg	[5]	A	-18	-48					
H1-6"	[1]	B	-8	-104	-29	0.165	9.601	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N2-3"	[1]	B	-8	-104	-29	0.074	9.735	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-3"	[1]	A	-8	-104	-29	0.077	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						

Warmest MDMT: 0 -7

Required Minimum Design Metal Temperature -28.0 °C
Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material



- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Ammonia Scrubber

Design Internal Pressure (for Hydrotest)	1.8001	MPa
Design Internal Temperature	100.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-28.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes



Complete Listing of Vessel Elements and Details:

Element#	1/16	
Element From Node	10	
Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	3200	mm
Skirt Outside Diameter	378	mm
Diameter of Skirt at Base	378	mm
Skirt Thickness	26	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	26	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing lug	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm

Element#	2/16	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	380	mm
Diameter of Skirt at Base	380	mm
Skirt Thickness	28	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins	
Dist. from "FROM" Node / Offset dist	500	mm
Height/Length of Insulation	500	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³



Element#	3/16	
Element From Node	30	
Element To Node	40	
Element Type	Elliptical	
Description	Bottom head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	333.35	mm
Element Thickness	22	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	26	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.8001	MPa
Design Temperature Internal Pressure	100	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	-83.337	mm
Height/Length of Liquid	133.34	mm
Liquid Density	0.0009185	kg/cm³
Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	-83.337	mm
Height/Length of Insulation	133.34	mm
Thickness of Insulation	80	mm
Density	0.0003516	kg/cm³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	N4-2"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	123.51	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	30	
Detail Type	Weight	
Detail ID	Vortex	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	49.031	N
Offset from Element Centerline	0	mm
Element From Node	30	
Detail Type	For./Mom.	
Detail ID	Guide lateral f	
Dist. from "FROM" Node / Offset dist	-83	mm
Force in X Direction	-11285	N
Force in Y Direction	0	N
Force in Z Direction	8296	N
Moment about X Axis	0	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	0	N-mm



Force/Moment Combination Method

SRSS

Element#	4/16
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	3500 mm
Inside Diameter	333.35 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.8001 MPa
Design Temperature Internal Pressure	100 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.37
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Packing
Detail ID	Bed#4
Dist. from "FROM" Node / Offset dist	2570 mm
Height of Packed Section	930 mm
Density	0.0001828 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.92000002
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1700 mm
Liquid Density	0.0009185 kg/cm ³
Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3500 mm
Thickness of Insulation	80 mm
Density	0.0003516 kg/cm ³
Element From Node	40
Detail Type	Nozzle
Detail ID	H4-6"
Dist. from "FROM" Node / Offset dist	2820 mm
Nozzle Diameter	6 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1080.1 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	40
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	1800 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300



Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	104.81 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	K2-2"
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	104.81 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	N1-2"
Dist. from "FROM" Node / Offset dist	1700 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	141.67 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	40
Detail Type	Weight
Detail ID	DLBS-4
Dist. from "FROM" Node / Offset dist	2570 mm
Miscellaneous Weight	98.062 N
Offset from Element Centerline	0 mm

Element From Node	40
Detail Type	Weight
Detail ID	TABS-4
Dist. from "FROM" Node / Offset dist	2570 mm
Miscellaneous Weight	49.031 N
Offset from Element Centerline	0 mm

Element From Node	40
Detail Type	Weight
Detail ID	H4 Davit
Dist. from "FROM" Node / Offset dist	2820 mm
Miscellaneous Weight	490.31 N
Offset from Element Centerline	0 mm

Element#	5/16
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	2000 mm
Inside Diameter	333.35 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.8001 MPa
Design Temperature Internal Pressure	100 °C
Design External Pressure	0.1034 MPa



Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Packing	
Detail ID	Bed#4	
Dist. from "FROM" Node / Offset dist	0.0008717	mm
Height of Packed Section	1140	mm
Density	0.0001828	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.92000002	
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2000	mm
Thickness of Insulation	80	mm
Density	0.0003516	kg/cm³
Element From Node	50	
Detail Type	Weight	
Detail ID	DLLD-4	
Dist. from "FROM" Node / Offset dist	1390	mm
Miscellaneous Weight	147.09	N
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	TALD-4	
Dist. from "FROM" Node / Offset dist	1390	mm
Miscellaneous Weight	49.031	N
Offset from Element Centerline	0	mm

Element#	6/16	
Element From Node	60	
Element To Node	70	
Element Type	Flange	
Description	Flange-1	
Distance "FROM" to "TO"	151.95	mm
Flange Inside Diameter	336.13	mm
Element Thickness	80	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	56	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.8001	MPa
Design Temperature Internal Pressure	100	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-105	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	227	MPa
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K03504	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	



Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	145.95	mm
Thickness of Insulation	80	mm
Density	0.0003516	kg/cm ³

Element#	7/16	
Element From Node	70	
Element To Node	80	
Element Type	Flange	
Description	Flange-2	
Distance "FROM" to "TO"	139.95	mm
Flange Inside Diameter	336.13	mm
Element Thickness	48	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	50	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.8001	MPa
Design Temperature Internal Pressure	100	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-105	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	136.9	mm
Thickness of Insulation	80	mm
Density	0.0003516	kg/cm ³

Element#	8/16	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	3200	mm
Inside Diameter	333.35	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.8001	MPa
Design Temperature Internal Pressure	100	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	239	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	



Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Packing	
Detail ID	Bed#3	
Dist. from "FROM" Node / Offset dist	600	mm
Height of Packed Section	2070	mm
Density	0.0002827	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.92000002	
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3200	mm
Thickness of Insulation	80	mm
Density	0.0003516	kg/cm ³
Element From Node	80	
Detail Type	Nozzle	
Detail ID	H3-6"	
Dist. from "FROM" Node / Offset dist	850	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1080.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	80	
Detail Type	Weight	
Detail ID	DLLD-3	
Dist. from "FROM" Node / Offset dist	2920	mm
Miscellaneous Weight	147.09	N
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	TALD-3	
Dist. from "FROM" Node / Offset dist	2920	mm
Miscellaneous Weight	49.031	N
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	DLBS-3	
Dist. from "FROM" Node / Offset dist	600.01	mm
Miscellaneous Weight	98.062	N
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	TABS-3	
Dist. from "FROM" Node / Offset dist	0.004651	mm
Miscellaneous Weight	49.031	N
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	H3 Davit	
Dist. from "FROM" Node / Offset dist	850.01	mm
Miscellaneous Weight	490.31	N
Offset from Element Centerline	0	mm



Element#	9/16
Element From Node	90
Element To Node	100
Element Type	Flange
Description	Flange-3
Distance "FROM" to "TO"	139.95 mm
Flange Inside Diameter	333.35 mm
Element Thickness	48 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	50 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.8001 MPa
Design Temperature Internal Pressure	100 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.37
Material Name	SA-105
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	227 MPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K03504
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	127 mm
Thickness of Insulation	80 mm
Density	0.0003516 kg/cm³

Element#	10/16
Element From Node	100
Element To Node	110
Element Type	Flange
Description	Flange-4
Distance "FROM" to "TO"	139.95 mm
Flange Inside Diameter	333.35 mm
Element Thickness	48 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	50 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.8001 MPa
Design Temperature Internal Pressure	100 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.37
Material Name	SA-105
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	100
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	127 mm
Thickness of Insulation	80 mm
Density	0.0003516 kg/cm³



Element#	11/16	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell-4	
Distance "FROM" to "TO"	3200	mm
Inside Diameter	333.35	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.8001	MPa
Design Temperature Internal Pressure	100	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.37	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	239	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	110	
Detail Type	Packing	
Detail ID	Bed#2	
Dist. from "FROM" Node / Offset dist	600	mm
Height of Packed Section	2070	mm
Density	0.0002827	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.92000002	
Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3200	mm
Thickness of Insulation	80	mm
Density	0.0003516	kg/cm³
Element From Node	110	
Detail Type	Nozzle	
Detail ID	H2-6"	
Dist. from "FROM" Node / Offset dist	849.91	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1080.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	110	
Detail Type	Weight	
Detail ID	DLLD-2	
Dist. from "FROM" Node / Offset dist	2920	mm
Miscellaneous Weight	147.09	N
Offset from Element Centerline	0	mm



Element From Node 110
Detail Type Weight
Detail ID TALD-2
Dist. from "FROM" Node / Offset dist 2920 mm
Miscellaneous Weight 49.031 N
Offset from Element Centerline 0 mm

Element From Node 110
Detail Type Weight
Detail ID DLBS-2
Dist. from "FROM" Node / Offset dist 600.01 mm
Miscellaneous Weight 98.062 N
Offset from Element Centerline 0 mm

Element From Node 110
Detail Type Weight
Detail ID TABS-2
Dist. from "FROM" Node / Offset dist 600.01 mm
Miscellaneous Weight 49.031 N
Offset from Element Centerline 0 mm

Element From Node 110
Detail Type Weight
Detail ID H2 Davit
Dist. from "FROM" Node / Offset dist 850.01 mm
Miscellaneous Weight 490.31 N
Offset from Element Centerline 0 mm

Element From Node 110
Detail Type Weight
Detail ID Clip weight
Dist. from "FROM" Node / Offset dist 1600 mm
Miscellaneous Weight 1961.2 N
Offset from Element Centerline 0 mm

Element From Node 110
Detail Type Weight
Detail ID MISC wt 7%
Dist. from "FROM" Node / Offset dist 1561.6 mm
Miscellaneous Weight 3922.4 N
Offset from Element Centerline 0 mm

Element# 12/16
Element From Node 120
Element To Node 130
Element Type Flange
Description Flange-5
Distance "FROM" to "TO" 145.95 mm
Flange Inside Diameter 333.35 mm
Element Thickness 54 mm
Internal Corrosion Allowance 3 mm
Nominal Thickness 56 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 1.8001 MPa
Design Temperature Internal Pressure 100 °C
Design External Pressure 0.1034 MPa
Design Temperature External Pressure 100 °C
Effective Diameter Multiplier 1.37
Material Name SA-105
Allowable Stress, Ambient 138 MPa
Allowable Stress, Operating 138 MPa
Allowable Stress, Hydrotest 225 MPa
Material Density 0.00775 kg/cm³
P Number Thickness 32.004 mm
Yield Stress, Operating 227 MPa
UCS-66 Chart Curve Designation A



External Pressure Chart Name	CS-2
UNS Number	K03504
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	120
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	127 mm
Thickness of Insulation	80 mm
Density	0.0003516 kg/cm ³

Element#	13/16
Element From Node	130
Element To Node	140
Element Type	Flange
Description	Flange-6
Distance "FROM" to "TO"	145.95 mm
Flange Inside Diameter	336.13 mm
Element Thickness	54 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	56 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.8001 MPa
Design Temperature Internal Pressure	100 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.37
Material Name	SA-105
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	130
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	127 mm
Thickness of Insulation	80 mm
Density	0.0003516 kg/cm ³

Element#	14/16
Element From Node	140
Element To Node	150
Element Type	Cylinder
Description	Shell-5
Distance "FROM" to "TO"	3570 mm
Inside Diameter	333.35 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.8001 MPa
Design Temperature Internal Pressure	100 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.37
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	239 MPa



UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	140
Detail Type	Packing
Detail ID	Bed#1
Dist. from "FROM" Node / Offset dist	600.02 mm
Height of Packed Section	2070 mm
Density	0.0002827 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.9200002
Element From Node	140
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3570 mm
Thickness of Insulation	80 mm
Density	0.0003516 kg/cm ³
Element From Node	140
Detail Type	Nozzle
Detail ID	H1-6"
Dist. from "FROM" Node / Offset dist	850.01 mm
Nozzle Diameter	6 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1043.7 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	140
Detail Type	Nozzle
Detail ID	N2-3"
Dist. from "FROM" Node / Offset dist	3400 mm
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	383.72 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	140
Detail Type	Weight
Detail ID	DLLD-1
Dist. from "FROM" Node / Offset dist	2920 mm
Miscellaneous Weight	147.09 N
Offset from Element Centerline	0 mm
Element From Node	140
Detail Type	Weight
Detail ID	TALD-1
Dist. from "FROM" Node / Offset dist	2920 mm
Miscellaneous Weight	49.031 N
Offset from Element Centerline	0 mm
Element From Node	140
Detail Type	Weight
Detail ID	TABS-1
Dist. from "FROM" Node / Offset dist	600.01 mm



Miscellaneous Weight	49.031	N
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	BSLD-1	
Dist. from "FROM" Node / Offset dist	600.01	mm
Miscellaneous Weight	98.062	N
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	DLMP	
Dist. from "FROM" Node / Offset dist	3500	mm
Miscellaneous Weight	98.062	N
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	TAMP	
Dist. from "FROM" Node / Offset dist	3500	mm
Miscellaneous Weight	49.031	N
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	DLFP	
Dist. from "FROM" Node / Offset dist	3400	mm
Miscellaneous Weight	49.031	N
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	H1 davit	
Dist. from "FROM" Node / Offset dist	850.01	mm
Miscellaneous Weight	490.31	N
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	Trunnion 1	
Dist. from "FROM" Node / Offset dist	2570	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	250	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	Trunnion 2	
Dist. from "FROM" Node / Offset dist	2570	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	250	mm
Element From Node	140	
Detail Type	For./Mom.	
Detail ID	N3 FY	
Dist. from "FROM" Node / Offset dist	3570	mm
Force in X Direction	2750	N
Force in Y Direction	-4123.9	N
Force in Z Direction	2750	N
Moment about X Axis	385021	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	385021	N-mm
Force/Moment Combination Method	SRSS	



Element Type	Flange
Description	Flange-7
Distance "FROM" to "TO"	145.95 mm
Flange Inside Diameter	333.35 mm
Element Thickness	54 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	56 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.8001 MPa
Design Temperature Internal Pressure	100 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.37
Material Name	SA-105
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	227 MPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K03504
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	150
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	127 mm
Thickness of Insulation	80 mm
Density	0.0003516 kg/cm³

Element#	16/16
Element From Node	160
Element To Node	170
Element Type	Flange
Description	Flange-8
Distance "FROM" to "TO"	52 mm
Flange Outside Diameter	535 mm
Element Thickness	52 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	54 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.8001 MPa
Design Temperature Internal Pressure	100 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.37
Material Name	SA-105
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	160
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	35.052 mm
Thickness of Insulation	80 mm
Density	0.0003516 kg/cm³
Element From Node	160
Detail Type	Nozzle
Detail ID	N3-3"
Dist. from "FROM" Node / Offset dist	0 mm



Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	220.28 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-2	...	3200	...	3200	...
Skirt-1	...	4200	...	1000	...
Bottom head	...	4250	...	50	...
Shell-1	...	7750	...	3500	...
Shell-2	...	9750	...	2000	...
Flange-1	...	9906.45	...	151.948	...
Flange-2	...	10046.4	...	139.948	...
Shell-3	...	13246.4	...	3200	...
Flange-3	...	13390.8	...	139.948	...
Flange-4	...	13530.8	...	139.948	...
Shell-4	...	16730.8	...	3200	...
Flange-5	...	16881.2	...	145.948	...
Flange-6	...	17027.2	...	145.948	...
Shell-5	...	20597.2	...	3570	...
Flange-7	...	20747.6	...	145.948	...
Flange-8	...	20799.6	...	52	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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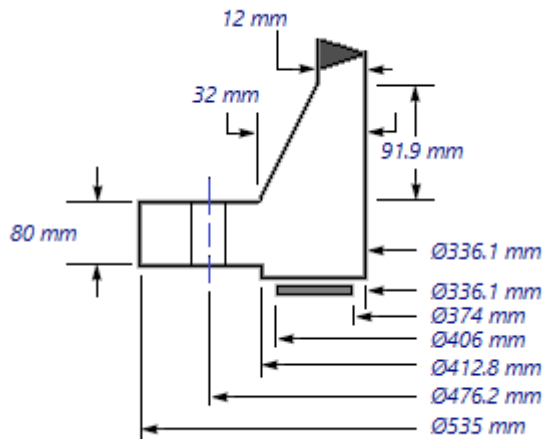


Flange Input Data Values

Description: Flange-1 :

Flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	1.80	MPa
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	336.131	mm
Flange Outside Diameter	A	535.000	mm
Flange Thickness	t	80.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	31.9596	mm
Length of Hub	h	91.9480	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	476.250	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	412.750	mm
Flange Face Inside Diameter	Fid	336.131	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	406.000	mm
Gasket Inside Diameter	Gi	374.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		18426.10	N
Bending Moment on Flange		35555700.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot (336.131/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8) + Ca$$

$$= 5.2491 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (171.0655 + 0.6 \cdot 9.0)$$

$$= 7.038 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (168.0655 + 0.6 \cdot 12.0)$$

$$= 9.448 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	342.131 mm
Corroded Large Hub,	g1Cor = g1-ci	28.960 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	38.100 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	391.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 391.7452^2 \cdot 1.8$$

$$= 216949.188 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1274 \cdot \pi \cdot 391.7452 \cdot 3.0 \cdot 1.8$$

$$= 94732.133 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 342.131^2 \cdot 1.8001$$

$$= 165476.203 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 216949 - 165476$$

$$= 51472.984 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(216949 + 94732 + 0, 0)$$

$$= 311681.281 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1274 \cdot \pi \cdot 391.745 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 604759.312 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((311681 + 18426 + 4 \cdot 35555700 / 391.75) / 172, 604759 / 172)$$

$$= 4029.541 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 80.0 / (3.0 + 0.5)$$

$$= 187.943 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 476.25 \cdot \sin(\pi / 20)$$

$$= 74.502 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{74.502}{(2 \cdot 25.4 + 80.0)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4029.541	7109.663	
Radial Distance between Hub and Bolts:	34.925	38.100	
Radial Distance between Bolts and Edge:	26.987	29.375	
Circ. Spacing between the Bolts:	57.150	74.502	187.943

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi(406.0 + 374.0))$$

$$= 7.238 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 7109.663$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 311681 - 216949$$

$$= 94732.11 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (476.25 - 391.7452) / 2$$

$$= 42.2524 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((476.25 - 342.131) / 2 + 42.2524)$$

$$= 54.6560 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (476.25 - 342.131 - 28.96) / 2$$

$$= 52.5797 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 96.434, Bb = 80.0, Cc = 91.948 \text{ and } Ddg = 18.98 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 96.434 \cdot 80.0^3 \cdot (1/3 - 0.210 \cdot (80.0/96.434) \cdot (1 - 1/12 \cdot (80.0/96.434)^4))$$

$$= 8196037.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.95 \cdot 18.98^3 \cdot (1/3 - 0.105 \cdot (18.98/91.95) \cdot (1 - 1/192 \cdot (18.98/91.95)^4))$$

$$= 195928.344 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 500.153 + 11.956$$

$$= 8391966.000 \text{ mm}^4$$



Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 96.434 \cdot 80.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{80.0}{96.434} \right) \left(1 - \frac{1}{12} \left(\frac{80.0}{96.434} \right)^4 \right) \right)$$

$$= 8196037.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(8391966, 8196037)$$

$$= 8391966.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.239 \cdot 9.0^2 \cdot 55.49 \cdot 342.131 / 0.049$$

$$= 6100025.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	165476.	52.5797	1.0000	8704216.
Face Pressure, Mt	51473.	54.6560	1.0000	2814446.
Gasket Load, Mg	94732.	42.2524	1.0000	4004283.
Gasket Seating, Matm	1222736.	42.2524	1.0000	51684496.
Moment Load, Moe	...	...	...	14154979.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 35555700 \left[\frac{1}{(0.384 \cdot 8391966 + 6100025)} \right] \cdot \left(\frac{52.6}{(476.2 - 2 \cdot 52.6)} \right) +$$

$$18426 \cdot 52.58$$

$$= 14154979.000 \text{ N-mm}$$

Total Moment for Operation, Mo 29677928. N-mm
Total Moment for Gasket seating, Mg 51684496. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 55.490 mm
Hub Ratio, h/h0 = HL / H0 1.657
Thickness Ratio, g1/g0 = (g1Cor/goCor) 3.218

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.599
Factor V per 4.16.5 0.049
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 535.0 / 342.131$$

$$= 1.564$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.564^2 \cdot \log_{10}(1.564))) / (1.564^2 - 1) / (1.564 - 1)$$

$$= 4.517$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.564^2(1 + 8.55246 \cdot \log_{10}(1.564)) - 1) / ((1.0472 + 1.9448 \cdot 1.564^2) \cdot (1.564 - 1))$$

$$= 1.683$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.564^2(1 + 8.55246 \cdot \log_{10}(1.564)) - 1) / (1.36136(1.564^2 - 1) \cdot (1.564 - 1))$$

$$= 4.964$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.564^2 + 1) / (1.564^2 - 1)$$

$$= 2.384$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.964 \cdot 9.0^2 \cdot 55.49 / 0.049$$

$$= 452181.719 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.599 / 55.49$$

$$= 0.011 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (80.0 \cdot 0.011 + 1) / 1.683 + 80.0^3 / 452181.719$$

$$= 2.239$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 29677924 / 342.131) / (2.2395 \cdot 28.9596^2)$$

$$= 46.17 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 51684488 / 342.131) / (2.2395 \cdot 28.9596^2)$$

$$= 80.41 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (2.1513 \cdot 29677924 / 342.131) / (2.2395 \cdot 80.0^2)$$

$$= 13.02 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (2.1513 \cdot 51684488 / 342.131) / (2.2395 \cdot 80.0^2)$$

$$= 22.67 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (4.5172 \cdot 29677924 / (80.0^2 \cdot 342.131)) - 2.3838 \cdot 13$$

$$= 30.18 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (4.5172 \cdot 51684488 / (80.0^2 \cdot 342.131)) - 2.3838 \cdot 23$$

$$= 52.55 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (46 + \max(13, 30)) / 2$$

$$= 38.17 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (80 + \max(23, 53)) / 2$$

$$= 66.48 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (W_{m1} + F_a + 4M_e / G) / A_b$$

$$= 693010 / 7109.6631$$

$$= 97.48 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (604759 / 7109.6631)$$

$$= 85.07 \text{ MPa}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	46.17	207.00	80.41	207.00
Radial Flange	13.02	138.00	22.67	138.00
Tangential Flange	30.18	138.00	52.55	138.00
Maximum Average	38.17	138.00	66.48	138.00
Bolting	97.48	172.00	85.07	172.00

Minimum Required Flange Thickness 45.390 mm

Estimated M.A.W.P. (Operating) 5 MPa

Estimated Finished Weight of Flange at given Thk. 102.0 kgm

Estimated Unfinished Weight of Forging at given Thk 181.3 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	453.55	499.99	0.907	453.55	499.99	0.907
Tangential Flange	St	296.44	374.99	0.791	296.44	374.99	0.791
Tangential Flange	Sto	113.39	250.00	0.454	113.39	250.00	0.454
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	424.30	749.99	0.566	424.30	749.99	0.566
Combined	Sh + Sto	566.94	749.99	0.756	566.94	749.99	0.756
Radial Flange	Sr	127.86	499.99	0.256	127.86	499.99	0.256
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; -1.2374/6.1137) \\ = 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 4.2952/8.0077) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.4446/2.0876) \\ = 0.2500$$

In this case, the hub ratio $h/h0$, 1.6570 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	970.2 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	355.4832 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	227.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.181 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (406.0^2 - 374.0^2)$$

$$= 19603.5 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 19603.523 / (20 \cdot 355.483)$$

$$= 474.3 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(474.257; 507.5)$$

$$= 474.3 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(474.257; 290.0)$$

$$= 474.3 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(474.257; 970.167)$$

$$= 474.3 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 474.257 \geq 140.0(19603.523 / (20 \cdot 355.483))$$

$$:: 474.3 \geq 386.0 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 474.257 \geq (97.0 \cdot 19603.523 + \pi/4 \cdot 1.8 \cdot 374.0^2) / (0.7 \cdot 355.483 \cdot 20)$$

$$:: 474.3 \geq 421.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 474.257 \leq 380.0(19603.523 / (20 \cdot 355.483))$$

$$:: 474.3 \leq 1047.8 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 474.257 \leq 970.167(1.0/0.181)$$

$$:: 474.3 \leq 5372.1 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 474.257 \cdot 0.2 \cdot 355.483 \cdot 25.4$$

$$= 856713.1 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 MPa

Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51684488 / 1.0 \cdot 1.0 \cdot 0.049 / (1.092 \cdot 202713 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.445 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 29677924 / 1.0 \cdot 1.0 \cdot 0.049 / (1.092 \cdot 198163 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.261 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51684488 / 1.0 \cdot 1.0 \cdot 0.049 / (2.239 \cdot 202713 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.217 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 29677924 / 1.0 \cdot 1.0 \cdot 0.049 / (2.239 \cdot 198163 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.127 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	45.31	207.00	75.91	207.00
Radial Flange	15.14	138.00	25.37	138.00
Tangential Flange	27.49	138.00	46.06	138.00
Maximum Average	36.40	138.00	60.99	138.00
Bolting	97.48	172.00	85.07	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 43.688 mm

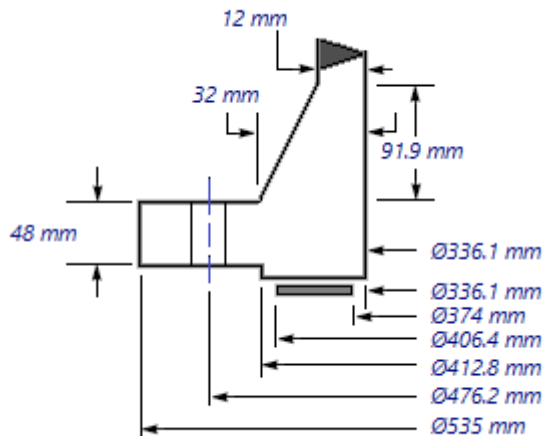
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Flange Input Data Values

Description: Flange-2 :

Flange-2

Flange Type	Integral Weld Neck		
Design Pressure	P	1.80	MPa
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	336.131	mm
Flange Outside Diameter	A	535.000	mm
Flange Thickness	t	48.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	31.9596	mm
Length of Hub	h	91.9480	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	476.250	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	412.750	mm
Flange Face Inside Diameter	Fid	336.131	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	406.400	mm
Gasket Inside Diameter	Gi	374.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		17268.70	N
Bending Moment on Flange		34827400.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot (336.131/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8) + Ca$$

$$= 5.2491 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (171.0655 + 0.6 \cdot 9.0)$$

$$= 7.038 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (168.0655 + 0.6 \cdot 12.0)$$

$$= 9.448 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	342.131 mm
Corroded Large Hub,	g1Cor = g1-ci	28.960 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	38.100 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.200 mm
Basic Gasket Width,	bo = N/2	8.100 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.172 mm
Gasket Reaction Diameter,	G = Go - 2 * b	392.056 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 392.0564^2 \cdot 1.8$$

$$= 217293.984 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1718 \cdot \pi \cdot 392.0564 \cdot 3.0 \cdot 1.8$$

$$= 95398.094 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 342.131^2 \cdot 1.8001$$

$$= 165476.203 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 217294 - 165476$$

$$= 51817.793 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(217294 + 95398 + 0, 0)$$

$$= 312692.094 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1718 \cdot \pi \cdot 392.056 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 609010.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((312692 + 17269 + 4 \cdot 34827400 / 392.06) / 172, 609011 / 172)$$

$$= 3983.825 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 48.0 / (3.0 + 0.5)$$

$$= 133.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 476.25 \cdot \sin(\pi / 20)$$

$$= 74.502 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{74.502}{(2 \cdot 25.4 + 48.0)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3983.825	7109.663	
Radial Distance between Hub and Bolts:	34.925	38.100	
Radial Distance between Bolts and Edge:	26.987	29.375	
Circ. Spacing between the Bolts:	57.150	74.502	133.086

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi(406.4 + 374.0))$$

$$= 7.234 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 7109.663$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 312692 - 217294$$

$$= 95398.12 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (476.25 - 392.0564) / 2$$

$$= 42.0968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((476.25 - 342.131) / 2 + 42.0968)$$

$$= 54.5782 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (476.25 - 342.131 - 28.96) / 2$$

$$= 52.5797 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 96.434, Bb = 48.0, Cc = 91.948 \text{ and } Ddg = 18.98 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 96.434 \cdot 48.0^3 \cdot (1/3 - 0.210 \cdot (48.0/96.434) \cdot (1 - 1/12 \cdot (48.0/96.434)^4))$$

$$= 2445893.750 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.95 \cdot 18.98^3 \cdot (1/3 - 0.105 \cdot (18.98/91.95) \cdot (1 - 1/192 \cdot (18.98/91.95)^4))$$

$$= 195928.344 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 149.258 + 11.956$$

$$= 2641822.250 \text{ mm}^4$$



Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 96.434 \cdot 48.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{48.0}{96.434} \right) \left(1 - \frac{1}{12} \left(\frac{48.0}{96.434} \right)^4 \right) \right)$$

$$= 2445893.750 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(2641822, 2445894)$$

$$= 2641822.2 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.147 \cdot 9.0^2 \cdot 55.49 \cdot 342.131 / 0.049$$

$$= 3123020.750 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	165476.	52.5797	1.0000	8704216.
Face Pressure, Mt	51818.	54.5782	1.0000	2829266.
Gasket Load, Mg	95398.	42.0968	1.0000	4017584.
Gasket Seating, Matm	1222736.	42.0968	1.0000	51494156.
Moment Load, Moe	...	...	...	15807388.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 34827400 \left[\frac{1}{(0.384 \cdot 2641822 + 3123021)} \right] \cdot \left(\frac{52.6}{(476.2 - 2 \cdot 52.6)} \right) +$$

$$17269 \cdot 52.58$$

$$= 15807388.000 \text{ N-mm}$$

Total Moment for Operation, Mo	31358458. N-mm
Total Moment for Gasket seating, Mg	51494156. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	55.490 mm
Hub Ratio, h/h0 = HL / H0	1.657
Thickness Ratio, g1/g0 = (g1Cor/goCor)	3.218

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.599
Factor V per 4.16.5	0.049
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 535.0 / 342.131$$

$$= 1.564$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.564^2 \cdot \log_{10}(1.564))) / (1.564^2 - 1)) / (1.564 - 1)$$

$$= 4.517$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.564^2 (1 + 8.55246 \cdot \log_{10}(1.564)) - 1) / ((1.0472 + 1.9448 \cdot 1.564^2) \cdot (1.564 - 1))$$

$$= 1.683$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.564^2 (1 + 8.55246 \cdot \log_{10}(1.564)) - 1) / (1.36136 (1.564^2 - 1) \cdot (1.564 - 1))$$

$$= 4.964$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.564^2 + 1) / (1.564^2 - 1)$$

$$= 2.384$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.964 \cdot 9.0^2 \cdot 55.49 / 0.049$$

$$= 452181.719 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.599 / 55.49$$

$$= 0.011 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (48.0 \cdot 0.011 + 1) / 1.683 + 48.0^3 / 452181.719$$

$$= 1.147$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 31358454 / 342.131) / (1.1465 \cdot 28.9596^2)$$

$$= 95.29 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 51494156 / 342.131) / (1.1465 \cdot 28.9596^2)$$

$$= 156.48 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.6908 \cdot 31358454 / 342.131) / (1.1465 \cdot 48.0^2)$$

$$= 58.65 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.6908 \cdot 51494156 / 342.131) / (1.1465 \cdot 48.0^2)$$

$$= 96.31 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.5172 \cdot 31358454 / (48.0^2 \cdot 342.131)) - 2.3838 \cdot 59$$

$$= 39.84 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.5172 \cdot 51494156 / (48.0^2 \cdot 342.131)) - 2.3838 \cdot 96$$

$$= 65.42 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (95 + \max(59, 40)) / 2$$

$$= 76.97 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (156 + \max(96, 65)) / 2$$

$$= 126.39 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm_1 + Fa + 4Me/G) / Ab$$

$$= 685147 / 7109.6631$$

$$= 96.38 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm_2 / Ab)$$

$$= (609011 / 7109.6631)$$

$$= 85.67 \text{ MPa}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	95.29	207.00	156.48	207.00
Radial Flange	58.65	138.00	96.31	138.00
Tangential Flange	39.84	138.00	65.42	138.00
Maximum Average	76.97	138.00	126.39	138.00
Bolting	96.38	172.00	85.67	172.00

Minimum Required Flange Thickness 45.288 mm

Estimated M.A.W.P. (Operating) 5 MPa

Estimated Finished Weight of Flange at given Thk. 68.2 kgm

Estimated Unfinished Weight of Forging at given Thk 147.6 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	209.03	374.99	0.557	209.03	374.99	0.557
Tangential Flange	Sto	125.00	250.00	0.500	125.00	250.00	0.500
Combined	Sh/f + St	709.02	749.99	0.945	709.02	749.99	0.945
Combined	Sr + St	516.75	749.99	0.689	516.75	749.99	0.689
Combined	Sh + Sto	624.99	749.99	0.833	624.99	749.99	0.833
Radial Flange	Sr	307.72	499.99	0.615	307.72	499.99	0.615
Maximum Stress Ratio				1.000			1.000



Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{\text{num}} / F1_{\text{den}})$$

$$= \max(1; -1.2374/6.1137)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_2) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; f_3)$$

$$= \max(0.25; 4.2952/8.0077) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; 0.4446/2.0876)$$

$$= 0.2500$$

In this case, the hub ratio h/h_0 , 1.6570 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	S_{bmax}	507.5 MPa
Minimum permissible bolt stress	S_{bmin}	290.0 MPa
Max. permissible bolt stress prior to flange damage	S_{fmax}	549.6 MPa
Target assembly gasket stress	S_{gT}	172.0 MPa
Maximum permissible gasket stress	S_{gmax}	380.0 MPa
Minimum gasket operating stress	S_{gmin-O}	97.0 MPa
Minimum gasket seating stress	S_{gmin-S}	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Φ_{hig}	0.700
Root Area of a Single Bolt	Abs	355.4832 mm ²
Max. permissible single flange rotation for gasket	Θ_{tagmax}	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	227.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.199 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (406.4^2 - 374.0^2)$$

$$= 19858.7 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 19858.748 / (20 \cdot 355.483)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(480.431; 507.5)$$

$$= 480.4 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(480.431; 290.0)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(480.431; 549.578)$$

$$= 480.4 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 480.431 \geq 140.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \geq 391.0 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 480.431 \geq (97.0 \cdot 19858.748 + \pi/4 \cdot 1.8 \cdot 374.0^2) / (0.7 \cdot 355.483 \cdot 20)$$

$$:: 480.4 \geq 426.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 480.431 \leq 380.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \leq 1061.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 480.431 \leq 549.578(1.0/0.199)$$

$$:: 480.4 \leq 2760.5 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 480.431 \cdot 0.2 \cdot 355.483 \cdot 25.4$$

$$= 867866.9 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 MPa

Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.09 \cdot 202713 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.444 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 31358454 / 1.0 \cdot 1.0 \cdot 0.049 / (1.09 \cdot 198163 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.277 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.147 \cdot 202713 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.422 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 31358454 / 1.0 \cdot 1.0 \cdot 0.049 / (1.147 \cdot 198163 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.263 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	87.75	207.00	138.07	207.00
Radial Flange	64.45	138.00	101.40	138.00
Tangential Flange	35.36	138.00	55.63	138.00
Maximum Average	76.10	138.00	119.73	138.00
Bolting	96.38	172.00	85.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 43.561 mm

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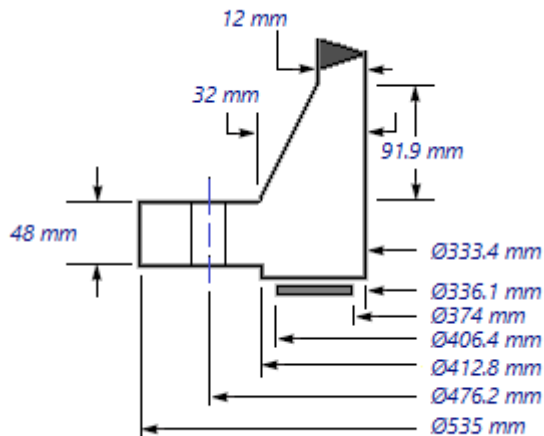


Flange Input Data Values

Description: Flange-3 :

Flange-3

Flange Type	Integral Weld Neck		
Design Pressure	P	1.80	MPa
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	333.350	mm
Flange Outside Diameter	A	535.000	mm
Flange Thickness	t	48.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	31.9596	mm
Length of Hub	h	91.9480	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	476.250	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	412.750	mm
Flange Face Inside Diameter	Fid	336.131	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	406.400	mm
Gasket Inside Diameter	Gi	374.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		12354.10	N
Bending Moment on Flange		18612800.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot (333.35/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8) + Ca$$

$$= 5.2308 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (169.675 + 0.6 \cdot 9.0)$$

$$= 7.094 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (166.675 + 0.6 \cdot 12.0)$$

$$= 9.524 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	339.350 mm
Corroded Large Hub,	g1Cor = g1-ci	28.960 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	39.490 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.200 mm
Basic Gasket Width,	bo = N/2	8.100 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.172 mm
Gasket Reaction Diameter,	G = Go - 2 * b	392.056 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 392.0564^2 \cdot 1.8$$

$$= 217293.984 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1718 \cdot \pi \cdot 392.0564 \cdot 3.0 \cdot 1.8$$

$$= 95398.094 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 339.35^2 \cdot 1.8001$$

$$= 162796.969 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 217294 - 162797$$

$$= 54497.016 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(217294 + 95398 + 0, 0)$$

$$= 312692.094 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1718 \cdot \pi \cdot 392.056 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 609010.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((312692 + 12354 + 4 \cdot 18612800/392.06) / 172, 609011/172)$$

$$= 3541.125 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 48.0/(3.0 + 0.5)$$

$$= 133.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 476.25 \cdot \sin(\pi/20)$$

$$= 74.502 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{74.502/(2 \cdot 25.4 + 48.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3541.125	7109.663	
Radial Distance between Hub and Bolts:	34.925	39.490	
Radial Distance between Bolts and Edge:	26.987	29.375	
Circ. Spacing between the Bolts:	57.150	74.502	133.086

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi(406.4 + 374.0))$$

$$= 7.234 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 7109.6631$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 312692 - 217294$$

$$= 95398.12 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (476.25 - 392.0564) / 2$$

$$= 42.0968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((476.25 - 339.35) / 2 + 42.0968)$$

$$= 55.2734 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (476.25 - 339.35 - 28.96) / 2$$

$$= 53.9702 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 97.825, Bb = 48.0, Cc = 91.948 \text{ and } Ddg = 18.98 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 97.825 \cdot 48.0^3 \cdot (1/3 - 0.210 \cdot (48.0/97.825) \cdot (1 - 1/12 \cdot (48.0/97.825)^4))$$

$$= 2496835.750 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.95 \cdot 18.98^3 \cdot (1/3 - 0.105 \cdot (18.98/91.95) \cdot (1 - 1/192 \cdot (18.98/91.95)^4))$$

$$= 195928.344 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 152.366 + 11.956$$

$$= 2692764.250 \text{ mm}^4$$



Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 97.825 \cdot 48.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{48.0}{97.825} \right) \left(1 - \frac{1}{12} \left(\frac{48.0}{97.825} \right)^4 \right) \right)$$

$$= 2496835.750 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip, \text{term1}, Ip, \text{term2})$$

$$= \max(2692764, 2496836)$$

$$= 2692764.2 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.155 \cdot 9.0^2 \cdot 55.264 \cdot 339.35 / 0.049$$

$$= 3116023.250 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	162797.	53.9702	1.0000	8789748.
Face Pressure, Mt	54497.	55.2734	1.0000	3013457.
Gasket Load, Mg	95398.	42.0968	1.0000	4017584.
Gasket Seating, Matm	1222736.	42.0968	1.0000	51494156.
Moment Load, Moe	...	...	...	8858471.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 18612800 \left[\frac{1}{(0.384 \cdot 2692764 + 3116023)} \right] \cdot \left(\frac{54.0}{(476.2 - 2 \cdot 54.0)} \right) +$$

$$12354 \cdot 53.97$$

$$= 8858471.000 \text{ N-mm}$$

Total Moment for Operation, Mo	24679262. N-mm
Total Moment for Gasket seating, Mg	51494156. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	55.264 mm
Hub Ratio, h/h0 = HL / H0	1.664
Thickness Ratio, g1/g0 = (g1Cor/goCor)	3.218

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.598
Factor V per 4.16.5	0.049
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 535.0 / 339.35$$

$$= 1.577$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.577^2 \cdot \log_{10}(1.577))) / (1.577^2 - 1) / (1.577 - 1)$$

$$= 4.440$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.577^2(1 + 8.55246 \cdot \log_{10}(1.577)) - 1) / ((1.0472 + 1.9448 \cdot 1.577^2) \cdot (1.577 - 1))$$

$$= 1.678$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.577^2(1 + 8.55246 \cdot \log_{10}(1.577)) - 1) / (1.36136(1.577^2 - 1) \cdot (1.577 - 1))$$

$$= 4.879$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.577^2 + 1) / (1.577^2 - 1)$$

$$= 2.346$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.879 \cdot 9.0^2 \cdot 55.264 / 0.049$$

$$= 443756.438 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.598 / 55.264$$

$$= 0.011 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (48.0 \cdot 0.011 + 1) / 1.678 + 48.0^3 / 443756.438$$

$$= 1.155$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 24679260 / 339.35) / (1.155 \cdot 28.9596^2)$$

$$= 75.05 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 51494156 / 339.35) / (1.155 \cdot 28.9596^2)$$

$$= 156.60 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.6928 \cdot 24679260 / 339.35) / (1.155 \cdot 48.0^2)$$

$$= 46.25 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.6928 \cdot 51494156 / 339.35) / (1.155 \cdot 48.0^2)$$

$$= 96.49 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.4395 \cdot 24679260 / (48.0^2 \cdot 339.35)) - 2.3464 \cdot 46$$

$$= 31.58 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.4395 \cdot 51494156 / (48.0^2 \cdot 339.35)) - 2.3464 \cdot 96$$

$$= 65.89 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (75 + \max(46, 32)) / 2$$

$$= 60.65 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (157 + \max(96, 66)) / 2$$

$$= 126.55 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm_1 + Fa + 4Me/G) / Ab$$

$$= 514868 / 7109.6631$$

$$= 72.42 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm_2 / Ab)$$

$$= (609011 / 7109.6631)$$

$$= 85.67 \text{ MPa}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	75.05	207.00	156.60	207.00
Radial Flange	46.25	138.00	96.49	138.00
Tangential Flange	31.58	138.00	65.89	138.00
Maximum Average	60.65	138.00	126.55	138.00
Bolting	72.42	172.00	85.67	172.00

Minimum Required Flange Thickness 45.339 mm

Estimated M.A.W.P. (Operating) 5 MPa

Estimated Finished Weight of Flange at given Thk. 68.6 kgm

Estimated Unfinished Weight of Forging at given Thk 149.2 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	210.37	374.99	0.561	210.37	374.99	0.561
Tangential Flange	Sto	125.00	250.00	0.500	125.00	250.00	0.500
Combined	Sh/f + St	710.36	749.99	0.947	710.36	749.99	0.947
Combined	Sr + St	518.46	749.99	0.691	518.46	749.99	0.691
Combined	Sh + Sto	624.99	749.99	0.833	624.99	749.99	0.833
Radial Flange	Sr	308.08	499.99	0.616	308.08	499.99	0.616
Maximum Stress Ratio				1.000			1.000



Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; -1.2797/6.1518) \\ = 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_2) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; f_3) \\ = \max(0.25; 4.3059/8.0644) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; 0.4414/2.1013) \\ = 0.2500$$

In this case, the hub ratio h/h_0 , 1.6638 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	S_{bmax}	507.5 MPa
Minimum permissible bolt stress	S_{bmin}	290.0 MPa
Max. permissible bolt stress prior to flange damage	S_{fmax}	549.1 MPa
Target assembly gasket stress	S_{gT}	172.0 MPa
Maximum permissible gasket stress	S_{gmax}	380.0 MPa
Minimum gasket operating stress	S_{gmin-O}	97.0 MPa
Minimum gasket seating stress	S_{gmin-S}	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Φ_{hig}	0.700
Root Area of a Single Bolt	Abs	355.4832 mm ²
Max. permissible single flange rotation for gasket	Θ_{tagmax}	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	227.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.198 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (406.4^2 - 374.0^2)$$

$$= 19858.7 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 19858.748 / (20 \cdot 355.483)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(480.431; 507.5)$$

$$= 480.4 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(480.431; 290.0)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(480.431; 549.148)$$

$$= 480.4 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 480.431 \geq 140.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \geq 391.0 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 480.431 \geq (97.0 \cdot 19858.748 + \pi/4 \cdot 1.8 \cdot 374.0^2) / (0.7 \cdot 355.483 \cdot 20)$$

$$:: 480.4 \geq 426.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 480.431 \leq 380.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \leq 1061.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 480.431 \leq 549.148(1.0/0.198)$$

$$:: 480.4 \leq 2776.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 480.431 \cdot 0.2 \cdot 355.483 \cdot 25.4$$

$$= 867866.9 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 MPa

Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.098 \cdot 202713 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.441 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 24679260 / 1.0 \cdot 1.0 \cdot 0.049 / (1.098 \cdot 198163 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.216 \quad (\text{should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.155 \cdot 202713 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.420 \quad (\text{should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 24679260 / 1.0 \cdot 1.0 \cdot 0.049 / (1.155 \cdot 198163 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.206 \quad (\text{should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	68.68	207.00	138.24	207.00
Radial Flange	50.51	138.00	101.66	138.00
Tangential Flange	27.88	138.00	56.11	138.00
Maximum Average	59.60	138.00	119.95	138.00
Bolting	72.42	172.00	85.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 43.637 mm

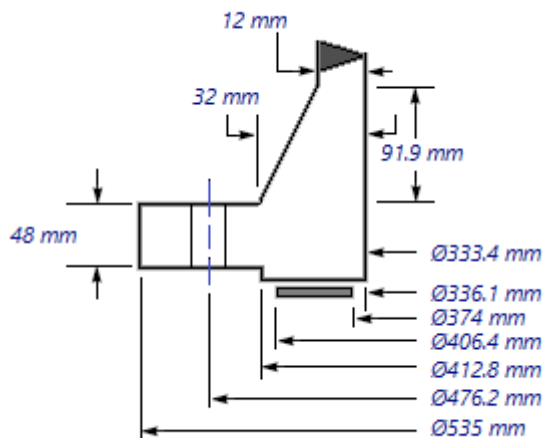
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Flange Input Data Values

Description: Flange-4 :

Flange-4

Flange Type	Integral Weld Neck		
Design Pressure	P	1.80	MPa
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	333.350	mm
Flange Outside Diameter	A	535.000	mm
Flange Thickness	t	48.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	31.9596	mm
Length of Hub	h	91.9480	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	476.250	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	412.750	mm
Flange Face Inside Diameter	Fid	336.131	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	406.400	mm
Gasket Inside Diameter	Gi	374.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		11710.50	N
Bending Moment on Flange		18011600.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot (333.35/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8) + Ca$$

$$= 5.2308 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (169.675 + 0.6 \cdot 9.0)$$

$$= 7.094 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (166.675 + 0.6 \cdot 12.0)$$

$$= 9.524 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	339.350 mm
Corroded Large Hub,	g1Cor = g1-ci	28.960 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	39.490 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.200 mm
Basic Gasket Width,	bo = N/2	8.100 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.172 mm
Gasket Reaction Diameter,	G = Go - 2 * b	392.056 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 392.0564^2 \cdot 1.8$$

$$= 217293.984 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1718 \cdot \pi \cdot 392.0564 \cdot 3.0 \cdot 1.8$$

$$= 95398.094 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 339.35^2 \cdot 1.8001$$

$$= 162796.969 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 217294 - 162797$$

$$= 54497.016 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(217294 + 95398 + 0, 0)$$

$$= 312692.094 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1718 \cdot \pi \cdot 392.056 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 609010.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((312692 + 11711 + 4 \cdot 18011600/392.06) / 172, 609011/172)$$

$$= 3541.125 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 48.0/(3.0 + 0.5)$$

$$= 133.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 476.25 \cdot \sin(\pi / 20)$$

$$= 74.502 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{74.502}{(2 \cdot 25.4 + 48.0)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3541.125	7109.663	
Radial Distance between Hub and Bolts:	34.925	39.490	
Radial Distance between Bolts and Edge:	26.987	29.375	
Circ. Spacing between the Bolts:	57.150	74.502	133.086

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi(406.4 + 374.0))$$

$$= 7.234 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 7109.663$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 312692 - 217294$$

$$= 95398.12 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (476.25 - 392.0564) / 2$$

$$= 42.0968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((476.25 - 339.35) / 2 + 42.0968)$$

$$= 55.2734 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (476.25 - 339.35 - 28.96) / 2$$

$$= 53.9702 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 97.825, Bb = 48.0, Cc = 91.948 \text{ and } Ddg = 18.98 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 97.825 \cdot 48.0^3 \cdot (1/3 - 0.210 \cdot (48.0/97.825) \cdot (1 - 1/12 \cdot (48.0/97.825)^4))$$

$$= 2496835.750 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.95 \cdot 18.98^3 \cdot (1/3 - 0.105 \cdot (18.98/91.95) \cdot (1 - 1/192 \cdot (18.98/91.95)^4))$$

$$= 195928.344 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 152.366 + 11.956$$

$$= 2692764.250 \text{ mm}^4$$



Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 97.825 \cdot 48.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{48.0}{97.825} \right) \left(1 - \frac{1}{12} \left(\frac{48.0}{97.825} \right)^4 \right) \right)$$

$$= 2496835.750 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip, \text{term1}, Ip, \text{term2})$$

$$= \max(2692764, 2496836)$$

$$= 2692764.2 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.155 \cdot 9.0^2 \cdot 55.264 \cdot 339.35 / 0.049$$

$$= 3116023.250 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	162797.	53.9702	1.0000	8789748.
Face Pressure, Mt	54497.	55.2734	1.0000	3013457.
Gasket Load, Mg	95398.	42.0968	1.0000	4017584.
Gasket Seating, Matm	1222736.	42.0968	1.0000	51494156.
Moment Load, Moe	...	...	...	8559135.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 18011600 \left[\frac{1}{(0.384 \cdot 2692764 + 3116023)} \right] \cdot \left(\frac{54.0}{(476.2 - 2 \cdot 54.0)} \right) +$$

$$11711 \cdot 53.97$$

$$= 8559135.000 \text{ N-mm}$$

Total Moment for Operation, Mo 24379926. N-mm
Total Moment for Gasket seating, Mg 51494156. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 55.264 mm
Hub Ratio, h/h0 = HL / H0 1.664
Thickness Ratio, g1/g0 = (g1Cor/goCor) 3.218

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.598
Factor V per 4.16.5 0.049
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 535.0 / 339.35$$

$$= 1.577$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot (1.577^2 \cdot \log_{10}(1.577))) / (1.577^2 - 1)) / (1.577 - 1)$$

$$= 4.440$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 / (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.577^2 / (1 + 8.55246 \cdot \log_{10}(1.577)) - 1) / ((1.0472 + 1.9448 \cdot 1.577^2) \cdot (1.577 - 1))$$

$$= 1.678$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 / (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 \cdot (K^2 - 1) \cdot (K - 1))$$

$$= (1.577^2 / (1 + 8.55246 \cdot \log_{10}(1.577)) - 1) / (1.36136 \cdot (1.577^2 - 1) \cdot (1.577 - 1))$$

$$= 4.879$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.577^2 + 1) / (1.577^2 - 1)$$

$$= 2.346$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 4.879 \cdot 9.0^2 \cdot 55.264 / 0.049 \\
 &= 443756.438 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.598 / 55.264 \\
 &= 0.011 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3 / d \\
 &= (48.0 \cdot 0.011 + 1) / 1.678 + 48.0^3 / 443756.438 \\
 &= 1.155
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2) \\
 &= (1.0 \cdot 24379926 / 339.35) / (1.155 \cdot 28.9596^2) \\
 &= 74.14 \text{ MPa}
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2) \\
 &= (1.0 \cdot 51494156 / 339.35) / (1.155 \cdot 28.9596^2) \\
 &= 156.60 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2) \\
 &= (1.6928 \cdot 24379926 / 339.35) / (1.155 \cdot 48.0^2) \\
 &= 45.68 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2) \\
 &= (1.6928 \cdot 51494156 / 339.35) / (1.155 \cdot 48.0^2) \\
 &= 96.49 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro} \\
 &= (4.4395 \cdot 24379926 / (48.0^2 \cdot 339.35)) - 2.3464 \cdot 46 \\
 &= 31.20 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra} \\
 &= (4.4395 \cdot 51494156 / (48.0^2 \cdot 339.35)) - 2.3464 \cdot 96 \\
 &= 65.89 \text{ MPa}
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2 \\
 &= (74 + \max(46, 31)) / 2 \\
 &= 59.91 \text{ MPa}
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2 \\
 &= (157 + \max(96, 66)) / 2 \\
 &= 126.55 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= (W_{m1} + F_a + 4M_e / G) / A_b \\
 &= 508094 / 7109.6631 \\
 &= 71.47 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (W_{m2} / A_b) \\
 &= (609011 / 7109.6631) \\
 &= 85.67 \text{ MPa}
 \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	74.14	207.00	156.60	207.00
Radial Flange	45.68	138.00	96.49	138.00
Tangential Flange	31.20	138.00	65.89	138.00
Maximum Average	59.91	138.00	126.55	138.00
Bolting	71.47	172.00	85.67	172.00

Minimum Required Flange Thickness 45.339 mm

Estimated M.A.W.P. (Operating) 5 MPa

Estimated Finished Weight of Flange at given Thk. 68.6 kgm

Estimated Unfinished Weight of Forging at given Thk 149.2 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	210.37	374.99	0.561	210.37	374.99	0.561
Tangential Flange	Sto	125.00	250.00	0.500	125.00	250.00	0.500
Combined	Sh/f + St	710.36	749.99	0.947	710.36	749.99	0.947
Combined	Sr + St	518.46	749.99	0.691	518.46	749.99	0.691
Combined	Sh + Sto	624.99	749.99	0.833	624.99	749.99	0.833
Radial Flange	Sr	308.08	499.99	0.616	308.08	499.99	0.616
Maximum Stress Ratio				1.000			1.000



Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{\text{num}} / F1_{\text{den}})$$

$$= \max(1; -1.2797/6.1518)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_2) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; f_3)$$

$$= \max(0.25; 4.3059/8.0644) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; 0.4414/2.1013)$$

$$= 0.2500$$

In this case, the hub ratio h/h_0 , 1.6638 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	S_{bmax}	507.5 MPa
Minimum permissible bolt stress	S_{bmin}	290.0 MPa
Max. permissible bolt stress prior to flange damage	S_{fmax}	549.1 MPa
Target assembly gasket stress	S_{gT}	172.0 MPa
Maximum permissible gasket stress	S_{gmax}	380.0 MPa
Minimum gasket operating stress	S_{gmin-O}	97.0 MPa
Minimum gasket seating stress	S_{gmin-S}	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Φ_{hig}	0.700
Root Area of a Single Bolt	A_{bs}	355.4832 mm ²
Max. permissible single flange rotation for gasket	Θ_{tagmax}	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	227.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.198 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (406.4^2 - 374.0^2)$$

$$= 19858.7 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 19858.748 / (20 \cdot 355.483)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(480.431; 507.5)$$

$$= 480.4 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(480.431; 290.0)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(480.431; 549.148)$$

$$= 480.4 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 480.431 \geq 140.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \geq 391.0 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 480.431 \geq (97.0 \cdot 19858.748 + \pi/4 \cdot 1.8 \cdot 374.0^2) / (0.7 \cdot 355.483 \cdot 20)$$

$$:: 480.4 \geq 426.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 480.431 \leq 380.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \leq 1061.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetafmax})$$

$$:: 480.431 \leq 549.148(1.0/0.198)$$

$$:: 480.4 \leq 2776.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 480.431 \cdot 0.2 \cdot 355.483 \cdot 25.4$$

$$= 867866.9 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 MPa

Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.098 \cdot 202713 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.441 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 24379926 / 1.0 \cdot 1.0 \cdot 0.049 / (1.098 \cdot 198163 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.214 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.155 \cdot 202713 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.420 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 24379926 / 1.0 \cdot 1.0 \cdot 0.049 / (1.155 \cdot 198163 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.203 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	67.83	207.00	138.24	207.00
Radial Flange	49.88	138.00	101.66	138.00
Tangential Flange	27.53	138.00	56.11	138.00
Maximum Average	58.85	138.00	119.95	138.00
Bolting	71.47	172.00	85.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 43.637 mm

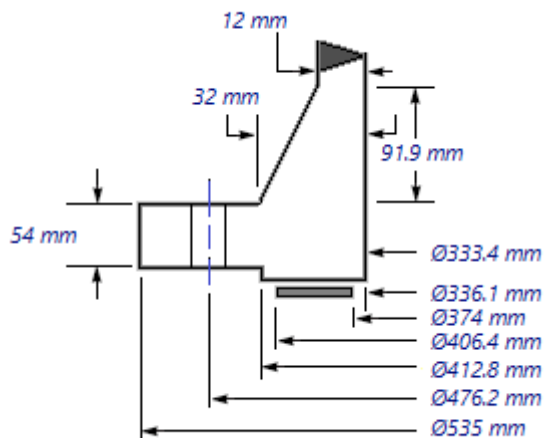
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Flange Input Data Values

Description: Flange-5 :

Flange-5

Flange Type	Integral Weld Neck		
Design Pressure	P	1.80	MPa
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	333.350	mm
Flange Outside Diameter	A	535.000	mm
Flange Thickness	t	54.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	31.9596	mm
Length of Hub	h	91.9480	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	476.250	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	412.750	mm
Flange Face Inside Diameter	Fid	336.131	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	406.400	mm
Gasket Inside Diameter	Gi	374.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		6796.05	N
Bending Moment on Flange		6062000.50	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot (333.35/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8) + Ca$$

$$= 5.2308 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (169.675 + 0.6 \cdot 9.0)$$

$$= 7.094 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (166.675 + 0.6 \cdot 12.0)$$

$$= 9.524 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	339.350 mm
Corroded Large Hub,	g1Cor = g1-ci	28.960 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	39.490 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.200 mm
Basic Gasket Width,	bo = N/2	8.100 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.172 mm
Gasket Reaction Diameter,	G = Go - 2 * b	392.056 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 392.0564^2 \cdot 1.8$$

$$= 217293.984 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1718 \cdot \pi \cdot 392.0564 \cdot 3.0 \cdot 1.8$$

$$= 95398.094 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 339.35^2 \cdot 1.8001$$

$$= 162796.969 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 217294 - 162797$$

$$= 54497.016 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(217294 + 95398 + 0, 0)$$

$$= 312692.094 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1718 \cdot \pi \cdot 392.056 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 609010.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((312692 + 6796 + 4 \cdot 6062001/392.06) / 172, 609011/172)$$

$$= 3541.125 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 54.0/(3.0 + 0.5)$$

$$= 143.371 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 476.25 \cdot \sin(\pi/20)$$

$$= 74.502 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{74.502}{(2 \cdot 25.4 + 54.0)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3541.125	7109.663	
Radial Distance between Hub and Bolts:	34.925	39.490	
Radial Distance between Bolts and Edge:	26.987	29.375	
Circ. Spacing between the Bolts:	57.150	74.502	143.371

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi(406.4 + 374.0))$$

$$= 7.234 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 7109.6631$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 312692 - 217294$$

$$= 95398.12 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (476.25 - 392.0564) / 2$$

$$= 42.0968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((476.25 - 339.35) / 2 + 42.0968)$$

$$= 55.2734 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (476.25 - 339.35 - 28.96) / 2$$

$$= 53.9702 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 97.825, Bb = 54.0, Cc = 91.948 \text{ and } Ddg = 18.98 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 97.825 \cdot 54.0^3 \cdot (1/3 - 0.210 \cdot (54.0/97.825) \cdot (1 - 1/12 \cdot (54.0/97.825)^4))$$

$$= 3362809.500 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.95 \cdot 18.98^3 \cdot (1/3 - 0.105 \cdot (18.98/91.95) \cdot (1 - 1/192 \cdot (18.98/91.95)^4))$$

$$= 195928.344 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 205.211 + 11.956$$

$$= 3558737.750 \text{ mm}^4$$



Flange Inertia Term [Ip,term2]:

$$= A_r \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{A_r} \right) \left(1 - \frac{1}{12} \left(\frac{t}{A_r} \right)^4 \right) \right)$$

$$= 97.825 \cdot 54.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{54.0}{97.825} \right) \left(1 - \frac{1}{12} \left(\frac{54.0}{97.825} \right)^4 \right) \right)$$

$$= 3362809.500 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(I_{p,term1}, I_{p,term2})$$

$$= \max(3558738, 3362810)$$

$$= 3558737.8 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot g_o^2 \cdot h_o \cdot B / V$$

$$= 0.0874 \cdot 1.299 \cdot 9.0^2 \cdot 55.264 \cdot 339.35 / 0.049$$

$$= 3505423.500 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	162797.	53.9702	1.0000	8789748.
Face Pressure, Mt	54497.	55.2734	1.0000	3013457.
Gasket Load, Mg	95398.	42.0968	1.0000	4017584.
Gasket Seating, Matm	1222736.	42.0968	1.0000	51494156.
Moment Load, Moe	...	...	...	2923470.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 6062001 \left[\frac{1}{(0.384 \cdot 3558738 + 3505424)} \right] \cdot (54.0 / (476.2 - 2 \cdot 54.0)) +$$

$$6796 \cdot 53.97$$

$$= 2923469.750 \text{ N-mm}$$

Total Moment for Operation, Mo	18744260. N-mm
Total Moment for Gasket seating, Mg	51494156. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	55.264 mm
Hub Ratio, h/h0 = HL / H0	1.664
Thickness Ratio, g1/g0 = (g1Cor/goCor)	3.218

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.598
Factor V per 4.16.5	0.049
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 535.0 / 339.35$$

$$= 1.577$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.577^2 \cdot \log_{10}(1.577))) / (1.577^2 - 1) / (1.577 - 1)$$

$$= 4.440$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.577^2(1 + 8.55246 \cdot \log_{10}(1.577)) - 1) / ((1.0472 + 1.9448 \cdot 1.577^2) \cdot (1.577 - 1))$$

$$= 1.678$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.577^2(1 + 8.55246 \cdot \log_{10}(1.577)) - 1) / (1.36136(1.577^2 - 1) \cdot (1.577 - 1))$$

$$= 4.879$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.577^2 + 1) / (1.577^2 - 1)$$

$$= 2.346$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.879 \cdot 9.0^2 \cdot 55.264 / 0.049$$

$$= 443756.438 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.598 / 55.264$$

$$= 0.011 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (54.0 \cdot 0.011 + 1) / 1.678 + 54.0^3 / 443756.438$$

$$= 1.299$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 18744258 / 339.35) / (1.2994 \cdot 28.9596^2)$$

$$= 50.67 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 51494156 / 339.35) / (1.2994 \cdot 28.9596^2)$$

$$= 139.20 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (1.7794 \cdot 18744258 / 339.35) / (1.2994 \cdot 54.0^2)$$

$$= 25.93 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (1.7794 \cdot 51494156 / 339.35) / (1.2994 \cdot 54.0^2)$$

$$= 71.24 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (4.4395 \cdot 18744258 / (54.0^2 \cdot 339.35)) - 2.3464 \cdot 26$$

$$= 23.22 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (4.4395 \cdot 51494156 / (54.0^2 \cdot 339.35)) - 2.3464 \cdot 71$$

$$= 63.80 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (51 + \max(26, 23)) / 2$$

$$= 38.30 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (139 + \max(71, 64)) / 2$$

$$= 105.22 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (W_{m1} + F_a + 4M_e / G) / A_b$$

$$= 381311 / 7109.6631$$

$$= 53.64 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (609011 / 7109.6631)$$

$$= 85.67 \text{ MPa}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	50.67	207.00	139.20	207.00
Radial Flange	25.93	138.00	71.24	138.00
Tangential Flange	23.22	138.00	63.80	138.00
Maximum Average	38.30	138.00	105.22	138.00
Bolting	53.64	172.00	85.67	172.00

Minimum Required Flange Thickness 45.339 mm

Estimated M.A.W.P. (Operating) 7 MPa

Estimated Finished Weight of Flange at given Thk. 75.0 kgm

Estimated Unfinished Weight of Forging at given Thk 155.6 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	229.15	374.99	0.611	229.15	374.99	0.611
Tangential Flange	Sto	125.00	250.00	0.500	125.00	250.00	0.500
Combined	Sh/f + St	729.14	749.99	0.972	729.14	749.99	0.972
Combined	Sr + St	485.03	749.99	0.647	485.03	749.99	0.647
Combined	Sh + Sto	624.99	749.99	0.833	624.99	749.99	0.833
Radial Flange	Sr	255.88	499.99	0.512	255.88	499.99	0.512
Maximum Stress Ratio				1.000			1.000



Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{\text{num}} / F1_{\text{den}})$$

$$= \max(1; -1.2797/6.1518)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_2) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; f_3)$$

$$= \max(0.25; 4.3059/8.0644) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; 0.4414/2.1013)$$

$$= 0.2500$$

In this case, the hub ratio h/h_0 , 1.6638 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, $S_{f\text{max}}$, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	$S_{b\text{max}}$	507.5 MPa
Minimum permissible bolt stress	$S_{b\text{min}}$	290.0 MPa
Max. permissible bolt stress prior to flange damage	$S_{f\text{max}}$	617.8 MPa
Target assembly gasket stress	S_{gT}	172.0 MPa
Maximum permissible gasket stress	$S_{g\text{max}}$	380.0 MPa
Minimum gasket operating stress	$S_{g\text{min-O}}$	97.0 MPa
Minimum gasket seating stress	$S_{g\text{min-S}}$	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Φ_{ig}	0.700
Root Area of a Single Bolt	A_{bs}	355.4832 mm ²
Max. permissible single flange rotation for gasket	$\Theta_{tag\text{max}}$	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	227.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.198 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (406.4^2 - 374.0^2)$$

$$= 19858.7 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 19858.748 / (20 \cdot 355.483)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(480.431; 507.5)$$

$$= 480.4 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(480.431; 290.0)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(480.431; 617.773)$$

$$= 480.4 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 480.431 \geq 140.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \geq 391.0 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 480.431 \geq (97.0 \cdot 19858.748 + \pi/4 \cdot 1.8 \cdot 374.0^2) / (0.7 \cdot 355.483 \cdot 20)$$

$$:: 480.4 \geq 426.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 480.431 \leq 380.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \leq 1061.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 480.431 \leq 617.773(1.0/0.198)$$

$$:: 480.4 \leq 3123.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 480.431 \cdot 0.2 \cdot 355.483 \cdot 25.4$$

$$= 867866.9 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 MPa

Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.098 \cdot 202713 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.441 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 18744258 / 1.0 \cdot 1.0 \cdot 0.049 / (1.098 \cdot 198163 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.164 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.299 \cdot 202713 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.373 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 18744258 / 1.0 \cdot 1.0 \cdot 0.049 / (1.299 \cdot 198163 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.139 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	46.59	207.00	124.54	207.00
Radial Flange	28.41	138.00	75.95	138.00
Tangential Flange	20.30	138.00	54.26	138.00
Maximum Average	37.50	138.00	100.24	138.00
Bolting	53.64	172.00	85.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 43.637 mm

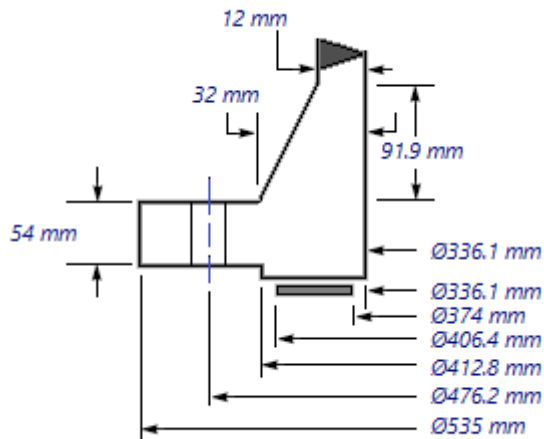
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Flange Input Data Values

Description: Flange-6 :

Flange-6

Flange Type	Integral Weld Neck		
Design Pressure	P	1.80	MPa
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	336.131	mm
Flange Outside Diameter	A	535.000	mm
Flange Thickness	t	54.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	31.9596	mm
Length of Hub	h	91.9480	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	476.250	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	412.750	mm
Flange Face Inside Diameter	Fid	336.131	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	406.400	mm
Gasket Inside Diameter	Gi	374.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		6155.75	N
Bending Moment on Flange		5682480.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot (336.131/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8) + Ca$$

$$= 5.2491 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (171.0655 + 0.6 \cdot 9.0)$$

$$= 7.038 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (168.0655 + 0.6 \cdot 12.0)$$

$$= 9.448 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	342.131 mm
Corroded Large Hub,	g1Cor = g1-ci	28.960 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	38.100 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.200 mm
Basic Gasket Width,	bo = N/2	8.100 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.172 mm
Gasket Reaction Diameter,	G = Go - 2 * b	392.056 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 392.0564^2 \cdot 1.8$$

$$= 217293.984 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1718 \cdot \pi \cdot 392.0564 \cdot 3.0 \cdot 1.8$$

$$= 95398.094 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 342.131^2 \cdot 1.8001$$

$$= 165476.203 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 217294 - 165476$$

$$= 51817.793 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(217294 + 95398 + 0, 0)$$

$$= 312692.094 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1718 \cdot \pi \cdot 392.056 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 609010.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((312692 + 6156 + 4 \cdot 5682480 / 392.06) / 172, 609011 / 172)$$

$$= 3541.125 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 54.0 / (3.0 + 0.5)$$

$$= 143.371 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 476.25 \cdot \sin(\pi / 20)$$

$$= 74.502 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{74.502}{(2 \cdot 25.4 + 54.0)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3541.125	7109.663	
Radial Distance between Hub and Bolts:	34.925	38.100	
Radial Distance between Bolts and Edge:	26.987	29.375	
Circ. Spacing between the Bolts:	57.150	74.502	143.371

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi (406.4 + 374.0))$$

$$= 7.234 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 7109.663$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 312692 - 217294$$

$$= 95398.12 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (476.25 - 392.0564) / 2$$

$$= 42.0968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((476.25 - 342.131) / 2 + 42.0968)$$

$$= 54.5782 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (476.25 - 342.131 - 28.96) / 2$$

$$= 52.5797 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 96.434, Bb = 54.0, Cc = 91.948 \text{ and } Ddg = 18.98 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 96.434 \cdot 54.0^3 \cdot (1/3 - 0.210 \cdot (54.0/96.434) \cdot (1 - 1/12 \cdot (54.0/96.434)^4))$$

$$= 3290639.500 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.95 \cdot 18.98^3 \cdot (1/3 - 0.105 \cdot (18.98/91.95) \cdot (1 - 1/192 \cdot (18.98/91.95)^4))$$

$$= 195928.344 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 200.807 + 11.956$$

$$= 3486567.750 \text{ mm}^4$$



Flange Inertia Term [Ip,term2]:

$$= A_r \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{A_r} \right) \left(1 - \frac{1}{12} \left(\frac{t}{A_r} \right)^4 \right) \right)$$

$$= 96.434 \cdot 54.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{54.0}{96.434} \right) \left(1 - \frac{1}{12} \left(\frac{54.0}{96.434} \right)^4 \right) \right)$$

$$= 3290639.500 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(I_{p,term1}, I_{p,term2})$$

$$= \max(3486568, 3290640)$$

$$= 3486567.8 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot g_o^2 \cdot h_o \cdot B / V$$

$$= 0.0874 \cdot 1.289 \cdot 9.0^2 \cdot 55.49 \cdot 342.131 / 0.049$$

$$= 3510179.500 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	165476.	52.5797	1.0000	8704216.
Face Pressure, Mt	51818.	54.5782	1.0000	2829266.
Gasket Load, Mg	95398.	42.0968	1.0000	4017584.
Gasket Seating, Matm	1222736.	42.0968	1.0000	51494156.
Moment Load, Moe	...	...	...	2655169.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 5682480 \left[\frac{1}{(0.384 \cdot 3486568 + 3510180)} \right] \cdot (52.6 / (476.2 - 2 \cdot 52.6)) +$$

$$6156 \cdot 52.58$$

$$= 2655169.000 \text{ N-mm}$$

Total Moment for Operation, Mo	18206238. N-mm
Total Moment for Gasket seating, Mg	51494156. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	55.490 mm
Hub Ratio, h/h0 = HL / H0	1.657
Thickness Ratio, g1/g0 = (g1Cor/goCor)	3.218

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.599
Factor V per 4.16.5	0.049
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 535.0 / 342.131$$

$$= 1.564$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.564^2 \cdot \log_{10}(1.564))) / (1.564^2 - 1) / (1.564 - 1)$$

$$= 4.517$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.564^2(1 + 8.55246 \cdot \log_{10}(1.564)) - 1) / ((1.0472 + 1.9448 \cdot 1.564^2) \cdot (1.564 - 1))$$

$$= 1.683$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.564^2(1 + 8.55246 \cdot \log_{10}(1.564)) - 1) / (1.36136(1.564^2 - 1) \cdot (1.564 - 1))$$

$$= 4.964$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.564^2 + 1) / (1.564^2 - 1)$$

$$= 2.384$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.964 \cdot 9.0^2 \cdot 55.49 / 0.049$$

$$= 452181.719 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.599 / 55.49$$

$$= 0.011 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (54.0 \cdot 0.011 + 1) / 1.683 + 54.0^3 / 452181.719$$

$$= 1.289$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 18206236 / 342.131) / (1.2887 \cdot 28.9596^2)$$

$$= 49.22 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 51494156 / 342.131) / (1.2887 \cdot 28.9596^2)$$

$$= 139.22 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.7772 \cdot 18206236 / 342.131) / (1.2887 \cdot 54.0^2)$$

$$= 25.16 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.7772 \cdot 51494156 / 342.131) / (1.2887 \cdot 54.0^2)$$

$$= 71.16 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.5172 \cdot 18206236 / (54.0^2 \cdot 342.131)) - 2.3838 \cdot 25$$

$$= 22.43 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.5172 \cdot 51494156 / (54.0^2 \cdot 342.131)) - 2.3838 \cdot 71$$

$$= 63.45 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (49 + \max(25, 22)) / 2$$

$$= 37.19 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (139 + \max(71, 63)) / 2$$

$$= 105.19 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm_1 + Fa + 4Me/G) / Ab$$

$$= 376801 / 7109.6631$$

$$= 53.00 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm_2 / Ab)$$

$$= (609011 / 7109.6631)$$

$$= 85.67 \text{ MPa}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	49.22	207.00	139.22	207.00
Radial Flange	25.16	138.00	71.16	138.00
Tangential Flange	22.43	138.00	63.45	138.00
Maximum Average	37.19	138.00	105.19	138.00
Bolting	53.00	172.00	85.67	172.00

Minimum Required Flange Thickness 45.288 mm

Estimated M.A.W.P. (Operating) 7 MPa

Estimated Finished Weight of Flange at given Thk. 74.6 kgm

Estimated Unfinished Weight of Forging at given Thk 153.9 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	227.89	374.99	0.608	227.89	374.99	0.608
Tangential Flange	Sto	125.00	250.00	0.500	125.00	250.00	0.500
Combined	Sh/f + St	727.88	749.99	0.971	727.88	749.99	0.971
Combined	Sr + St	483.44	749.99	0.645	483.44	749.99	0.645
Combined	Sh + Sto	624.99	749.99	0.833	624.99	749.99	0.833
Radial Flange	Sr	255.55	499.99	0.511	255.55	499.99	0.511
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; -1.2374/6.1137) \\ = 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 4.2952/8.0077) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.4446/2.0876) \\ = 0.2500$$

In this case, the hub ratio $h/h0$, 1.6570 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	617.7 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	355.4832 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	227.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.199 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (406.4^2 - 374.0^2)$$

$$= 19858.7 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 19858.748 / (20 \cdot 355.483)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(480.431; 507.5)$$

$$= 480.4 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(480.431; 290.0)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(480.431; 617.708)$$

$$= 480.4 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 480.431 \geq 140.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \geq 391.0 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 480.431 \geq (97.0 \cdot 19858.748 + \pi/4 \cdot 1.8 \cdot 374.0^2) / (0.7 \cdot 355.483 \cdot 20)$$

$$:: 480.4 \geq 426.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 480.431 \leq 380.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \leq 1061.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 480.431 \leq 617.708(1.0/0.199)$$

$$:: 480.4 \leq 3102.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 480.431 \cdot 0.2 \cdot 355.483 \cdot 25.4$$

$$= 867866.9 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 MPa

Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.09 \cdot 202713 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.444 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 18206236 / 1.0 \cdot 1.0 \cdot 0.049 / (1.09 \cdot 198163 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.160 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.289 \cdot 202713 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.376 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 18206236 / 1.0 \cdot 1.0 \cdot 0.049 / (1.289 \cdot 198163 \cdot 9.0^2 \cdot 55.49 \cdot 0.3)$$

$$= 0.136 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	45.24	207.00	124.48	207.00
Radial Flange	27.55	138.00	75.81	138.00
Tangential Flange	19.59	138.00	53.89	138.00
Maximum Average	36.40	138.00	100.14	138.00
Bolting	53.00	172.00	85.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 43.586 mm

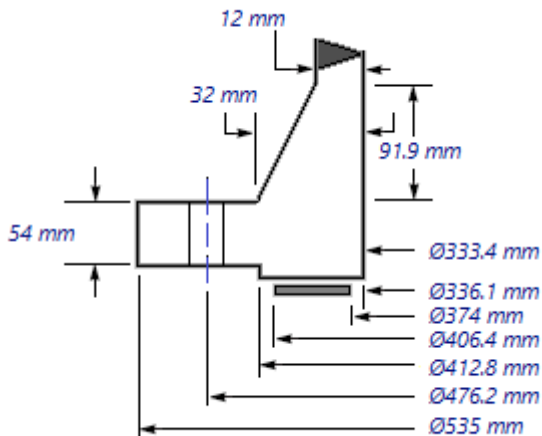
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Flange Input Data Values

Description: Flange-7 :

Flange-7

Flange Type	Integral Weld Neck		
Design Pressure	P	1.80	MPa
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	333.350	mm
Flange Outside Diameter	A	535.000	mm
Flange Thickness	t	54.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	31.9596	mm
Length of Hub	h	91.9480	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	476.250	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	412.750	mm
Flange Face Inside Diameter	Fid	336.131	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	406.400	mm
Gasket Inside Diameter	Gi	374.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		747.31	N
Bending Moment on Flange		42695.50	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot (333.35/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8) + Ca$$

$$= 5.2308 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (169.675 + 0.6 \cdot 9.0)$$

$$= 7.094 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (166.675 + 0.6 \cdot 12.0)$$

$$= 9.524 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	339.350 mm
Corroded Large Hub,	g1Cor = g1-ci	28.960 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	39.490 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.200 mm
Basic Gasket Width,	bo = N/2	8.100 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.172 mm
Gasket Reaction Diameter,	G = Go - 2 * b	392.056 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 392.0564^2 \cdot 1.8$$

$$= 217293.984 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1718 \cdot \pi \cdot 392.0564 \cdot 3.0 \cdot 1.8$$

$$= 95398.094 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 339.35^2 \cdot 1.8001$$

$$= 162796.969 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 217294 - 162797$$

$$= 54497.016 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(217294 + 95398 + 0, 0)$$

$$= 312692.094 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1718 \cdot \pi \cdot 392.056 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 609010.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((312692 + 747 + 4 \cdot 42696 / 392.06) / 172, 609011 / 172)$$

$$= 3541.125 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 54.0 / (3.0 + 0.5)$$

$$= 143.371 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 476.25 \cdot \sin(\pi/20)$$

$$= 74.502 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{74.502/(2 \cdot 25.4 + 54.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3541.125	7109.663	
Radial Distance between Hub and Bolts:	34.925	39.490	
Radial Distance between Bolts and Edge:	26.987	29.375	
Circ. Spacing between the Bolts:	57.150	74.502	143.371

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi(406.4 + 374.0))$$

$$= 7.234 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 7109.6631$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 312692 - 217294$$

$$= 95398.12 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (476.25 - 392.0564) / 2$$

$$= 42.0968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((476.25 - 339.35) / 2 + 42.0968)$$

$$= 55.2734 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (476.25 - 339.35 - 28.96) / 2$$

$$= 53.9702 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 97.825, Bb = 54.0, Cc = 91.948 \text{ and } Ddg = 18.98 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 97.825 \cdot 54.0^3 \cdot (1/3 - 0.210 \cdot (54.0/97.825) \cdot (1 - 1/12 \cdot (54.0/97.825)^4))$$

$$= 3362809.500 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.95 \cdot 18.98^3 \cdot (1/3 - 0.105 \cdot (18.98/91.95) \cdot (1 - 1/192 \cdot (18.98/91.95)^4))$$

$$= 195928.344 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 205.211 + 11.956$$

$$= 3558737.750 \text{ mm}^4$$



Flange Inertia Term [Ip,term2]:

$$= A_r \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{A_r} \right) \left(1 - \frac{1}{12} \left(\frac{t}{A_r} \right)^4 \right) \right)$$

$$= 97.825 \cdot 54.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{54.0}{97.825} \right) \left(1 - \frac{1}{12} \left(\frac{54.0}{97.825} \right)^4 \right) \right)$$

$$= 3362809.500 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(I_{p,term1}, I_{p,term2})$$

$$= \max(3558738, 3362810)$$

$$= 3558737.8 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot g_o^2 \cdot h_o \cdot B / V$$

$$= 0.0874 \cdot 1.299 \cdot 9.0^2 \cdot 55.264 \cdot 339.35 / 0.049$$

$$= 3505423.500 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	162797.	53.9702	1.0000	8789748.
Face Pressure, Mt	54497.	55.2734	1.0000	3013457.
Gasket Load, Mg	95398.	42.0968	1.0000	4017584.
Gasket Seating, Matm	1222736.	42.0968	1.0000	51494156.
Moment Load, Moe	...	...	...	58355.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 42696 [3505424 / (0.384 \cdot 3558738 + 3505424)] \cdot (54.0 / (476.2 - 2 \cdot 54.0)) +$$

$$747 \cdot 53.97$$

$$= 58354.980 \text{ N-mm}$$

Total Moment for Operation, Mo	15879145. N-mm
Total Moment for Gasket seating, Mg	51494156. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	55.264 mm
Hub Ratio, h/h0 = HL / H0	1.664
Thickness Ratio, g1/g0 = (g1Cor/goCor)	3.218

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.598
Factor V per 4.16.5	0.049
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 535.0 / 339.35$$

$$= 1.577$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.577^2 \cdot \log_{10}(1.577))) / (1.577^2 - 1)) / (1.577 - 1)$$

$$= 4.440$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.577^2 (1 + 8.55246 \cdot \log_{10}(1.577)) - 1) / ((1.0472 + 1.9448 \cdot 1.577^2) \cdot (1.577 - 1))$$

$$= 1.678$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.577^2 (1 + 8.55246 \cdot \log_{10}(1.577)) - 1) / (1.36136 (1.577^2 - 1) \cdot (1.577 - 1))$$

$$= 4.879$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.577^2 + 1) / (1.577^2 - 1)$$

$$= 2.346$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.879 \cdot 9.0^2 \cdot 55.264 / 0.049$$

$$= 443756.438 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.598 / 55.264$$

$$= 0.011 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (54.0 \cdot 0.011 + 1) / 1.678 + 54.0^3 / 443756.438$$

$$= 1.299$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 15879144 / 339.35) / (1.2994 \cdot 28.9596^2)$$

$$= 42.93 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 51494156 / 339.35) / (1.2994 \cdot 28.9596^2)$$

$$= 139.20 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (1.7794 \cdot 15879144 / 339.35) / (1.2994 \cdot 54.0^2)$$

$$= 21.97 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (1.7794 \cdot 51494156 / 339.35) / (1.2994 \cdot 54.0^2)$$

$$= 71.24 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (4.4395 \cdot 15879144 / (54.0^2 \cdot 339.35)) - 2.3464 \cdot 22$$

$$= 19.67 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (4.4395 \cdot 51494156 / (54.0^2 \cdot 339.35)) - 2.3464 \cdot 71$$

$$= 63.80 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SH_o + \max(SR_o, ST_o)) / 2$$

$$= (43 + \max(22, 20)) / 2$$

$$= 32.45 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (SH_a + \max(SR_a, ST_a)) / 2$$

$$= (139 + \max(71, 64)) / 2$$

$$= 105.22 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (W_{m1} + F_a + 4M_e / G) / A_b$$

$$= 313875 / 7109.6631$$

$$= 44.15 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (609011 / 7109.6631)$$

$$= 85.67 \text{ MPa}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	42.93	207.00	139.20	207.00
Radial Flange	21.97	138.00	71.24	138.00
Tangential Flange	19.67	138.00	63.80	138.00
Maximum Average	32.45	138.00	105.22	138.00
Bolting	44.15	172.00	85.67	172.00

Minimum Required Flange Thickness 45.339 mm

Estimated M.A.W.P. (Operating) 7 MPa

Estimated Finished Weight of Flange at given Thk. 75.0 kgm

Estimated Unfinished Weight of Forging at given Thk 155.6 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	229.15	374.99	0.611	229.15	374.99	0.611
Tangential Flange	Sto	125.00	250.00	0.500	125.00	250.00	0.500
Combined	Sh/f + St	729.14	749.99	0.972	729.14	749.99	0.972
Combined	Sr + St	485.03	749.99	0.647	485.03	749.99	0.647
Combined	Sh + Sto	624.99	749.99	0.833	624.99	749.99	0.833
Radial Flange	Sr	255.88	499.99	0.512	255.88	499.99	0.512
Maximum Stress Ratio				1.000			1.000



Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{\text{num}} / F1_{\text{den}})$$

$$= \max(1; -1.2797/6.1518)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_2) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; f_3)$$

$$= \max(0.25; 4.3059/8.0644) \text{ if } h/h_0 < 0.35, \text{ else } \max(0.25; 0.4414/2.1013)$$

$$= 0.2500$$

In this case, the hub ratio h/h_0 , 1.6638 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	S_{bmax}	507.5 MPa
Minimum permissible bolt stress	S_{bmin}	290.0 MPa
Max. permissible bolt stress prior to flange damage	S_{fmax}	617.8 MPa
Target assembly gasket stress	S_{gT}	172.0 MPa
Maximum permissible gasket stress	S_{gmax}	380.0 MPa
Minimum gasket operating stress	S_{gmin-O}	97.0 MPa
Minimum gasket seating stress	S_{gmin-S}	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Φ_{hig}	0.700
Root Area of a Single Bolt	A_{bs}	355.4832 mm ²
Max. permissible single flange rotation for gasket	Θ_{tagmax}	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	227.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.198 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (406.4^2 - 374.0^2)$$

$$= 19858.7 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 19858.748 / (20 \cdot 355.483)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(480.431; 507.5)$$

$$= 480.4 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(480.431; 290.0)$$

$$= 480.4 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(480.431; 617.773)$$

$$= 480.4 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 480.431 \geq 140.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \geq 391.0 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 480.431 \geq (97.0 \cdot 19858.748 + \pi/4 \cdot 1.8 \cdot 374.0^2) / (0.7 \cdot 355.483 \cdot 20)$$

$$:: 480.4 \geq 426.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 480.431 \leq 380.0(19858.748 / (20 \cdot 355.483))$$

$$:: 480.4 \leq 1061.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetafmax})$$

$$:: 480.431 \leq 617.773(1.0/0.198)$$

$$:: 480.4 \leq 3123.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 480.431 \cdot 0.2 \cdot 355.483 \cdot 25.4$$

$$= 867866.9 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 MPa

Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.098 \cdot 202713 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.441 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 15879144 / 1.0 \cdot 1.0 \cdot 0.049 / (1.098 \cdot 198163 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.139 \quad (\text{should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 51494156 / 1.0 \cdot 1.0 \cdot 0.049 / (1.299 \cdot 202713 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.373 \quad (\text{should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 15879144 / 1.0 \cdot 1.0 \cdot 0.049 / (1.299 \cdot 198163 \cdot 9.0^2 \cdot 55.264 \cdot 0.3)$$

$$= 0.117 \quad (\text{should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	39.22	207.00	124.54	207.00
Radial Flange	23.92	138.00	75.95	138.00
Tangential Flange	17.09	138.00	54.26	138.00
Maximum Average	31.57	138.00	100.24	138.00
Bolting	44.15	172.00	85.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 43.637 mm

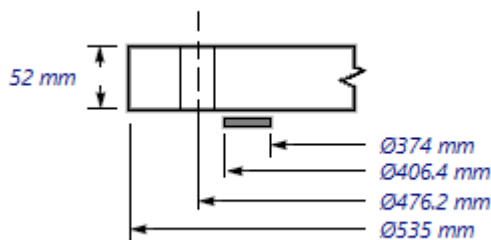
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Flange Input Data Values

Description: Flange-8 :

Flange-8

Flange Type		Blind	
Design Pressure	P	1.80	MPa
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	535.000	mm
Flange Thickness	t	52.0000	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Perimeter along the Center of the Bolts	L	1496.184	mm
Diameter of Bolt Circle	C	476.250	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	412.750	mm
Flange Face Inside Diameter	Fid	336.131	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	406.400	mm
Gasket Inside Diameter	Gi	374.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	16.200 mm
Basic Gasket Width,	$bo = N/2$	8.100 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.172 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	392.056 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 392.0564^2 \cdot 1.8$
 $= 217293.984 \text{ N}$



Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1718 \cdot \pi \cdot 392.0564 \cdot 3.0 \cdot 1.8$$

$$= 95398.094 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(217294 + 95398 + 0, 0)$$

$$= 312692.094 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1718 \cdot \pi \cdot 392.056 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 609010.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((312692 + 0 + 4 \cdot 0/392.06)/172, 609011/172)$$

$$= 3541.125 \text{ mm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3541.125	7109.663	
Radial Distance between Bolts and Edge:	26.987	29.375	
Circ. Spacing between the Bolts:	57.150	74.502	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi (406.4 + 374.0))$$

$$= 7.234 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 7109.6631$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 312692.09 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (476.25 - 392.0564) / 2$$

$$= 42.0968 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 312692 \cdot 42.0968 \cdot 1.0 / (49.0^2 \cdot 392.0564) + 0.3 \cdot 1.0 \cdot 1.8 \cdot 392.0564^2 / 49.0^2$$

$$= 61.14 \text{ MPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1222736 \cdot 42.0968 \cdot 1.0 / (49.0^2 \cdot 392.0564)$$

$$= 103.90 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 312692 / 7109.6631$$

$$= 43.99 \text{ MPa}$$

Bolt Stress, Seating [BSa]:



= (Wm2 / Ab)
= (609011/7109.6631)
= 85.67 MPa

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	61.14	138.00	103.90	138.00
Bolting	43.99	172.00	85.67	172.00

Reqd. Blind Flange Thickness at Center 45.518 mm
Reqd. Blind Flange Thickness at Gasket 45.518 mm

Estimated M.A.W.P. (Operating) 4 MPa
Estimated Finished Weight of Flange at given Thk. 90.6 kgm
Estimated Unfinished Weight of Forging at given Thk 90.6 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.888, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 0 °C
Min Metal Temp. at Required thickness (UCS 66.1) -7 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	54.29	138.00	92.26	138.00
Bolting	43.99	172.00	85.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 42.520 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2	...	...	26	1.6	378	...	1
Skirt-1	...	...	28	1.6	380	...	2
Bottom head		1.8166	26	3	333.35	138	3
Shell-1		1.8154	14	3	333.35	138	4
Shell-2		1.8001	12	3	333.35	138	5
Flange-1		1.8001	56	3	336.13	138	6
Flange-2		1.8001	50	3	336.13	138	7
Shell-3		1.8001	12	3	333.35	138	8
Flange-3		1.8001	50	3	333.35	138	9
Flange-4		1.8001	50	3	333.35	138	10
Shell-4		1.8001	12	3	333.35	138	11
Flange-5		1.8001	56	3	333.35	138	12
Flange-6		1.8001	56	3	336.13	138	13
Shell-5		1.8001	12	3	333.35	138	14
Flange-7		1.8001	56	3	333.35	138	15
Flange-8		1.8001	54	3	535	138	16

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2	...	...	No Calc	No Calc	26	No Calc	1
Skirt-1	...	...	No Calc	No Calc	28	No Calc	2
Bottom head		1.8001	15.6203	17.9775	22	5.18523	3
Shell-1		1.8001	8.59607	11.0351	14	5.24993	4
Shell-2		1.8001	7.09398	9.52392	12	5.23078	5
Flange-1		1.8001	4.85982	4.85982	80	45.3898	6
Flange-2		1.8001	4.67848	4.8167	48	45.2882	7
Shell-3		1.8001	7.09398	9.52392	12	5.23078	8
Flange-3		1.8001	5.38139	5.55109	48	45.339	9
Flange-4		1.8001	5.41593	5.58648	48	45.339	10
Shell-4		1.8001	7.09398	9.52392	12	5.23078	11
Flange-5		1.8001	6.64407	6.64407	54	45.339	12
Flange-6		1.8001	6.67002	6.67002	54	45.2882	13
Shell-5		1.8001	7.09398	9.52392	12	5.23078	14
Flange-7		1.8001	7.03224	7.03224	54	45.339	15
Flange-8		1.8001	4.06267	4.57545	52	45.5183	16

Minimum

4

5

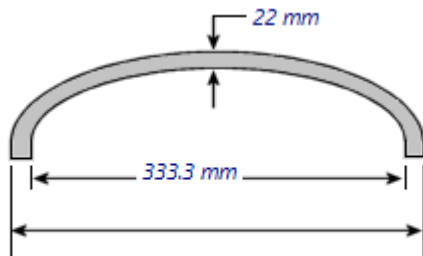
MAWP: 4 MPa, limited by: Flange-8.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	1.817 MPa	Int. Design Temperature		100 °C
Material		SA-516 70	Ext. Design Temperature		100 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	333.350 mm	Actual Stress		16.0 MPa
Spec. Min. Thk t,e, or sa		22.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		221496.2 mm²	SG of Contents		0.920
Element Volume		9.2 ltr	Weight of Contents		83.1 N
Empty Weight		416.9 N	Operating Weight		500.0 N
			Comp. All. Ope. Stress		147.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 4.167 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 2.083 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.817 \cdot 339.3496 \cdot 0.977) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1.817)$$

$$= 2.1852 + 3.0000 = 5.1852 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.017 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 19.0) / (0.977 \cdot 339.3496 + 0.2 \cdot 19.0)$$

$$= 16 - 0 = 16 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 22.0) / (1.0 \cdot 333.3496 + 0.2 \cdot 22.0)$$

$$= 18 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.817 \cdot (0.977 \cdot 339.3496 + 0.2 \cdot 19.0)) / (2 \cdot 1.0 \cdot 19.0)$$

$$= 16.032 \text{ MPa}$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (1.817 \cdot 169.6748) / (138.0 \cdot 1.0 - 0.6 \cdot 1.817) + 3.0$$

$$= 5.251 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.016 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 23.0) / (169.6748 + 0.6 \cdot 23.0)$$

$$= 17 - 0 = 17 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (339.35 / (2 \cdot 86.337))^2) / 6$$

$$= 0.977036$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 27.989 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 22.0$, $t_r = 4.915$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.259$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 26.0$, $t_r = 5.106$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.222$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.079 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	26.000 mm	Basic MDMT	0 °C
Stress/Thk. Ratio	0.222	Temperature Difference	104 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 4.167 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 2.083 mm

EXTERNAL PRESSURE RESULTS for: Bottom head

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
19.000	377.35	19.86	0.0069932	122.62

$$MAEP = B / (K_0 \cdot Do / t) = 122.622 / (0.9 \cdot 19.8605) = 6.8602 \text{ MPa}$$



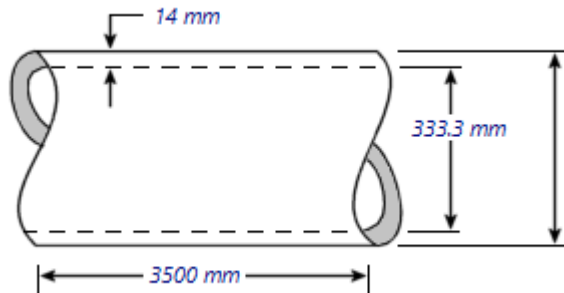
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
0.977	377.35	386.23	0.0003596	35.95

$$MAEP = B / (K0 * Do / t) = 35.9521 / (0.9 * 386.2293) = 0.1034 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:			
Design Pressure P	1.815 MPa	Int. Design Temperature	100 °C
Material	SA-516 70	Ext. Design Temperature	100 °C
Length	3500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	333.350 mm	Actual Stress	29.1 MPa
Spec. Min. Thk t _e , or s _a	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	3973246.0 mm ²	SG of Contents	0.920
Element Volume	305.5 ltr	Weight of Contents	1337.9 N
Empty Weight	4063.8 N	Operating Weight	5401.7 N
		Comp. All. Ope. Stress	147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.815 \cdot 169.6748) / (138.0 - 0.6 \cdot 1.815)$$

$$= 2.2499 + 3.0000 = 5.2499 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.015 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 11.0) / (169.6748 + 0.6 \cdot 11.0)$$

$$= 9 - 0 = 9 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (166.6748 + 0.6 \cdot 14.0)$$

$$= 11 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.815 \cdot (169.6748 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0)$$

$$= 29.092 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 4.031 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 12.133, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 1.0, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.078 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	361.35	5577.78	32.85	15.4360	0.0010569	86.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*86.9837)/(3*32.85) = 3.5305 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.160	361.35	5577.78	114.36	15.4360	0.0000887	8.87

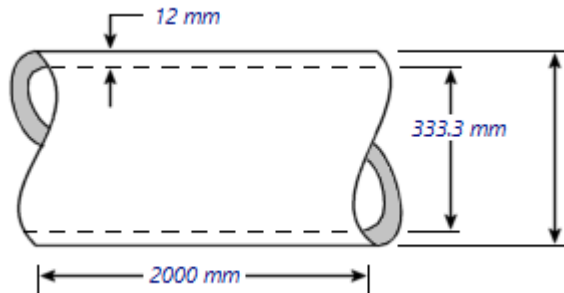
$$MAEP = (4*B)/(3*(Do/t)) = (4*8.871)/(3*114.3635) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	361.35	189984.86	32.85	50.0000	0.0010433	86.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*86.5939)/(3*32.85) = 3.5147 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	1.800 MPa	Int. Design Temperature		100 °C
Material		SA-516 70	Ext. Design Temperature		100 °C
Length		2000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	333.350 mm	Actual Stress		35.0 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		2245293.8 mm²	SG of Contents		0.000
Element Volume		174.6 ltr	Weight of Contents		0.0 N
Empty Weight		1979.0 N	Operating Weight		1979.0 N
			Comp. All. Ope. Stress		146.1 MPa
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot 169.6748) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8)$$

$$= 2.2308 + 3.0000 = 5.2308 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (169.6748 + 0.6 \cdot 9.0)$$

$$= 7 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (166.6748 + 0.6 \cdot 12.0)$$

$$= 10 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.8 \cdot (169.6748 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 35.017 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 3.475 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 9.258, c = 3.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 1.0$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.063 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	357.35	5577.78	39.71	15.6087	0.0007061	70.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*70.59)/(3*39.7055) = 2.3705 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.126	357.35	5577.78	114.33	15.6087	0.0000887	8.87

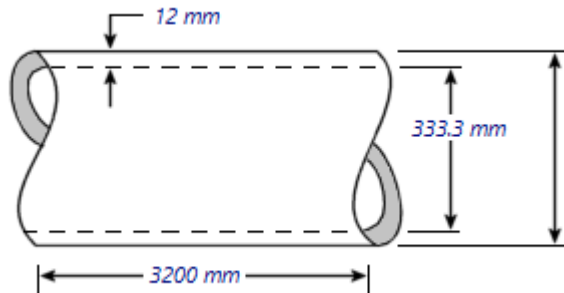
$$MAEP = (4*B)/(3*(Do/t)) = (4*8.8681)/(3*114.3281) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	357.35	127380.07	39.71	50.0000	0.0007009	70.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*70.0775)/(3*39.7055) = 2.3532 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	1.800 MPa	Int. Design Temperature		100 °C
Material		SA-516 70	Ext. Design Temperature		100 °C
Length		3200.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	333.350 mm	Actual Stress		35.0 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		3592470.0 mm²	SG of Contents		0.000
Element Volume		279.3 ltr	Weight of Contents		0.0 N
Empty Weight		3166.4 N	Operating Weight		3166.4 N
			Comp. All. Ope. Stress		146.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot 169.6748) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 1.8)$$

$$= 2.2308 + 3.0000 = 5.2308 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (169.6748 + 0.6 \cdot 9.0)$$

$$= 7 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (166.6748 + 0.6 \cdot 12.0)$$

$$= 10 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.8 \cdot (169.6748 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 35.017 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 3.475 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 7.268, c = 3.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.808$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -34 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.063 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.808	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	357.35	3200.00	39.71	8.9548	0.0007326	73.24

$$MAEP = (4*B) / (3*(Do/t)) = (4*73.2417) / (3*39.7055) = 2.4595 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.922	357.35	3200.00	122.29	8.9548	0.0000949	9.49

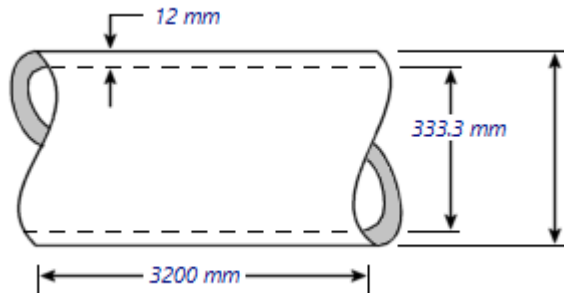
$$MAEP = (4*B) / (3*(Do/t)) = (4*9.4859) / (3*122.2884) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	357.35	74438.49	39.71	50.0000	0.0007009	70.08

$$MAEP = (4*B) / (3*(Do/t)) = (4*70.0775) / (3*39.7055) = 2.3532 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	1.800 MPa	Int. Design Temperature		100 °C
Material		SA-516 70	Ext. Design Temperature		100 °C
Length		3200.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	333.350 mm	Actual Stress		35.0 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		3592470.0 mm²	SG of Contents		0.000
Element Volume		279.3 ltr	Weight of Contents		0.0 N
Empty Weight		3166.4 N	Operating Weight		3166.4 N
			Comp. All. Ope. Stress		146.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot 169.6748) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 1.8)$$

$$= 2.2308 + 3.0000 = 5.2308 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (169.6748 + 0.6 \cdot 9.0)$$

$$= 7 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (166.6748 + 0.6 \cdot 12.0)$$

$$= 10 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.8 \cdot (169.6748 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 35.017 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 3.475 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 5.085, c = 3.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.565$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.063 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.565	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	357.35	3200.00	39.71	8.9548	0.0007326	73.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*73.2417)/(3*39.7055) = 2.4595 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.922	357.35	3200.00	122.29	8.9548	0.0000949	9.49

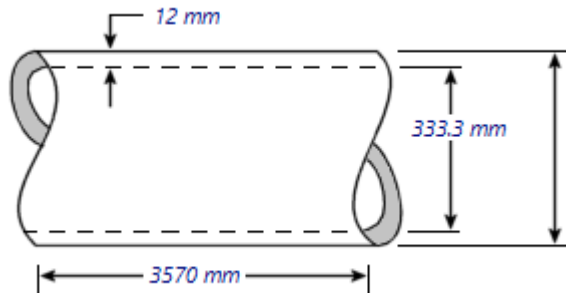
$$MAEP = (4*B)/(3*(Do/t)) = (4*9.4859)/(3*122.2884) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	357.35	74438.49	39.71	50.0000	0.0007009	70.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*70.0775)/(3*39.7055) = 2.3532 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:			
Design Pressure P	1.800 MPa	Int. Design Temperature	100 °C
Material	SA-516 70	Ext. Design Temperature	100 °C
Length	3570.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	333.350 mm	Actual Stress	35.0 MPa
Spec. Min. Thk t, e, or sa	12.0000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	4007849.0 mm ²	SG of Contents	0.000
Element Volume	311.6 ltr	Weight of Contents	0.0 N
Empty Weight	3532.5 N	Operating Weight	3532.5 N
		Comp. All. Ope. Stress	146.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.8 \cdot 169.6748) / (138.0 \cdot 1.0 - 0.6 \cdot 1.8)$$

$$= 2.2308 + 3.0000 = 5.2308 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (169.6748 + 0.6 \cdot 9.0)$$

$$= 7 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (166.6748 + 0.6 \cdot 12.0)$$

$$= 10 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.8 \cdot (169.6748 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 35.017 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 3.475 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 5.085, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr * E / (tg - c) = 0.565$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.063 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.565	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 140 to 150 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	357.35	3570.00	39.71	9.9902	0.0007213	72.11

$$MAEP = (4*B)/(3*(Do/t)) = (4*72.1125)/(3*39.7055) = 2.4216 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.002	357.35	3570.00	119.03	9.9902	0.0000923	9.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.2328)/(3*119.0259) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	357.35	82402.83	39.71	50.0000	0.0007009	70.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*70.0775)/(3*39.7055) = 2.3532 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.281 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.340 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	5.948 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.469 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	2.574 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2.574 MPa
App 27-4	= M.A.W.P.	4.063 MPa

Vertical Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	46.9	234.0	0.200	6.11
Shell-1	76.4	234.0	0.327	6.11
Shell-2	88.0	234.0	0.376	6.08
Shell-3	87.7	234.0	0.375	6.05
Shell-4	87.2	234.0	0.373	6.02
Shell-5	86.7	234.0	0.371	5.98

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4-2"	Nozzle	138.00	138.00	1.000
H4-6"	Nozzle	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
N1-2"	Nozzle	138.00	138.00	1.000
H3-6"	Nozzle	138.00	138.00	1.000
H2-6"	Nozzle	118.00	118.00	1.000
H1-6"	Nozzle	138.00	138.00	1.000
N2-3"	Nozzle	138.00	138.00	1.000
N3-3"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Flange-3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Flange-7	138.00	138.00	1.000
Flange-8	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):



Description	Ambient	Operating	Ratio
N4-2"	21.66	225.00	0.096
H4-6"	50.86	225.00	0.226
K1-2"	12.87	225.00	0.057
K2-2"	12.89	225.00	0.057
N1-2"	19.13	225.00	0.085
H3-6"	50.55	225.00	0.225
H2-6"	50.27	215.90	0.233
H1-6"	49.98	225.00	0.222
N2-3"	24.79	225.00	0.110
N3-3"	26.23	225.00	0.117

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.281 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.340 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	5.948 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.469 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	2.574 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2.574 MPa
App 27-4	= M.A.W.P.	4.063 MPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1.8 • 1.0
 = 2.340 MPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	22.1	234.0	0.094	2.50
Shell-1	40.1	234.0	0.171	2.50
Shell-2	48.0	234.0	0.205	2.47
Shell-3	47.6	234.0	0.203	2.45
Shell-4	46.9	234.0	0.200	2.41
Shell-5	46.2	234.0	0.198	2.38

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4-2"	Nozzle	138.00	138.00	1.000
H4-6"	Nozzle	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
N1-2"	Nozzle	138.00	138.00	1.000
H3-6"	Nozzle	138.00	138.00	1.000
H2-6"	Nozzle	118.00	118.00	1.000
H1-6"	Nozzle	138.00	138.00	1.000
N2-3"	Nozzle	138.00	138.00	1.000
N3-3"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Flange-3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Flange-7	138.00	138.00	1.000
Flange-8	138.00	138.00	1.000



Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4-2"	8.88	225.00	0.039
H4-6"	20.70	225.00	0.092
K1-2"	5.25	225.00	0.023
K2-2"	5.27	225.00	0.023
N1-2"	7.80	225.00	0.035
H3-6"	20.38	225.00	0.091
H2-6"	20.10	215.90	0.093
H1-6"	19.81	225.00	0.088
N2-3"	9.76	225.00	0.043
N3-3"	10.32	225.00	0.046

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	4200	...	...	No Calc	No Calc
20	30	4200	...	...	No Calc	No Calc
30	40	No Calc	377.35	19	0.0069932	122.622
40	50	5577.78	361.35	11	0.0010569	86.9837
50	60	5577.78	357.35	9	0.00070606	70.59
60	70	No Calc	...	77	No Calc	No Calc
70	80	No Calc	...	45	No Calc	No Calc
80	90	3200	357.35	9	0.00073258	73.2417
90	100	No Calc	...	45	No Calc	No Calc
100	110	No Calc	...	45	No Calc	No Calc
110	120	3200	357.35	9	0.00073258	73.2417
120	130	No Calc	...	51	No Calc	No Calc
130	140	No Calc	...	51	No Calc	No Calc
140	150	3570	357.35	9	0.00072129	72.1125
150	160	No Calc	...	51	No Calc	No Calc
160	170	No Calc	...	49	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	22	3.97701	0.10342	6.86018
40	50	14	6.15966	0.10342	3.53054
50	60	12	6.12565	0.10342	2.37045
60	70	80	45.3898	0.10342	No Calc
70	80	48	45.2882	0.10342	No Calc
80	90	12	5.92219	0.10342	2.4595
90	100	48	45.339	0.10342	No Calc
100	110	48	45.339	0.10342	No Calc
110	120	12	5.92219	0.10342	2.4595
120	130	54	45.339	0.10342	No Calc
130	140	54	45.2882	0.10342	No Calc
140	150	12	6.00228	0.10342	2.42158
150	160	54	45.339	0.10342	No Calc
160	170	52	45.5422	0.10342	No Calc

Minimum

2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	4200	No Calc	No Calc	No Calc
20	30	4200	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	5577.78	189985	No Calc	No Calc
50	60	5577.78	127380	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	3200	74438.5	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc



100	110	No Calc	No Calc	No Calc	No Calc
110	120	3200	74438.5	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc
140	150	3570	82402.8	No Calc	No Calc
150	160	No Calc	No Calc	No Calc	No Calc
160	170	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	999.637	...	955.754	...	...
20	30	239.983	...	226.27	...	...
30	40	42.519	9.21424	37.6129	9.63939	...
40	50	414.422	305.517	328.43	316.614	...
50	60	201.813	174.581	152.675	180.922	...
60	70	101.987	15.5167	97.7274	15.8106	...
70	80	70.3502	12.7582	66.8338	13.0521	...
80	90	322.901	279.33	244.28	289.476	...
90	100	70.78	12.5493	67.2925	12.8408	...
100	110	70.78	12.5493	67.2925	12.8408	...
110	120	322.901	279.33	244.28	289.476	...
120	130	77.1754	13.0921	73.5404	13.3836	...
130	140	76.6775	13.3099	73.0125	13.6038	...
140	150	360.237	311.627	272.524	322.947	...
150	160	77.1754	13.0921	73.5404	13.3836	...
160	170	94.0852	...	94.0852	...	...
Total		3543	1452.47	3075	1503.99	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	50	...	...	...	Tailing lug
20	Insl	20.3713	...	750	...	Ins
30	Liqd	8.47168	...	-41.6687	...	Liquid 30
30	Insl	9.03789	...	-16.6689	...	Ins: 110
30	Nozl	12.5956	...	-197.344	...	N4-2"
30	Wght	5.00009	...	...	...	Vortex
30	Forc	...	...	-83	...	Guide lateral f
40	Pack	14.8537	...	3035	...	Bed#4
40	Liqd	136.435	...	850	...	Liquid 40
40	Insl	136.661	...	1750	...	Ins: 110
40	Nozl	110.148	291.681	2820	...	H4-6"
40	Nozl	10.6883	269.862	1800	...	K1-2"
40	Nozl	10.6883	269.862	500	...	K2-2"
40	Nozl	14.4474	-291.681	1700	...	N1-2"
40	Wght	10.0002	...	2570	...	DL_BS-4
40	Wght	5.00009	...	2570	...	TA_BS-4
40	Wght	50.0009	...	2820	...	H4 Davit
50	Pack	18.2078	...	570.001	...	Bed#4
50	Insl	77.3843	...	1000	...	Ins: 110
50	Wght	15.0003	...	1390	...	DL_LD-4
50	Wght	5.00009	...	1390	...	TA_LD-4
60	Insl	6.41299	...	72.974	...	Ins: 110
70	Insl	7.09971	...	68.45	...	Ins: 110
80	Pack	51.1279	...	1635	...	Bed#3
80	Insl	123.815	...	1600	...	Ins: 110
80	Nozl	110.148	291.681	849.995	...	H3-6"
80	Wght	15.0003	...	2920	...	DL_LD-3
80	Wght	5.00009	...	2920	...	TA_LD-3
80	Wght	10.0002	...	600.005	...	DL_BS-3
80	Wght	5.00009	...	0.0046512	...	TA_BS-3
80	Wght	50.0009	...	850.005	...	H3 Davit
90	Insl	7.1561	...	63.5	...	Ins: 110
100	Insl	7.1561	...	63.5	...	Ins: 110
110	Pack	51.1279	...	1635	...	Bed#2
110	Insl	123.815	...	1600	...	Ins: 110
110	Nozl	110.148	291.681	849.908	...	H2-6"
110	Wght	15.0003	...	2920.01	...	DL_LD-2



110	Wght	5.00009	...	2920.01	...	TA_LD-2
110	Wght	10.0002	...	600.008	...	DL_BS-2
110	Wght	5.00009	...	600.008	...	TA_BS-2
110	Wght	50.0009	...	850.007	...	H2 Davit
110	Wght	200	...	1600.01	...	Clip weight
110	Wght	400	...	1561.61	...	Misc wt 7%
120	Insl	6.98917	...	63.5	...	Ins: 110
130	Insl	6.93136	...	63.5	...	Ins: 110
140	Pack	51.1279	...	1635.03	...	Bed#1
140	Insl	138.131	...	1785	...	Ins: 110
140	Nozl	106.439	241.681	850.013	...	H1-6"
140	Nozl	39.1307	291.681	3400.01	...	N2-3"
140	Wght	15.0003	...	2920.01	...	DL_LD-1
140	Wght	5.00009	...	2920.01	...	TA_LD-1
140	Wght	5.00009	...	600.011	...	TA_BS-1
140	Wght	10.0002	...	600.011	...	BS_LD-1
140	Wght	10.0002	...	3500.01	...	DL_MP
140	Wght	5.00009	...	3500.01	...	TA_MP
140	Wght	5.00009	...	3400.01	...	DL_FP
140	Wght	50.0009	...	850.011	...	H1 davit
140	Wght	50	250	2570.01	...	Trunnion 1
140	Wght	50	-250	2570.01	...	Trunnion 2
140	Forc	...	...	3570.01	...	N3 FY
150	Insl	6.98917	...	63.5	...	Ins: 110
160	Insl	8.05236	...	17.526	...	Ins: 110
160	Nozl	22.4634	...	177.006	...	N3-3"

Total Weight of Each Detail Type:

Packing	186.4
Liquid	144.9
Insulation	686.0
Nozzles	546.9
Weights	1115.0

Sum of the Detail Weights 2679.3 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	3543.4	3543.4	3543.4	3543.4	3543.4	3543.4
Nozzles	546.9	546.9	546.9	546.9	546.9	546.9
Wld Weights	1000.0	1000.0	1000.0	1000.0	1000.0	1000.0
Packing	...	...	...	...	186.4	186.4
Insulation	...	...	686.0	686.0	686.0	686.0
Empty Weights	...	...	...	115.0	115.0	...
Test Weights	...	115.0	...	...	...	...
Ope Weights	...	...	...	...	...	115.0
Ope. Liquid	...	...	...	...	...	144.9
Test Liquid	...	1451.6	...	...	...	...
Totals	5090.3	6656.9	5776.3	5891.3	6077.8	6222.7

Field Installation Options:

* Packing installed after lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	5090.3 kgm
Shop Test Wt.	- Fabricated weight + water (full)	6656.9 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	5776.3 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	5891.3 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	6077.8 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	6222.7 kgm
Field Test Wt.	- Empty weight + test medium (full)	7342.9 kgm

Weights used in Vortex Shedding Calculations:



Mass of the Upper 1/3 of the Vertical Vessel (operating) 2353.7 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water) 2843.6 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty) 2522.7 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	3800070	3277309
20	30	1193805	1017876
30	40	221496	172817
40	50	3973246	3665370
50	60	2245293	2094497
60	70	386083	181575
70	80	335660	149895
80	90	3592470	3351195
90	100	336319	148655
100	110	336319	148655
110	120	3592470	3351195
120	130	346404	154939
130	140	345745	156231
140	150	4007849	3738677
150	160	346404	154939
160	170	315561	...

Total 25375198.0 21763826.0 mm²
Total 25.4 21.8 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	1049.64	1049.64	1049.64	...	1049.64
20	30	260.354	260.354	260.354	...	260.354
30	40	69.1526	77.6242	78.3612	...	69.1526
40	50	762.056	913.345	1067.39	330427	776.91
50	60	299.198	317.406	473.673	...	317.406
60	70	108.4	108.4	123.907	...	108.4
70	80	77.4499	77.4499	90.2003	...	77.4499
80	90	641.866	692.994	921.025	315175	692.994
90	100	77.9361	77.9361	90.4778	...	77.9361
100	110	77.9361	77.9361	90.4778	...	77.9361
110	120	1241.87	1292.99	1521.03	315175	1292.99
120	130	84.1646	84.1646	97.2486	...	84.1646
130	140	83.6089	83.6089	96.9107	...	83.6089
140	150	848.939	900.067	1160.38	364322	900.067
150	160	84.1646	84.1646	97.2486	...	84.1646
160	170	124.601	124.601	124.601	...	124.601

Empty Support Force + the Sum of the Y forces 61894.27 N
Operating Support Force + the Sum of the Y forces 65143.51 N
Hydro Support Force + the Sum of the Y forces 76128.47 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	6077.78	6222.68	7342.91
20	30	5028.14	5173.05	6293.27
30	40	4767.78	4912.69	6032.92
40	50	4698.63	4835.07	5954.56



50	60	3921.72	3921.72	4887.17
60	70	3604.32	3604.32	4413.5
70	80	3495.92	3495.92	4289.59
80	90	3418.47	3418.47	4199.39
90	100	2725.47	2725.47	3278.37
100	110	2647.54	2647.54	3187.89
110	120	2569.6	2569.6	3097.41
120	130	1276.61	1276.61	1576.39
130	140	1192.44	1192.44	1479.14
140	150	1108.83	1108.83	1382.23
150	160	208.765	208.765	221.85
160	170	124.601	124.601	124.601

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	1325100	1325100	1325100
20	30	1325100	1325100	1325100
30	40	1325100	1325100	1325100
40	50	1325100	1325100	1325100
50	60	994673	994673	994673
60	70	994673	994673	994673
70	80	994673	994673	994673
80	90	994673	994673	994673
90	100	679498	679498	679498
100	110	679498	679498	679498
110	120	679498	679498	679498
120	130	364322	364322	364322
130	140	364322	364322	364322
140	150	364322	364322	364322
150	160	...	...	...
160	170	...	...	...



Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N4-2"	9.34	10.20	100	600	GR 1.1	6.129	7.882	9.341	7.882
H4-6"	4.67	5.10	100	300	GR 1.1	...	...	...	...
K1-2"	4.67	5.10	100	300	GR 1.1	...	...	...	...
K2-2"	4.67	5.10	100	300	GR 1.1	...	...	...	...
N1-2"	9.34	10.20	100	600	GR 1.1	6.129	7.882	9.341	7.882
H3-6"	4.67	5.10	100	300	GR 1.1	...	...	...	...
H2-6"	4.67	5.10	100	300	GR 1.1	...	...	...	...
H1-6"	4.67	5.10	100	300	GR 1.1	...	...	...	...
N2-3"	9.34	10.20	100	600	GR 1.1	5.320	8.691	9.341	8.691
N3-3"	4.67	5.10	100	300	GR 1.1	...	...	...	...
Min Rating	4.670	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N4-2"	77.851	0.50	495002	3749	4.063	7.882	Pass
N1-2"	77.851	0.50	495000	3749	4.063	7.882	Pass
N2-3"	111.125	0.50	1267201	5998	4.063	8.691	Pass

Evaluating Flange on Nozzle [N4-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(495002) + 4 \cdot \text{abs}(3749) 77.851 \leq \pi \cdot 77.851^3 [(9.341 - 4.063) + .500 \cdot 9.341]$$

$$= 0.90878E+07 \leq 0.14751E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9.341, 0.500 \cdot 9.341 + 9.341 - 9087821.0 / (\pi \cdot 77.851^3))$$

$$= \min(9.341, 7.882)$$

$$= 7.882 \text{ MPa}$$

Evaluating Flange on Nozzle [N1-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(495000) + 4 \cdot \text{abs}(3749) 77.851 \leq \pi \cdot 77.851^3 [(9.341 - 4.063) + .500 \cdot 9.341]$$

$$= 0.90878E+07 \leq 0.14751E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9.341, 0.500 \cdot 9.341 + 9.341 - 9087787.0 / (\pi \cdot 77.851^3))$$

$$= \min(9.341, 7.882)$$

$$= 7.882 \text{ MPa}$$

Evaluating Flange on Nozzle [N2-3"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1267201) + 4 \cdot \text{abs}(5998) 111.125 \leq \pi \cdot 111.125^3 [(9.341 - 4.063) + .500 \cdot 9.341]$$

$$= 0.22942E+08 \leq 0.42902E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-3" [Pmax]:



$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9.341, 0.500 \cdot 9.341 + 9.341 - 22942268.0 / (\pi \cdot 111.125^3)) \\ &= \min(9.341, 8.691) \\ &= 8.691 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 0.5169 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.51664 Hz.

The Natural Frequency for the Vessel (Filled) is 0.47167 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	14006.2	-1162989
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	3889.09	14434217
150	160	...	...
160	170	...	...

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	14006.2	-1162989
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	3889.09	14434217
150	160	...	...
160	170	...	...

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
30	-83.00	-11285.		8296.			
140	3570.01	2750.	-4124.	2750.	385021.		385021.



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Dynamic Gust-Effect Factor, Operating Case [Gf]:

$$= 0.925[(1+1.7 \cdot I_{zbar} \cdot \sqrt{(gQ^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_{zbar})]$$

$$= 0.925[(1+1.7 \cdot 0.193 \cdot \sqrt{(3.4^2 \cdot 0.922^2 + 4.029^2 \cdot 1.223^2)}) / (1+1.7 \cdot 3.4 \cdot 0.193)]$$

$$= 1.275$$

Natural Frequency of Vessel (Operating)	0.517 Hz
Natural Frequency of Vessel (Empty)	0.517 Hz
Natural Frequency of Vessel (Test)	0.472 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.700
Structure Height to Diameter ratio	42.393
Height to top of Structure	20799.635 mm

This is classified as a flexible structure. Dynamic analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
Exposure Category: C from Table 26.11-1
α = 9.5: Zg = 274320.0 mm

Effective Height [z]:
= Centroid Height + Vessel Base Elevation
= 1600.0 + 300.0 = 1900.0 mm
= 6.234 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
Because z (6.234 ft.) < 15 ft.

$$= 2.01(15 / Zg)^{2/\alpha}$$

$$= 2.01(15/900.0)^{2/9.5}$$

$$= 0.849$$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
= 1, per Table 26.6-1



As there is No Hill Present: [Kzt]:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 $= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 $= 16.0 \text{ psf } [0.766] \text{ kPa}$

Force on the first element [F]:
 $= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$
 $= 16.0 \cdot 1.275 \cdot 0.7 \cdot 17.837$
 $= 254.7 \text{ lbs. } [1132.8] \text{ N}$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	1900.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-1	4000.0	0.000	0.000	0.000	0.849	1.000	0.766
Bottom head	4525.0	0.000	0.000	0.000	0.849	1.000	0.766
Shell-1	6300.0	0.000	0.000	0.000	0.908	1.000	0.766
Shell-2	9050.0	0.000	0.000	0.000	0.980	1.000	0.766
Flange-1	10130.5	0.000	0.000	0.000	1.004	1.000	0.766
Flange-2	10276.4	0.000	0.000	0.000	1.007	1.000	0.766
Shell-3	11946.4	0.000	0.000	0.000	1.039	1.000	0.766
Flange-3	13620.9	0.000	0.000	0.000	1.068	1.000	0.766
Flange-4	13760.8	0.000	0.000	0.000	1.071	1.000	0.766
Shell-4	15430.8	0.000	0.000	0.000	1.097	1.000	0.766
Flange-5	17108.3	0.000	0.000	0.000	1.121	1.000	0.766
Flange-6	17254.2	0.000	0.000	0.000	1.123	1.000	0.766
Shell-5	19112.2	0.000	0.000	0.000	1.147	1.000	0.766
Flange-7	20974.7	0.000	0.000	0.000	1.170	1.000	0.766
Flange-8	21073.6	0.000	0.000	0.000	1.171	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
D - Average outer insulated diameter of vessel, mm
Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
Dr - Average outer insulated of top half of vessel, mm
f - Natural frequency of vibration (Hertz)
f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
L - Total height of structure mm
Lc - Total length of conical section(s) of vessel, mm
tb - Uncorroded plate thickness at bottom of vessel, mm
V30 - Design Wind Speed provided by user, km/hr
Vc - Critical wind velocity, km/hr
Vw - Maximum wind speed at top of structure, km/hr
W - Total corroded weight of structure, N
Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
Z - Maximum amplitude of vibration at top of vessel, mm
Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
:: $0.0/20799.635 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)



$$:: [1.61/68.24^2] \cdot 10^4 = 3.466$$

Compute the vibration possibility. If $V_p > 0.000004$ no chance. $[V_p]$:

$$= W / (L \cdot Dr^2)$$

$$= 56428 / (20799.63 \cdot 470.947^2)$$

$$= 0.12232E-04$$

Since V_p is > 0.000004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	300.00	0.77	0.00
Vortex	0.00	4500.00	0.77	0.00
DLBS-4	0.00	7120.00	0.77	0.00
TABS-4	0.00	7120.00	0.77	0.00
H4 Davit	0.00	7370.00	0.77	0.00
DLLD-4	0.00	9440.00	0.77	0.00
TALD-4	0.00	9440.00	0.77	0.00
DLLD-3	0.00	13266.40	0.77	0.00
TALD-3	0.00	13266.40	0.77	0.00
DLBS-3	0.00	10946.40	0.77	0.00
TABS-3	0.00	10346.40	0.77	0.00
H3 Davit	0.00	11196.40	0.77	0.00
DLLD-2	0.00	16750.80	0.77	0.00
TALD-2	0.00	16750.80	0.77	0.00
DLBS-2	0.00	14430.80	0.77	0.00
TABS-2	0.00	14430.80	0.77	0.00
H2 Davit	0.00	14680.80	0.77	0.00
Clip weight	0.00	15430.80	0.77	0.00
Misc wt 7%	0.00	15392.40	0.77	0.00
DLLD-1	0.00	20247.20	0.77	0.00
TALD-1	0.00	20247.20	0.77	0.00
TABS-1	0.00	17927.20	0.77	0.00
BSLD-1	0.00	17927.20	0.77	0.00
DLMP	0.00	20827.20	0.77	0.00
TAMP	0.00	20827.20	0.77	0.00
DLFP	0.00	20727.20	0.77	0.00
H1 davit	0.00	18177.20	0.77	0.00
Trunnion 1	0.00	19897.20	0.77	0.00
Trunnion 2	0.00	19897.20	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 0.51664 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1900	517.86	1657151	0.76606	1246.08
20	30	4000	630.2	630200	0.76606	473.873
30	40	4525	736.169	36808.4	0.76606	27.6777
40	50	6300	714.249	2499871	0.76606	1879.75
50	60	9050	708.769	1417537	0.76606	1065.9
60	70	10130.5	671.044	101964	0.76606	76.6706
70	80	10276.4	674.925	94454.5	0.76606	71.0241
80	90	11946.4	708.769	2268060	0.76606	1705.45
90	100	13620.9	655.609	91751.2	0.76606	68.9914
100	110	13760.8	655.609	91751.2	0.76606	68.9914
110	120	15430.8	708.769	2268060	0.76606	1705.45
120	130	17108.3	647.431	94491.3	0.76606	71.0518
130	140	17254.2	651.241	95047.4	0.76606	71.4699
140	150	19112.2	708.769	2530304	0.76606	1902.64
150	160	20974.7	647.431	94491.3	0.76606	71.0518
160	170	21073.6	880.708	45796.8	0.76606	34.4365



The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(68.24030.75)$
 $= 0.475$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/0.517)$
 $= 1.936$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.475, 1.936)$ per 12.8.2
 $= 0.807$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.157 / (2.0/1.25)$
 $= 0.098$

Maximum value of Cs per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / Ie))$
 $= 0.104 / (0.807(2.0/1.25))$
 $= 0.081$

Minimum value of Cs per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot Ie, 0.030)$
 $= \max(0.044 \cdot 0.1568 \cdot 1.25, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= Cs \cdot W$ (Equation 12.8-1):
 $= 0.0805 \cdot 61019.6$
 $= 4913.300$ N

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$



$$= 0.2 \cdot 0.157 \cdot 61020$$

$$= 1913.6 \text{ N}$$

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.7178

The Natural Frequency for the Vessel (Ope...) is 0.51664 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1600	10292.7	25.979	25.838	-2210.32
20	30	3700	2553.03	27.1996	27.0412	-2533.11
30	40	4225	761.183	10.1855	9.02048	-2613.17
40	50	6000	8956.26	218.917	181.55	-2637.04
50	60	8750	3112.48	145.456	136.258	-2917.91
60	70	9830.47	1062.97	60.6746	60.2935	-3015.51
70	80	9976.42	759.474	44.4625	44.1828	-3048.85
80	90	11646.4	6795.5	518.998	477.649	-3072.67
90	100	13320.9	764.241	73.5179	73.0455	-3285.77
100	110	13460.8	764.241	74.8496	74.3683	-3309.74
110	120	15130.8	12679.1	1518.08	1448.59	-3333.71
120	130	16808.3	825.317	118.376	117.602	-3731.32
130	140	16954.2	819.869	119.354	118.573	-3757.21
140	150	18812.2	8826.06	1536.16	1439.35	-3782.92
150	160	20674.7	825.317	168.935	167.814	64.1987
160	170	20773.6	1221.84	252.158	250.485	38.3168



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1600	17895.3	17895.3	137982656	137982656
20	30	3700	17895.3	17895.3	80956192	80956192
30	40	4225	17895.3	17895.3	63191080	63191080
40	50	6000	3889.09	3889.09	64145968	64145968
50	60	8750	3889.09	3889.09	50528640	50528640
60	70	9830.47	3889.09	3889.09	42729808	42729808
70	80	9976.42	3889.09	3889.09	42138628	42138628
80	90	11646.4	3889.09	3889.09	41594136	41594136
90	100	13320.9	3889.09	3889.09	29126504	29126504
100	110	13460.8	3889.09	3889.09	28582012	28582012
110	120	15130.8	3889.09	3889.09	28037520	28037520
120	130	16808.3	3889.09	3889.09	15569886	15569886
130	140	16954.2	3889.09	3889.09	15002052	15002052
140	150	18812.2	3889.09	3889.09	14434217	14434217
150	160	20674.7	...	...	...	...
160	170	20773.6	...	...	...	...

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1600	10540.5	4913.3	113875440	76486240
20	30	3700	9294.42	4887.32	82126696	60798888
30	40	4225	8820.55	4860.12	73065528	55923196
40	50	6000	8792.87	4849.94	72625008	55680344
50	60	8750	6913.12	4631.02	45128384	39081944
60	70	9830.47	5847.22	4485.56	32336548	29941470
70	80	9976.42	5770.55	4424.89	31453546	29264234
80	90	11646.4	5699.52	4380.43	30650612	28647838
90	100	13320.9	3994.08	3861.43	15116587	15438144
100	110	13460.8	3925.08	3787.91	14562225	14902671
110	120	15130.8	3856.09	3713.06	14017523	14377584
120	130	16808.3	2150.65	2194.98	4393159	4911001
130	140	16954.2	2079.6	2076.61	4084336	4599160
140	150	18812.2	2008.13	1957.25	3785917	4304673
150	160	20674.7	105.488	421.093	11110.5	55707.5
160	170	20773.6	34.4365	252.158	895.656	6558.37

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	17895.3	1.88147	6.96753	0.0040057
20	30	17895.3	9.07363	11.3704	0.0047685
30	40	17895.3	11.4899	11.61	0.0048148
40	50	3889.09	22.5606	38.1903	0.010162
50	60	3889.09	49.1681	61.691	0.013253
60	70	3889.09	62.6982	63.7058	0.013266
70	80	3889.09	64.6346	65.5644	0.013294
80	90	3889.09	88.5111	114.457	0.017044
90	100	3889.09	115.65	116.844	0.017063
100	110	3889.09	118.038	119.233	0.017083
110	120	3889.09	147.666	177.948	0.019394
120	130	3889.09	179.364	180.779	0.019403
130	140	3889.09	182.196	183.612	0.019412
140	150	3889.09	218.522	253.646	0.019686
150	160	...	255.083	256.519	0.019686
160	170	...	257.031	257.543	0.019686

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)

Allowable deflection : 103.998 Actual deflection : 257.543 mm

**** Warning: Allowable Deflection Criteria Exceeded ****

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	10540.5	1.60393	6.09745	0.003615
20	30	9294.42	8.01282	10.1358	0.0044454
30	40	8820.55	10.2472	10.3594	0.0044988
40	50	8792.87	21.0419	36.5594	0.010037
50	60	6913.12	47.3196	59.3789	0.012639
60	70	5847.22	60.3393	61.3001	0.012649
70	80	5770.55	62.1856	63.0719	0.01267
80	90	5699.52	84.5409	107.937	0.015091
90	100	3994.08	108.993	110.05	0.015101
100	110	3925.08	111.107	112.164	0.015111
110	120	3856.09	136.863	162.308	0.016056
120	130	2150.65	163.48	164.652	0.016058
130	140	2079.6	165.823	166.995	0.016061
140	150	2008.13	195.811	224.754	0.016221
150	160	105.488	225.938	227.121	0.016221
160	170	34.4365	227.543	227.965	0.016221

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	4.80291	30.0182
20	30	5.84481	36.5301
30	40	6.82762	42.6726



40	50	6.62432	41.402
50	60	6.5735	41.0844
60	70	6.22362	38.8976
70	80	6.25962	39.1226
80	90	6.5735	41.0844
90	100	6.08047	38.0029
100	110	6.08047	38.0029
110	120	6.5735	41.0844
120	130	6.00462	37.5289
130	140	6.03996	37.7497
140	150	6.5735	41.0844
150	160	6.00462	37.5289
160	170	8.16815	51.0509

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)

Allowable deflection : 103.998 Actual deflection : 227.965 mm

**** Warning: Allowable Deflection Criteria Exceeded ****

Total Deflection in the Operating Condition + Applied Forces:

Allowable deflection : 103.998 Actual deflection : 485.508 mm

**** Warning: Allowable Deflection Criteria Exceeded ****

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	28751.9	26982.5	2368985	2231207
20	30	30963.5	29194.2	2539987	2403293
30	40	24560	21390	2062517	1824902
40	50	15277.2	12107.2	1277308	1029172
50	60	13019.4	9849.36	1087624	836711
60	70	13019.4	9849.36	1087624	836711
70	80	13019.4	9849.36	1087624	836711
80	90	13019.4	9849.36	1087624	836711
90	100	13019.4	9849.36	1087624	836711
100	110	13019.4	9849.36	1087624	836711
110	120	13019.4	9849.36	1087624	836711
120	130	13019.4	9849.36	1087624	836711
130	140	13019.4	9849.36	1087624	836711
140	150	13019.4	9849.36	1087624	836711
150	160	13019.4	9849.36	1087624	836711
160	170	13019.4	9849.36	1087624	836711



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	115.918	196.556	-147.277	-147.277
20	30	82.7985	140.397	-147.277	-147.277
30	40	165.597	280.795	-147.277	-147.277
40	50	165.597	280.795	-147.277	-147.277
50	60	165.597	280.795	-146.106	-146.106
60	70	165.597	269.995	-146.106	-146.106
70	80	165.597	269.995	-146.106	-146.106
80	90	165.597	280.795	-146.106	-146.106
90	100	165.597	269.995	-146.106	-146.106
100	110	165.597	269.995	-146.106	-146.106
110	120	165.597	280.795	-146.106	-146.106
120	130	165.597	269.995	-146.106	-146.106
130	140	165.597	269.995	-146.106	-146.106
140	150	165.597	280.795	-146.106	-146.106
150	160	165.597	269.995	-146.106	-146.106
160	170	165.597	269.995	-146.106	-146.106



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	7.67767	-0.54072	9.98098
40	50	13.5233	-0.87601	17.5802
50	60	16.6084	-1.05312	21.591
60	70	...	...	...
70	80	...	...	...
80	90	16.6084	-1.05312	21.591
90	100	...	...	...
100	110	...	...	...
110	120	16.6084	-1.05312	21.591
120	130	...	...	...
130	140	...	...	...
140	150	16.6084	-1.05312	21.591
150	160	...	...	...
160	170	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-2.20898	-2.26164	-2.66879	0.5937	0.5937
20	30	-1.68904	-1.73772	-2.11402	0.55119	0.55119
30	40	-2.18592	-2.18592	-2.18592	0.72589	0.72589
40	50	-3.80588	-3.80588	-3.80588	1.28713	1.28713
50	60	-3.90479	-3.90479	-3.90479	1.18841	1.18841
60	70	-3.58875	-3.58875	-3.58875	1.18841	1.18841
70	80	-3.48082	-3.48082	-3.48082	1.18841	1.18841
80	90	-3.40371	-3.40371	-3.40371	1.18841	1.18841
90	100	-2.7137	-2.7137	-2.7137	0.81185	0.81185
100	110	-2.6361	-2.6361	-2.6361	0.81185	0.81185
110	120	-2.5585	-2.5585	-2.5585	0.81185	0.81185
120	130	-1.27109	-1.27109	-1.27109	0.43528	0.43528
130	140	-1.18729	-1.18729	-1.18729	0.43528	0.43528
140	150	-1.10405	-1.10405	-1.10405	0.43528	0.43528
150	160	-0.20786	-0.20786	-0.20786	...	...
160	170	-0.12406	-0.12406	-0.12406	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.5937	51.0212	34.2692	25.5106	...
20	30	0.55119	34.1616	25.2901	17.0808	...
30	40	0.72589	40.0252	30.6347	20.0126	...
40	50	1.28713	70.5438	54.0847	35.2719	...
50	60	1.18841	53.9182	46.6941	26.9591	...
60	70	1.18841	38.6348	35.7733	19.3174	...



70	80	1.18841	37.5799	34.9641	18.7899	...
80	90	1.18841	36.6205	34.2277	18.3103	...
90	100	0.81185	18.0609	18.4451	9.03045	...
100	110	0.81185	17.3986	17.8053	8.69928	...
110	120	0.81185	16.7478	17.178	8.37388	...
120	130	0.43528	5.24883	5.86753	2.62441	...
130	140	0.43528	4.87986	5.49495	2.43993	...
140	150	0.43528	4.52331	5.14311	2.26166	...
150	160	...	0.013275	0.066558	0.0066373	...
160	170	...	0.0010701	0.0078358	0.00053505	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	32.5803
20	30	...	...	...	24.0704
30	40	...	...	...	29.1717
40	50	...	...	...	51.5034
50	60	...	...	...	44.5291
60	70	...	...	...	34.1311
70	80	...	...	...	33.3607
80	90	...	...	...	32.6599
90	100	...	...	...	17.6136
100	110	...	...	...	17.0025
110	120	...	...	...	16.4037
120	130	...	...	...	5.6136
130	140	...	...	...	5.2586
140	150	...	...	...	4.9242
150	160	...	...	...	0.066116
160	170	...	...	...	0.0077838

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.15285	-0.081924
20	30	-0.14127	-0.086775
30	40	-0.19281	-0.12218
40	50	-0.34064	-0.21783
50	60	-0.41873	-0.29628
60	70	-0.41873	-0.30619
70	80	-0.41873	-0.30957
80	90	-0.41873	-0.31199
90	100	-0.41873	-0.33363
100	110	-0.41873	-0.33606
110	120	-0.41873	-0.3385
120	130	-0.41873	-0.37887
130	140	-0.41873	-0.3815
140	150	-0.41873	-0.38411
150	160	...	0.0065186
160	170	...	0.0038906

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	61.8223	61.8223	58.2268	58.2268
20	30	33.6747	33.6747	31.8625	31.8625
30	40	34.616	34.616	30.628	30.628



40	50	62.3078	62.3078	50.2035	50.2035
50	60	60.3703	60.3703	46.4429	46.4429
60	70	51.0524	51.0524	39.2747	39.2747
70	80	50.3461	50.3461	38.7313	38.7313
80	90	49.6956	49.6956	38.2308	38.2308
90	100	34.7996	34.7996	26.7713	26.7713
100	110	34.149	34.149	26.2709	26.2709
110	120	33.4985	33.4985	25.7704	25.7704
120	130	18.6025	18.6025	14.3109	14.3109
130	140	17.9241	17.9241	13.789	13.789
140	150	17.2456	17.2456	13.2671	13.2671
150	160	...	...	...	...
160	170	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
EP = Longitudinal Stress due to External Pressure
HP = Longitudinal Stress due to Hydrotest Pressure
NP = No Pressure
EW = Longitudinal Stress due to Weight (No Liquid)
OW = Longitudinal Stress due to Weight (Operating)
HW = Longitudinal Stress due to Weight (Hydrotest)
WI = Bending Stress due to Wind Moment (Operating)
EQ = Bending Stress due to Earthquake Moment (Operating)
EE = Bending Stress due to Earthquake Moment (Empty)
HI = Bending Stress due to Wind Moment (Hydrotest)
HE = Bending Stress due to Earthquake Moment (Hydrotest)
WE = Bending Stress due to Wind Moment (Empty) (no CA)
WF = Bending Stress due to Wind Moment (Filled) (no CA)
CW = Longitudinal Stress due to Weight (Empty) (no CA)
VO = Bending Stress due to Vortex Shedding Loads (Ope)
VE = Bending Stress due to Vortex Shedding Loads (Emp)
VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
FW = Axial Stress due to Vertical Forces for the Wind Case
FS = Axial Stress due to Vertical Forces for the Seismic Case
BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.08	115.92	-115.80	147.28	0.9582	0.7863
20	66.56	82.80	-70.22	147.28	0.8038	0.4768
30	72.99	165.60	-77.75	147.28	0.4408	0.5279
40	129.99	165.60	-138.29	147.28	0.7850	0.9389
50	111.15	165.60	-119.80	146.11	0.6712	0.8200
80	83.68	165.60	-91.33	146.11	0.5053	0.6251
110	48.08	165.60	-54.04	146.11	0.2903	0.3698
140	20.68	165.60	-23.73	146.11	0.1249	0.1624

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	92.71	115.92	-97.29	147.28	0.7998	0.6606
20	56.52	82.80	-60.07	147.28	0.6826	0.4079
30	62.21	165.60	-66.82	147.28	0.3756	0.4537
40	111.07	165.60	-119.12	147.28	0.6708	0.8088
50	101.89	165.60	-110.29	146.11	0.6153	0.7549
80	79.83	165.60	-87.26	146.11	0.4821	0.5972
110	47.82	165.60	-53.61	146.11	0.2888	0.3669
140	21.12	165.60	-24.09	146.11	0.1275	0.1649

Analysis of Load Case 3 : NP+OW+WI+FW+BW



From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.02	115.92	-115.85	147.28	0.9578	0.7866
20	66.51	82.80	-70.27	147.28	0.8033	0.4771
30	72.99	165.60	-77.75	147.28	0.4408	0.5279
40	129.99	165.60	-138.29	147.28	0.7850	0.9389
50	111.15	165.60	-119.80	146.11	0.6712	0.8200
80	83.68	165.60	-91.33	146.11	0.5053	0.6251
110	48.08	165.60	-54.04	146.11	0.2903	0.3698
140	20.68	165.60	-23.73	146.11	0.1249	0.1624

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	94.34	115.92	-99.03	147.28	0.8139	0.6724
20	57.69	82.80	-61.34	147.28	0.6968	0.4165
30	63.67	165.60	-68.28	147.28	0.3845	0.4636
40	113.66	165.60	-121.70	147.28	0.6863	0.8264
50	104.05	165.60	-112.45	146.11	0.6283	0.7697
80	81.40	165.60	-88.83	146.11	0.4915	0.6080
110	48.59	165.60	-54.39	146.11	0.2934	0.3722
140	21.34	165.60	-24.31	146.11	0.1288	0.1664

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	23.44	196.56	-28.77	147.28	0.1192	0.1954
20	15.52	140.40	-19.75	147.28	0.1105	0.1341
30	18.55	280.79	-22.92	147.28	0.0661	0.1557
40	32.75	280.79	-40.36	147.28	0.1166	0.2741
50	24.24	280.79	-32.05	146.11	0.0863	0.2194
80	16.09	280.79	-22.90	146.11	0.0573	0.1568
110	6.63	280.79	-11.74	146.11	0.0236	0.0804
140	1.59	280.79	-3.80	146.11	0.0057	0.0260

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-3.26	147.28		0.0222
20		140.40	-2.67	147.28		0.0181
30		280.79	-2.91	147.28		0.0198
40		280.79	-5.09	147.28		0.0346
50		280.79	-5.09	146.11		0.0349
80		280.79	-4.59	146.11		0.0314
110		280.79	-3.37	146.11		0.0231
140		280.79	-1.54	146.11		0.0105

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.02	115.92	-115.85	147.28	0.9578	0.7866
20	66.51	82.80	-70.27	147.28	0.8033	0.4771
30	80.67	165.60	-70.07	147.28	0.4871	0.4758
40	143.52	165.60	-124.76	147.28	0.8667	0.8471
50	127.76	165.60	-103.19	146.11	0.7715	0.7063
80	100.29	165.60	-74.72	146.11	0.6056	0.5114
110	64.69	165.60	-37.43	146.11	0.3906	0.2562
140	37.29	165.60	-7.12	146.11	0.2252	0.0487

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	94.34	115.92	-99.03	147.28	0.8139	0.6724



20	57.69	82.80	-61.34	147.28	0.6968	0.4165
30	71.35	165.60	-60.61	147.28	0.4308	0.4115
40	127.18	165.60	-108.18	147.28	0.7680	0.7345
50	120.66	165.60	-95.85	146.11	0.7286	0.6560
80	98.00	165.60	-72.22	146.11	0.5918	0.4943
110	65.20	165.60	-37.78	146.11	0.3937	0.2586
140	37.94	165.60	-7.70	146.11	0.2291	0.0527

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.02	115.92	-115.85	147.28	0.9578	0.7866
20	66.51	82.80	-70.27	147.28	0.8033	0.4771
30	72.45	165.60	-78.29	147.28	0.4375	0.5316
40	129.12	165.60	-139.16	147.28	0.7797	0.9449
50	110.10	165.60	-120.85	146.11	0.6649	0.8272
80	82.63	165.60	-92.38	146.11	0.4990	0.6323
110	47.03	165.60	-55.09	146.11	0.2840	0.3770
140	19.63	165.60	-24.78	146.11	0.1185	0.1696

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	94.34	115.92	-99.03	147.28	0.8139	0.6724
20	57.69	82.80	-61.34	147.28	0.6968	0.4165
30	63.13	165.60	-68.83	147.28	0.3812	0.4673
40	112.78	165.60	-122.58	147.28	0.6811	0.8323
50	103.00	165.60	-113.51	146.11	0.6220	0.7769
80	80.34	165.60	-89.88	146.11	0.4852	0.6152
110	47.54	165.60	-55.44	146.11	0.2871	0.3794
140	20.28	165.60	-25.37	146.11	0.1225	0.1736

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	23.44	196.56	-28.77	147.28	0.1192	0.1954
20	15.52	140.40	-19.75	147.28	0.1105	0.1341
30	28.53	280.79	-12.94	147.28	0.1016	0.0879
40	50.33	280.79	-22.78	147.28	0.1793	0.1547
50	45.83	280.79	-10.46	146.11	0.1632	0.0716
80	37.69	280.79	-1.31	146.11	0.1342	0.0090
110	28.22	280.79		146.11	0.1005	
140	23.18	280.79		146.11	0.0826	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-3.26	147.28		0.0222
20		140.40	-2.67	147.28		0.0181
30	8.52	280.79		147.28	0.0303	
40	15.06	280.79		147.28	0.0536	
50	18.87	280.79		146.11	0.0672	
80	19.38	280.79		146.11	0.0690	
110	19.84	280.79		146.11	0.0707	
140	20.92	280.79		146.11	0.0745	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-2.80	147.28		0.0190
20		82.80	-2.24	147.28		0.0152
30	6.22	165.60		147.28	0.0375	
40	11.00	165.60		147.28	0.0665	
50	13.89	165.60		146.11	0.0839	



80	14.39	165.60	146.11	0.0869
110	14.86	165.60	146.11	0.0897
140	15.94	165.60	146.11	0.0963

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-2.12	147.28		0.0144
20		82.80	-1.64	147.28		0.0111
30	5.77	165.60		147.28	0.0349	
40	10.51	165.60		147.28	0.0634	
50	13.65	165.60		146.11	0.0825	
80	14.03	165.60		146.11	0.0847	
110	14.67	165.60		146.11	0.0886	
140	15.77	165.60		146.11	0.0953	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-2.86	147.28		0.0194
20		82.80	-2.29	147.28		0.0155
30	6.22	165.60		147.28	0.0375	
40	11.00	165.60		147.28	0.0665	
50	13.89	165.60		146.11	0.0839	
80	14.39	165.60		146.11	0.0869	
110	14.86	165.60		146.11	0.0897	
140	15.94	165.60		146.11	0.0963	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-2.80	147.28		0.0190
20		82.80	-2.24	147.28		0.0152
30	6.22	165.60		147.28	0.0375	
40	11.00	165.60		147.28	0.0665	
50	13.89	165.60		146.11	0.0839	
80	14.39	165.60		146.11	0.0869	
110	14.86	165.60		146.11	0.0897	
140	15.94	165.60		146.11	0.0963	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-2.86	147.28		0.0194
20		82.80	-2.29	147.28		0.0155
30		165.60	-2.91	147.28		0.0198
40		165.60	-5.09	147.28		0.0346
50		165.60	-5.09	146.11		0.0349
80		165.60	-4.59	146.11		0.0314
110		165.60	-3.37	146.11		0.0231
140		165.60	-1.54	146.11		0.0105

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	60.07	96.60	-64.76	122.73	0.6219	0.5277
20	32.40	69.00	-36.05	122.73	0.4696	0.2937
30	40.71	138.00	-29.97	122.73	0.2950	0.2442
40	73.09	138.00	-54.10	122.73	0.5297	0.4408
50	73.97	138.00	-49.15	121.76	0.5360	0.4037
80	63.78	138.00	-37.99	121.76	0.4622	0.3120
110	48.02	138.00	-20.60	121.76	0.3480	0.1692
140	32.80	138.00	-2.56	121.76	0.2377	0.0210



Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	60.07	96.60	-64.76	122.73	0.6219	0.5277
20	32.40	69.00	-36.05	122.73	0.4696	0.2937
30	32.49	138.00	-38.19	122.73	0.2355	0.3112
40	58.70	138.00	-68.49	122.73	0.4253	0.5581
50	56.30	138.00	-66.81	121.76	0.4080	0.5487
80	46.12	138.00	-55.65	121.76	0.3342	0.4571
110	30.36	138.00	-38.26	121.76	0.2200	0.3142
140	15.14	138.00	-20.22	121.76	0.1097	0.1661

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-2.86	147.28		0.0194
20		82.80	-2.29	147.28		0.0155
30	6.22	165.60		147.28	0.0375	
40	11.00	165.60		147.28	0.0665	
50	13.89	165.60		146.11	0.0839	
80	14.39	165.60		146.11	0.0869	
110	14.86	165.60		146.11	0.0897	
140	15.94	165.60		146.11	0.0963	

Absolute Maximum of the all of the Stress Ratio's 0.9582

Governing Element: Skirt-2

Governing Load Case 1 : NP+EW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-2.4	0.6	112.8	111.1	+ Tens
Skirt-2	...	-2.4	-0.6	-112.8	-115.8	- Comp
Skirt-1	...	-1.8	0.6	67.8	66.6	+ Tens
Skirt-1	...	-1.8	-0.6	-67.8	-70.2	- Comp
Bottom head	...	-2.4	0.7	74.6	73.0	+ Tens
Bottom head	...	-2.4	-0.7	-74.6	-77.7	- Comp
Shell-1	...	-4.1	1.3	132.9	130.0	+ Tens
Shell-1	...	-4.1	-1.3	-132.9	-138.3	- Comp
Shell-2	...	-4.3	1.2	114.3	111.2	+ Tens
Shell-2	...	-4.3	-1.2	-114.3	-119.8	- Comp
Shell-3	...	-3.8	1.2	86.3	83.7	+ Tens
Shell-3	...	-3.8	-1.2	-86.3	-91.3	- Comp
Shell-4	...	-3.0	0.8	50.2	48.1	+ Tens
Shell-4	...	-3.0	-0.8	-50.2	-54.0	- Comp
Shell-5	...	-1.5	0.4	21.8	20.7	+ Tens
Shell-5	...	-1.5	-0.4	-21.8	-23.7	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt-1

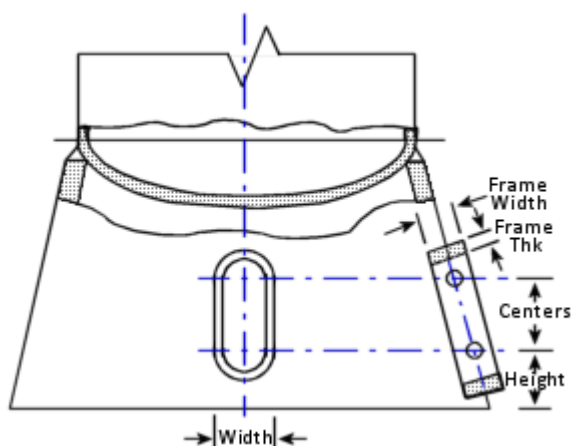
Total vessel (empty) weight	Fg	47442.938 N
Head Weight below section 2-2	DeltaFg	678.110 N
Weight of vessel contents (liquid)	Ff	1420.955 N
Vessel Gross Weight	Fg + Ff	48863.891 N
Skirt Mean Diameter	Dz	352.000 mm
Skirt Analysis Thickness (Corroded)	ez	26.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-4123.900 N
Moment at section 1-1	M1	136256608.000 N-mm
Applied Moment - Bottom of Skirt	Mb	163082896.000 N-mm
Weight at Bottom of Skirt	W	50726.867 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 358.350 mm
Head Corroded Thickness	eb 19.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 86.337 \text{ mm} \\
 K &= D_i / (2 * h_i) = 339.35 / (2 * 86.337) = 1.9653 \\
 R &= D_i (0.44 * K + 0.02) = 339.35 * (0.44 * 1.965 + 0.02) = 300.2263 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 339.35 * ((0.5 / 1.965) - 0.08) = 59.1894 \text{ mm}
 \end{aligned}$$



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	100.0000	0.0	375.00	0.00	80.000	16.000
2	50.8000	0.0	800.00	0.00	80.000	16.000
3	50.8000	90.0	800.00	0.00	80.000	16.000
4	50.8000	180.0	800.00	0.00	80.000	16.000
5	50.8000	270.0	800.00	0.00	80.000	16.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	425.0000	349.6000
2	3	276.4555	225.6555
3	4	276.4555	225.6555
4	5	276.4555	225.6555
1	5	507.0036	431.6036

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 276.4555 mm
 Minimum Ligament between Openings is: 225.6555 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(352.0 \cdot 26.4)} \text{ mm}$$

$$= 192.798 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 5

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
2	-8.2978	8.2978
3	81.7022	98.2978
4	171.7022	188.2978
5	261.7022	278.2978

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
0.000	-8.298
8.298	81.702
98.298	171.702
188.298	261.702
-81.702	0.000

Height of Weakest Section	H	799.970	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	12.307	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	528685632.000	mm ⁴
Principle Moment of Inertia	Iyy	603585984.000	mm ⁴
Distance to Extreme Fiber	ybar	189.200	mm
Distance to Extreme Fiber	xbar	201.507	mm
Minimum Section Modulus	W4	2794323.750	mm ³
Metal Area at the Cutting Plane	A4	36610.922	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(100.0/352.0)$$

$$= 0.2881 \text{ rad (16.5045 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 352.0/2$$

$$= 176.0 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.



Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
2560.0	2239.2	...	...
2560.0	920.2	...	...
2560.0	920.2	...	...
2560.0	920.2	...	...
2560.0	920.2	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{\left(\frac{r}{ez} \right)}$$

$$= 0.288 \cdot \sqrt{\left(\frac{176.0}{26.4} \right)}$$

$$= 0.744$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= ex(new) - ci - cext$$

$$= 28.0 - 0.8 - 0.8$$

$$= 26.4 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top}$$

$$= (.16308E+09 - 0.13626E+09) \cdot (1000.0 - 799.97) / 1000.0 + 0.13626E+09$$

$$= 0.14162E+09 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{bar} = 0.52869E+09 / 189.2 = 2794324 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{bar} = 0.60359E+09 / 201.507 = 2995367 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(2794324, 2995367) = 2794324 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\left(\frac{\min(I_{xx}, I_{yy})}{Area(A_4)} \right)}$$

$$= \sqrt{\left(\frac{\min(0.52869E+09, 0.60359E+09)}{36610.922} \right)}$$

$$= 120.169 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 120.169 / 1000.0$$

$$= 0.120$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{ex^2 + ey^2}$$

$$= \sqrt{(12.307^2 + -0.0^2)}$$

$$= 12.307 \text{ mm}$$

$$\text{Where } ex = 12.307 : ey = -0.0$$

Moment Increment at Centroid [ΔM4]:

$$= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$$

$$= \text{abs}(-4123.9 - 47442.94 - 1420.96) \cdot 12.307$$

$$= 652369.938 \text{ N-mm}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= -4123.9 - 47442.938 - 1420.955 + 4 \cdot 0.13626E+09 / 352.0$$

$$= 1494755 \text{ N}$$



Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -4123.9 - 47442.938 - 1420.955 - 4 \cdot 0.13626E+09/352.0$
 $= -1600730.6 \text{ N}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (1494755 + 678.11 + 1420.955) / (\pi \cdot 358.35 \cdot 19.0) + 1.8 \cdot 358.35 / (4 \cdot 19.0)$
 $= 78.47 \text{ MPa}$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-0.2E+07 + 678.11 + 1420.955) / (\pi \cdot 358.35 \cdot 19.0) + 1.8 \cdot 358.35 / (4 \cdot 19.0)$
 $= -66.26 \text{ MPa}$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (1420.955 + 678.11) / (\pi \cdot 358.35 \cdot 19.0) + 1.8 \cdot 358.35 / (4 \cdot 19.0)$
 $= 8.59 \text{ MPa}$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= 1494755 / (\pi \cdot 352.0 \cdot 26.4)$
 $= 51.20 \text{ MPa}$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -1600730.625 / (\pi \cdot 352.0 \cdot 26.4)$
 $= -54.84 \text{ MPa}$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:
 $= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5(358.35 + 19.0 - 352.0 + 26.4) / (59.19 + 19.0)$
 $= 0.6691$

$\gamma = \arccos(\cos(\gamma)) = \arccos(0.669) = 48.004 \text{ deg}$

Dimension [a]:
 $= 0.5 \cdot \sqrt{eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma)}$
 $= 0.5 \cdot \sqrt{19.0^2 + 26.4^2 + 2 \cdot 19.0 \cdot 26.4 \cdot 0.669}$
 $= 20.792 \text{ mm}$

Bending moment at verification points p and q [Mp & Mq]:
 $Mp = a \cdot Fzp = 20.79 \cdot 1494755.1 = 31.0911 \times 10^6 \text{ N-mm}$
 $Mq = a \cdot Fzq = 20.79 \cdot -1600730.6 = -33.2954 \times 10^6 \text{ N-mm}$

Correction factor [C]:
 $= 0.63 - 0.057(eb/ez)^2$
 $= 0.63 - 0.057(19.0/26.4)^2$
 $= 0.600$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:
 $= C \cdot 6 \cdot Mp / (\pi \cdot Db \cdot eb^2)$
 $= 0.6 \cdot 6 \cdot 31091080 / (\pi \cdot 358.35 \cdot 19.0^2)$
 $= 275.54 \text{ MPa}$



Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2)$$

$$= 0.6 \cdot 6 \cdot -0.3E+08 / (\pi \cdot 358.35 \cdot 19.0^2)$$

$$= -295.07 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2)$$

$$= 0.6 \cdot 6 \cdot 31091080 / (\pi \cdot 352.0 \cdot 26.4^2)$$

$$= 145.29 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2)$$

$$= 0.6 \cdot 6 \cdot -0.3E+08 / (\pi \cdot 352.0 \cdot 26.4^2)$$

$$= -155.59 \text{ MPa}$$

Geometry variables, 16.12-32 to 36:

$$e_b/D_b = 0.748/14.108 = 0.053$$

$$y = 125 \cdot e_b/D_b = 125 \cdot 19.0/358.35 = 6.6276$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 6.628 = 2.8745$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + P_h) \cdot D_b \cdot ((y / y_A) \cdot \alpha - 1) / (4 \cdot e_b)$$

$$= (1.8 + 0.0) \cdot 358.35 \cdot ((48.0/40.0) \cdot 2.874 - 1) / (4 \cdot 19.0)$$

$$= 20.79 \text{ MPa}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 78.473 - 275.537 + 20.792$$

$$= -176.27 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 78.473 + 275.537 - 20.792$$

$$= 333.22 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= -66.256 - -295.072 + 20.792$$

$$= 249.61 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= -66.256 + -295.072 - 20.792$$

$$= -382.12 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 8.586 + 275.537 + 20.792$$

$$= 304.91 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 8.586 - 275.537 - 20.792$$

$$= -287.74 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 8.586 + -295.072 + 20.792$$

$$= -265.69 \text{ MPa}$$

Stress [σ_{2qotot}]:



$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 8.586 - -295.072 - 20.792 \\
 &= 282.87 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3pitot}] &: \\
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= 51.205 - 145.293 \\
 &= -94.09 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3potot}] &: \\
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= 51.205 + 145.293 \\
 &= 196.50 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qitot}] &: \\
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -54.835 - -155.594 \\
 &= 100.76 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}] &: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -54.835 + -155.594 \\
 &= -210.43 \text{ MPa}
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 137.997(3 - (1/1.5)) \cdot (78.473/137.997)^2 \\
 &= 384.24 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 137.997(3 - (1/1.5)) \cdot (-66.256/137.997)^2 \\
 &= 392.79 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 137.997(3 - (1/1.5)) \cdot (8.586/137.997)^2 \\
 &= 413.64 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 137.997(3 - (1/1.5)) \cdot (8.586/137.997)^2 \\
 &= 413.64 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 137.997(3 - (1/1.5)) \cdot (51.205/137.997)^2 \\
 &= 401.33 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 137.997(3 - (1/1.5)) \cdot (-54.835/137.997)^2 \\
 &= 399.47 \text{ MPa}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

$$\begin{aligned}
 \text{Weakening Factor, 16.12-70(b) [psi1]:} \\
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 36610.922 / (\pi \cdot 352.0 \cdot 26.4)) \\
 &= 1.000
 \end{aligned}$$

$$\begin{aligned}
 \text{Weakening Factor, 16.12-70(c) [psi2]:} \\
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^23 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 2995367 / (\pi \cdot 352.0^2 \cdot 26.4)) \\
 &= 1.000
 \end{aligned}$$

Maximum Bending Moment [Mmax]:



$$= \pi/4 \cdot d_3^2 \cdot e_{a3} \cdot \sigma_{call}$$

$$= \pi/4 \cdot 352.0^2 \cdot 26.4 \cdot 147.3$$

$$= 378489504.0 \text{ N-mm}$$

Maximum Compressive Force [F_{c,max}]:

$$= \pi \cdot d_3 \cdot e_{a3} \cdot \sigma_{call}$$

$$= \pi \cdot 352.0 \cdot 26.4 \cdot 147.3$$

$$= 4299274.0 \text{ N}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$= F_4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_g)) / (\psi_2 \cdot M_{max})$$

$$= 52987.8 / (1.0 \cdot 4299274) + (\text{abs}(0.14162E+09) + 52987.8 \cdot \text{abs}(12.307)) /$$

$$(1.0 \cdot 0.37849E+09)$$

$$= 0.388 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	-176.27	384.24	Pass	46
Stot1po	333.22	384.24	Pass	47
Stot1qi	249.61	392.79	Pass	48
Stot1qo	-382.12	392.79	Pass	49
Stot2pi	304.91	413.64	Pass	50
Stot2po	-287.74	413.64	Pass	51
Stot2qi	-265.69	413.64	Pass	52
Stot2qo	282.87	413.64	Pass	53
Stot3pi	-94.09	401.33	Pass	54
Stot3po	196.50	401.33	Pass	55
Stot3qi	100.76	399.47	Pass	56
Stot3qo	-210.43	399.47	Pass	57

The stresses are satisfactory in the region without openings.
The stresses are satisfactory in the region with the opening(s).

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Shop/Field Installation Options :

Packing is installed in the Field after being lifted.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	13872.699 mm
Center of Gravity of the Liquid	5044.948 mm
Center of Gravity of the Insulation	12246.634 mm
Center of Gravity of the Nozzles	12775.685 mm
Center of Gravity of the Added Weights (Operating)	14233.676 mm
Center of Gravity of the Added Weights (Empty)	14233.676 mm
Center of Gravity of Bare Shell New and Cold	9055.754 mm
Center of Gravity of Bare Shell Corroded	8740.027 mm
Vessel CG in the Operating Condition	10679.342 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	10752.617 mm
Vessel CG in the Test Condition	11086.862 mm

Rigging Analysis Results:

- Lifting weight based on the user selected calculated empty weight

Total Effective Length of Vessel for this analysis	20781.63 mm
Total vessel weight (No Liquid)	Twt 59598.65 N
Impact weight multiplication factor	Imp 1.50
Design lifting weight, DWT = Imp * Twt	89397.97 N
Elevation of the Tailing Lug (bottom)	0.00 mm
Elevation of the Lifting Lug (top)	15000.00 mm
Design Reaction force at the tailing lug	25313.82 N
Design Reaction force at the lifting lug	64084.15 N
CG distance during field lifting operation	10752.62 mm
CG distance from tailing lug	10752.62 mm
CG distance from the nearer lifting lug/trunnion	4247.38 mm

Compressive Allowable Increase Factor 1.00

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	67.72	14801.79	122.73 (UG-23)
Shear	2.62	14801.79	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	0.0	0.0	24440.7	0.9
4250.00	61798456.0	48.4	4422.3	0.3
9901.95	36306544.0	32.8	13326.6	1.0
13381.84	31132916.0	28.6	25424.5	2.0
16867.73	37172660.0	33.6	17318.9	1.3
20729.62	49179.6	0.0	1890.8	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.5517 at elevation 14801.7881 mm - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.



--- Plot data successfully generated ...----

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	378.0000	mm
Skirt Thickness	STHK	26.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	56.0000	mm
Design Temperature of the Basing		82.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	MPa
Basing Yield Stress		242.60	MPa
Inside Diameter of Basing	DI, D4	198.0000	mm
Outside Diameter of Basing	DOU, D5	658.0000	mm

Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	284.00	MPa
Number of Bolts	RN, nb	8	
Diameter of Bolt Circle	DC, DBC	518.0000	mm
Bolt Allowable Shear Stress		170.400	MPa

Ultimate Comp. Strength of Concrete	FPC	41.3	MPa
Allowable Comp. Strength of Concrete	FC	22.5	MPa
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	24.0000	mm
Width of Gussets at Top Plate	TWDT	130.0000	mm
Width of Gussets at Base Plate	BWDT	130.0000	mm
Gusset Plate Elastic Modulus	E	199226.0	MPa
Gusset Plate Yield Stress	SY	242.6	MPa
Height of Chair	h1	300.0000	mm
Distance between Gussets	RG, b6	90.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	42.0000	mm
Radial Width of the Top Plate	TOPPTH, b5	140.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	58.0000	mm

External Corrosion Allowance	CA	3.0000	mm
------------------------------	----	--------	----

Dead Weight of Vessel	DW	63722.5	N
Operating Weight of Vessel	ROW	65143.5	N
Test Weight of Vessel	TW	72004.6	N
Earthquake Moment on Basing	EQMOM	76486232.0	N-mm
Wind Moment on Basing	WIMOM	113875440.0	N-mm

User Moment at the Base [Earthquake Case]		137982656.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		137982656.0	N-mm (UWIMOM)
Test Moment on Basing	TM	125929048.0	N-mm
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
-----------------------------------	------	--------	--

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:



Bolt Load, Earthquake Operating Case:
 $= ((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 0.21447E+09/518.0) - 63230) / 8$
 $= 199028.6 \text{ N}$

Bolt Load, Windload Empty Case:
 $= ((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 0.25186E+09/518.0) - 63723) / 8$
 $= 235042.4 \text{ N}$

Bolt Load, Windload Vortex Shedding Case:
 $= ((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 0/518.0) - 65144) / 8$
 $= -8142.9 \text{ N}$

Bolt Load, Test Moment Case:
 $= ((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 0.12593E+09/518.0) - 72005) / 8$
 $= 112503.3 \text{ N}$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded) Abss : 855.6120 mm²
 Area Available in all the Bolts Abss * RN : 6844.8960 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.211	0.317	1.253	2.636	0.457	0.776	588631.1	652353.7	132249.5
4	0.264	0.238	1.410	2.520	0.445	0.779	587136.3	650858.9	137962.4
6	0.251	0.244	1.372	2.549	0.448	0.779	587450.8	651173.3	136477.3
8	0.248	0.246	1.363	2.556	0.449	0.779	587535.3	651257.9	136112.2
10	0.245	0.245	1.355	2.562	0.449	0.778	587600.4	651322.9	135840.0

The Actual Stress in a Single Bolt [Sbolt]:
 $= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$
 $= 2 \cdot 587600 / (4.206 \cdot 518.0 \cdot 2.562)$
 $= 210.57 \text{ MPa, Should be } \leq 284.0 \text{ MPa}$

Thickness of the Band of Bolting Steel [T1]
 $= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$
 $= 8 \cdot 855.612 / (\pi \cdot 518.0)$
 $= 4.206 \text{ mm}$

Check the Bearing Stress in the Concrete [fc(max)]
 $= fc'[(2kd + t3) / (2kd)]$
 $= 6.946[(2 \cdot 0.2451 \cdot 518.0 + 230.0) / (2 \cdot 0.2451 \cdot 518.0)]$
 $= 13.238 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$

Values for table 10.3, l = 140.0, b = 101.709, l/b = 1.376475

Maximum Moment per unit width [Mmax]:
 $= \max(Mx, My) = \max(15969.844, 31850.426) = 31850.426 \text{ N}$

Reqd Thickness of Basing, Brownell & Young Method [T]:
 $= \sqrt{(6 \cdot Mmax / fallow)} + Ca$
 $= \sqrt{(6 \cdot 31850.426 / 160.1)} + 3.0$
 $= 37.549 \text{ mm}$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140.0, b = 90.0, b/l = 0.642857



As b/l (0.643) is less than 1, inverting $b/l = 1.556$.

Moment Term, based on geometry [Mo]:

$$= P/(4 \cdot \pi) [1.3(\ln((2\sin(\pi \cdot a/l)/(\pi \cdot g))) + 1)] - [(0.7 - g^2)P/(4\pi)]$$

$$= 242973.14/(4 \cdot \pi) [1.3(\ln((2 \cdot 140.0 \cdot \sin(\pi \cdot 70.0/140.0))/(\pi \cdot 35.0))) + 1] - [(0.7 - 0.065) \cdot 242973.14/(4 \cdot \pi)]$$

$$= 30542.7949 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / F_b) + C_a}$$

$$= \sqrt{(6 \cdot \text{Abs}(30542.79)/161.73) + 3.0}$$

$$= 36.6629 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{\text{avg}}) + C_a$$

$$= 242973.14/(124.11 \cdot 130.0) + 3.0$$

$$= 18.061 \text{ (not less than } 9.525 + 3.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 518.0/8 = 203.418 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a/(m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + C_a$$

$$= 1.76(242973 \cdot 70.0/(203.42 \cdot 258.0 \cdot 207))^{2/3} \cdot 189.01^{1/3} + C_a$$

$$= 13.619 + 1.6 = 15.219 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force}/(\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 28435.8/(855.612 \cdot 8)$$

$$= 4.2 \text{ MPa. Must be less than } 170.4 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	37.5491	mm
Actual Basing Thickness as entered by user	56.0000	mm
Required Thickness of Chair Cap	36.6629	mm
Actual Top Ring Thickness as entered by user	42.0000	mm
Required Gusset thickness, + CA	18.0607	mm
Actual Gusset Thickness as entered by user	24.0000	mm
Required Thickness of Skirt for Local Stress	15.2191	mm
Given Thickness of Skirt	26.0000	mm
Required Gusset Height to meet local stress	61.5572	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$$

$$= 115.852 \cdot (26.0 - 1.6)$$

$$= 2826.54 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 2827/[(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$

$$= 20.601 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (C_{\text{mwth}} + 2 \cdot (H_g + T_{\text{ta}}))$$

$$= 242973.1/(88.0 + 2 \cdot (202.0 + 42.0))$$

$$= 421.828 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (C_{\text{mwth}} \cdot (H_g + T_{\text{ta}}) + 0.6667 \cdot (H_g + T_{\text{ta}})^2)$$

$$= 242973.1 \cdot 70.0/(88.0 \cdot (244.0) + 0.6667 \cdot (244.0)^2)$$



= 278.079 N/mm

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{W_v^2 + W_h^2} \\ &= \sqrt{421.83^2 + 278.08^2} \\ &= 505.24 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 505.24 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 3.682 \text{ mm} \end{aligned}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 242973.1 / (2 \cdot 130.0) \\ &= 934.512 \text{ N/mm} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 242973.1 \cdot 70.0 / (2 \cdot 244.0 \cdot 130.0) \\ &= 268.098 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{W_v^2 + W_h^2} \\ &= \sqrt{934.51^2 + 268.1^2} \\ &= 972.208 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 972.21 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 7.086 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	20.6014	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	7.0860	mm
Required Gusset to Top Plate Double Fillet Weld Size	7.0860	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N4-2"

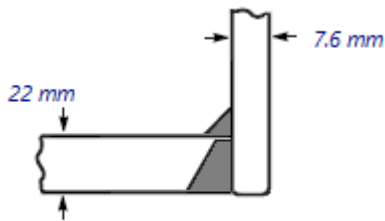
From: 30

Pressure for Reinforcement Calculations	P	1.8166	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	333.35	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		300.0	mm
Centerline Distance to Tangent		300.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.82 \cdot 339.35 \cdot 0.8851 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1.82)$
 $= 1.9796 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.82 \cdot 30.1625 / (138 \cdot 1.0 + 0.4 \cdot 1.82)$
 $= 0.3950 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2826 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	102.0684 mm
Parallel to Vessel Wall, opening length	d	51.0342 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	11.6135 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3950 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.1852 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.1852 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(5.1852, 4.5)]$
 $= 5.1852 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.395, 5.1852)$
 $= 5.1852 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	47.0,	Allowable	:	138.0 MPa	Passed
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Expansion :	0.0,	Allowable :	298.0 MPa	Passed
Occasional :	4.6,	Allowable :	183.5 MPa	Passed
Shear :	35.8,	Allowable :	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 7.645$, $t_r = 0.395$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.085$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 7.645$, $t_r = 0.395$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.085$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.82/10.20 = 0.178$

Weld Size Calculations, Description: N4-2"

Intermediate Calc. for nozzle/shell Welds T_{min} 4.6454 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.2518 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.4355 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 173.4354 mm

Input Echo, WRC107/537 Item 1, Description: N4-2" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	3.0000 mm
Vessel Diameter	Dv	600.029 mm
Vessel Thickness	Tv	22.000 mm
Design Temperature	T1	100.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	60.325	mm
Nozzle Thickness	Tn	7.645	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	1.817	MPa
Include Pressure Thrust		No	

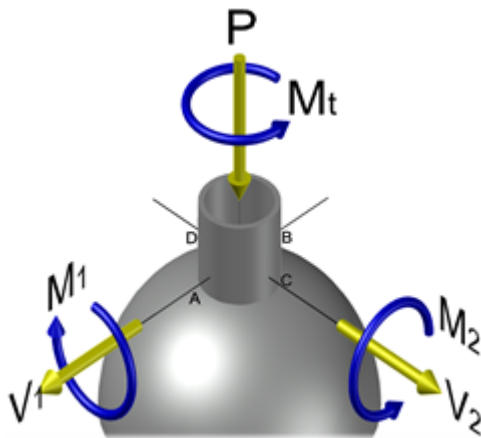
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3748.6	N
Longitudinal Shear (SUS)	(V1) V1	2499.0	N
Circumferential Shear (SUS)	(Vc) V2	2499.0	N
Circumferential Moment (SUS)	(Mc) M1	350019.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	350019.0	N-mm
Torsional Moment (SUS)	Mt	500025.8	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(312.515 (22.0 - 3.0))}$$

$$= 314.613 \text{ mm}$$

WRC 107 Stress Calculation for SUS tained loads:

Radial Load	P	-3748.6	N
Circumferential Shear	(VC) V2	2499.0	N
Longitudinal Shear	(VL) V1	2499.0	N
Circumferential Moment	(MC) M1	350019.0	N-mm
Longitudinal Moment	(ML) M2	350019.0	N-mm
Torsional Moment	MT	500025.8	N-mm

Dimensionless Param: U = 0.39 TAU = 5.99 RHO = 4.00 (4.09)



Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.07238	(A,B,C,D)
M(x) / P	SP 4	0.01801	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.20907	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.05615	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.20907	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.05615	(A,B,C,D)
N(y) * T / P	SP 4	0.30902	(A,B,C,D)
M(y) / P	SP 4	0.10370	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.25970	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	0.35892	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.25970	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	0.35892	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Rad. Bend. P		1.1	-1.1	1.1	-1.1	1.1	-1.1	1.1	-1.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-2.6	-2.6	2.6	2.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-4.2	4.2	4.2	-4.2
Rad. Memb. ML		-2.6	-2.6	2.6	2.6	0.0	0.0	0.0	0.0
Rad. Bend. ML		-4.2	4.2	4.2	-4.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-5.0	1.2	8.7	-2.0	-5.0	1.2	8.7	-2.0
Tang. Memb. P		3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Tang. Bend. P		6.5	-6.5	6.5	-6.5	6.5	-6.5	6.5	-6.5
Tang. Memb. MC		0.0	0.0	0.0	0.0	-3.3	-3.3	3.3	3.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-27.1	27.1	27.1	-27.1
Tang. Memb. ML		-3.3	-3.3	3.3	3.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-27.1	27.1	27.1	-27.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-20.7	20.6	40.0	-27.1	-20.7	20.6	40.0	-27.1
Shear VC		1.4	1.4	-1.4	-1.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Shear MT		4.6	4.6	4.6	4.6	4.6	4.6	4.6	4.6
Tot. Shear		6.0	6.0	3.2	3.2	3.2	3.2	6.0	6.0
Str. Int.		22.7	22.7	40.4	27.5	21.3	21.1	41.1	28.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		14.5	14.5	14.5	14.5	14.5	14.5	14.5	14.5
Rad. Pl (SUS)		-1.9	-1.9	3.4	3.4	-1.9	-1.9	3.4	3.4
Rad. Q (SUS)		-3.1	3.1	5.4	-5.4	-3.1	3.1	5.4	-5.4
Long. Pm (SUS)		14.5	14.5	14.5	14.5	14.5	14.5	14.5	14.5
Long. Pl (SUS)		-0.1	-0.1	6.5	6.5	-0.1	-0.1	6.5	6.5
Long. Q (SUS)		-20.6	20.6	33.5	-33.5	-20.6	20.6	33.5	-33.5



Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	1.4	1.4	-1.4	-1.4	-1.4	-1.4	1.4	1.4
Shear Q (SUS)	4.6	4.6	4.6	4.6	4.6	4.6	4.6	4.6
Pm (SUS)	14.5	14.5	14.5	14.5	14.5	14.5	14.5	14.5
Pm+Pl (SUS)	15.2	15.2	21.5	21.5	15.2	15.2	21.5	21.5
Pm+Pl+Q (Total)	19.7	36.8	54.9	25.9	17.0	35.6	55.7	27.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.52	138.00	Passed
Pm+Pl (SUS)	21.53	207.00	Passed
Pm+Pl+Q (TOTAL)	55.66	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: H4-6"

From: 40

Pressure for Reinforcement Calculations	P	1.8001	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	333.35	mm
Design Length of Section	L	5577.7793	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2870.00	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.4088	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

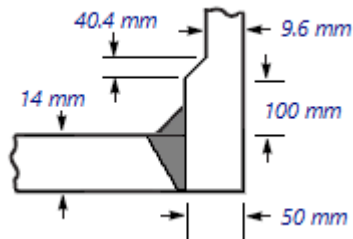
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

**Reinforcement CALCULATION, Description: H4-6"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 169.6748 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 2.2308 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.9449 and the shell thk. is 14.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.8 \cdot 84.1375 / (138 \cdot 1.0 + 0.4 \cdot 1.8)$
 $= 1.0918 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6362 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.399 / 0.5773$
 $= 69.9728 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

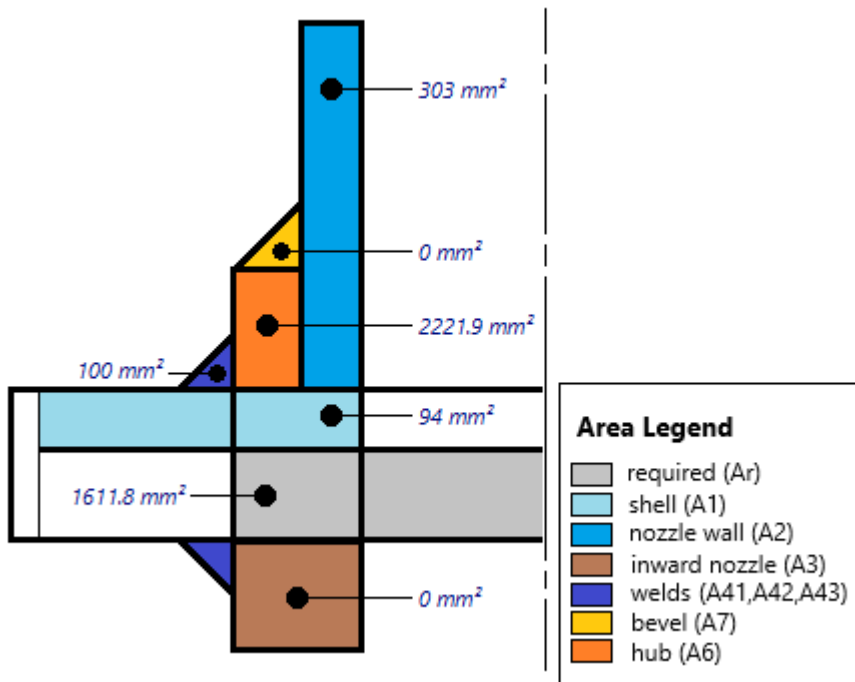
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.1452 mm
Parallel to Vessel Wall, opening length	d	155.0726 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1611.798	244.988	NA
Area in Shell	A1	94.001	1215.822	NA
Area in Nozzle Wall	A2	303.016	328.076	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	2221.934	2221.934	NA
TOTAL AREA AVAILABLE	Atot	2718.951	3865.832	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (155.0726 \cdot 10.3938 \cdot 1.0 + 2 \cdot 47.0 \cdot 10.3938 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 1611.798 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 155.073(1.0 \cdot 11.0 - 1.0 \cdot 10.394) - 2 \cdot 47.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 10.3938) \cdot (1 - 1.0)$$

$$= 94.001 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 27.5)(6.6 - 1.09)1.0$$

$$= 303.016 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



= 100.000 mm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(27.5, 250.0, 100.0)) \cdot (50.0 - 9.6) \cdot 1.0$$

$$= 2221.934 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.0918 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.0918 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 9.601$, $t_r = 1.092$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.165$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 2.231$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.203$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.80/5.10 = 0.353$

Weld Size Calculations, Description: H4-6"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 11.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1611.798 - 94.0006 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 11.0 - 10.3938)) \cdot 138)$$

$$= 217296.89 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (2524.9504 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$



$$= 362205.78 \text{ N}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (2524.9504 + 0.0 + 100.0 + (1034.0)) \cdot 138 \\ &= 504883.09 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (2524.9504 + 0.0 + 100.0 + 0.0 + (1034.0)) \cdot 138 \\ &= 504883.09 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 178719. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 80.8369) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138 \\ &= 1152916. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 168.275 \cdot (9.7346 - 0.0) \cdot 0.74 \cdot 138 \\ &= 262742. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (178719 + 1152916) = 1331634 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (178719 + 0 + 262742 + 0) = 441461 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (178719 + 262742 + 0) = 441461 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1331634 N	, must exceed W = 217296 N	or W1 = 362205 N
Path 2-2 = 441460 N	, must exceed W = 217296 N	or W2 = 504883 N
Path 3-3 = 441460 N	, must exceed W = 217296 N	or W3 = 504883 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 9 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 22.7951 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 286.7951 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 155.07 mm

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Input, Nozzle Desc: K1-2"

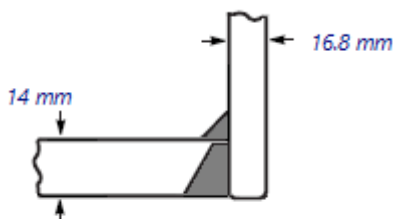
From: 40

Pressure for Reinforcement Calculations	P	1.8001	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	333.35	mm
Design Length of Section	L	5577.7793	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1850.00	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.8 \cdot 169.6748 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
= 2.2308 mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.9449 and the shell thk. is 14.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.8 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
= 0.3734 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3925	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.2308	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.2308	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(5.2308, 4.5)]
= 5.2308 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.3925, 5.2308)
= 5.2308 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 2.231, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c)$ = 0.203, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.80/5.10 = 0.353$

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 5.4307 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8057 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm

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Input, Nozzle Desc: K2-2"

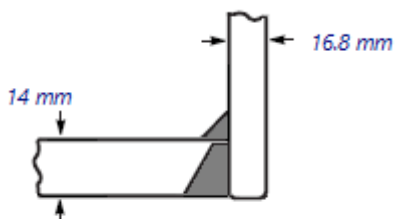
From: 40

Pressure for Reinforcement Calculations	P	1.8109	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	333.35	mm
Design Length of Section	L	5577.7793	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.81 \cdot 169.6748 / (138 \cdot 1.0 - 0.6 \cdot 1.81)$
= 2.2443 mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.9449 and the shell thk. is 14.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.81 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 1.81)$
= 0.3756 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3925	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.2443	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.2443	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(5.2443, 4.5)]
= 5.2443 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.3925, 5.2443)
= 5.2443 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 2.244, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.204$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.81/5.10 = 0.355$

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 5.4307 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8057 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm

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Input, Nozzle Desc: N1-2"

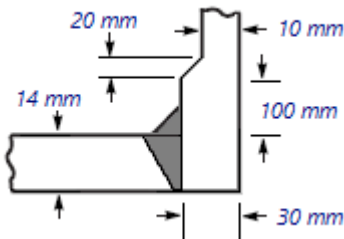
From: 40

Pressure for Reinforcement Calculations	P	1.8001	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	333.35	mm
Design Length of Section	L	5577.7793	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1750.00	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 169.6748 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 2.2308 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.9449 and the shell thk. is 14.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 0.3734 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3834 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 27.0$, less than the hub height, use (e-2)
 $= 67.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	132.8000 mm
Parallel to Vessel Wall	Rn+tn+t	66.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N1-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3834 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.2308 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.2308 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.5200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.52, \max(5.2308, 4.5)]$



= 5.2308 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.3834, 5.2308)
 = 5.2308 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	24.8,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	320.2 MPa	Passed
Occasional	:	3.3,	Allowable	:	183.5 MPa	Passed
Shear	:	19.8,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.373, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.053, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 2.231, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.203, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.80/10.20 = 0.176

Weld Size Calculations, Description: N1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 9.4764 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 273.4764 mm

Input Echo, WRC297 Item 1, Description: N1-2" :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm
Shell External Corrosion Allowance	0.0000 mm



Shell Diameter	333.350	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	1.80	MPa
Radial Load	P	-3748.56	N
Circumferential Shear	Vc	2499.04	N
Longitudinal Shear	Vl	2499.04	N
Circumferential Moment	Mc	350017.62	N-mm
Longitudinal Moment	Ml	350017.62	N-mm
Torsional Moment	Mt	500025.78	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (3.693) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 110.8 + 2 \cdot 1.65 \cdot \sqrt{175.175(14.0 - 3.0)}$$

$$= 255.659 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	350.350	mm
Nozzle Outside Diameter	(d) =	110.800	mm
Vessel Thickness used	(T) =	11.000	mm
Nozzle Thickness used	(t) =	27.000	mm
T / t	=	0.407	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.785$$

Nr/P	=	0.074
Mr/P	=	0.155
M0/P	=	0.045
N0/P	=	0.097
MrD/Mc	=	0.290
NrDL/Mc	=	0.096
M0d/Mc	=	0.090
N0DL/Mc	=	0.068
MrD/Ml	=	0.123
NrDL/Ml	=	0.077



MØD/M1 = 0.040
NØDL/M1 = 0.121

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	26	28	26	28
Outplane Stress Summary	28	26	47	20

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	28	-28	28	-28
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-19	19	19	-19
Inplane Pressure Stress	13	13	13	13
Inplane Stress Summary	23	4	65	-30

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	30	26	65	50

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-14	14	14	-14
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	13	13	13	13
Outplane Stress Summary	8	20	40	-4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to)	Cu Left	Cl Left	Du Right	Dl Right
---	------------	------------	-------------	-------------



circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	28	-28	28	-28
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-45	45	45	-45
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	26	28	26	28
Inplane Stress Summary	10	45	105	-40

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	10	45	106	41

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	1	3	7	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	0	-2	9	-1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	3	7	10	11

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	2	4	6	8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-6	6	6	-6
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-4	1	14	-5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	7	4	14	14

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	26	28	26	28	26	28	26	28
Circ. Pl (SUS)	0	0	6	6	0	0	4	4
Circ. Q (SUS)	2	-2	14	-14	-17	17	73	-73
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. Pl (SUS)	1	1	3	3	2	2	4	4
Long. Q (SUS)	9	-9	47	-47	-6	6	22	-22



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	26.0	28.0	26.0	28.0	26.0	28.0	26.0	28.0
Pm+P1 (SUS)	26.1	28.1	32.1	34.1	26.1	28.1	30.1	32.1
Pm+P1+Q (Total)	29.4	26.4	63.1	51.0	10.0	45.0	103.1	41.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28.00	138.00	Passed
Pm+P1 (SUS)	34.06	207.00	Passed
Pm+P1+Q (TOTAL)	103.14	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	3	1	3	1	3	1	3
Circ. P1 (SUS)	0	0	6	6	2	2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. P1 (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	1	-1	5	-5	-3	3	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0
Pm+P1 (SUS)	3.0	5.0	7.0	9.0	5.0	7.0	5.0	7.0
Pm+P1+Q (Total)	2.0	6.0	7.0	12.0	8.0	5.0	11.0	14.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+P1 (SUS)	9.00	207.00	Passed
Pm+P1+Q (TOTAL)	14.00	414.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: H3-6"

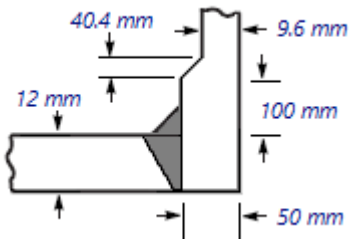
From: 80

Pressure for Reinforcement Calculations	P	1.8001	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	333.35	mm
Design Length of Section	L	3200.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6696.39	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.4088	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H3-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 169.6748 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 2.2308 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6323 and the shell thk. is 12.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.8 \cdot 84.1375 / (138 \cdot 1.0 + 0.4 \cdot 1.8)$
 $= 1.0918 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6362 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.399 / 0.5773$
 $= 69.9728 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

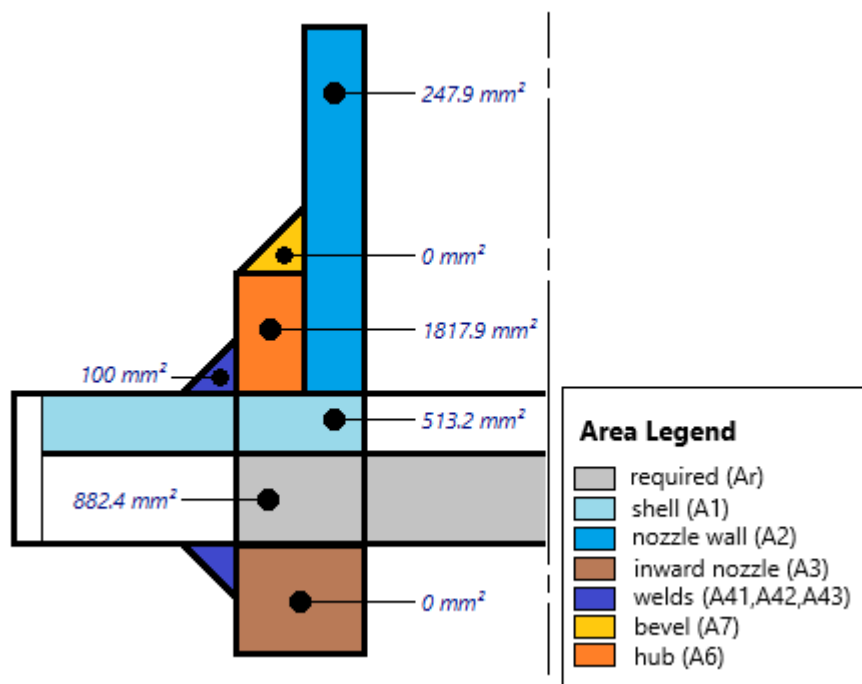
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.1452	mm
Parallel to Vessel Wall, opening length	d	155.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	882.443	226.576	NA
Area in Shell	A1	513.210	942.502	NA
Area in Nozzle Wall	A2	247.922	268.426	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1817.946	1817.946	NA
TOTAL AREA AVAILABLE	Atot	2679.078	3128.874	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (155.0726 \cdot 5.6905 \cdot 1.0 + 2 \cdot 47.0 \cdot 5.6905 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 882.443 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 155.073(1.0 \cdot 9.0 - 1.0 \cdot 5.691) - 2 \cdot 47.0$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 5.6905) \cdot (1 - 1.0)$$

$$= 513.210 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(6.6 - 1.09)1.0$$

$$= 247.922 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



= 100.000 mm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(22.5, 250.0, 100.0)) \cdot (50.0 - 9.6) \cdot 1.0$$

$$= 1817.946 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.0918 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.0918 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 9.601$, $t_r = 1.092$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.165$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.231$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.248$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.80/5.10 = 0.353$

Weld Size Calculations, Description: H3-6"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (882.4435 - 513.2098 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 9.0 - 5.6905)) 138)$$

$$= 93875.16 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (2065.8684 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$



$$= 298859.03 \text{ N}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (2065.8684 + 0.0 + 100.0 + (846.0)) \cdot 138 \\ &= 415594.97 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (2065.8684 + 0.0 + 100.0 + 0.0 + (846.0)) \cdot 138 \\ &= 415594.97 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 178719. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 80.8369) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138 \\ &= 1152916. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 168.275 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 138 \\ &= 242916. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (178719 + 1152916) = 1331634 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (178719 + 0 + 242916 + 0) = 421635 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (178719 + 242916 + 0) = 421635 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1331634 N	, must exceed W = 93875 N	or W1 = 298859 N
Path 2-2 = 421634 N	, must exceed W = 93875 N	or W2 = 415594 N
Path 3-3 = 421634 N	, must exceed W = 93875 N	or W3 = 415594 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 22.7951 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 284.7951 mm

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Input, Nozzle Desc: H2-6"

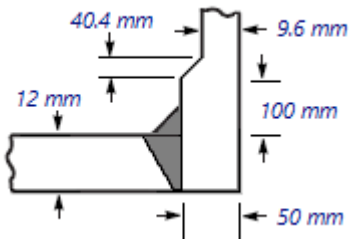
From: 110

Pressure for Reinforcement Calculations	P	1.8001	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	333.35	mm
Design Length of Section	L	3200.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		10180.70	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.4088	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H2-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 6.625 in.
 Actual Thickness Used in Calculation 0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 169.6748 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 2.2308 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.3937 and the shell thk. is 12.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.8 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 1.8)$
 $= 1.2757 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6362 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.399 / 0.5773$
 $= 69.9728 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

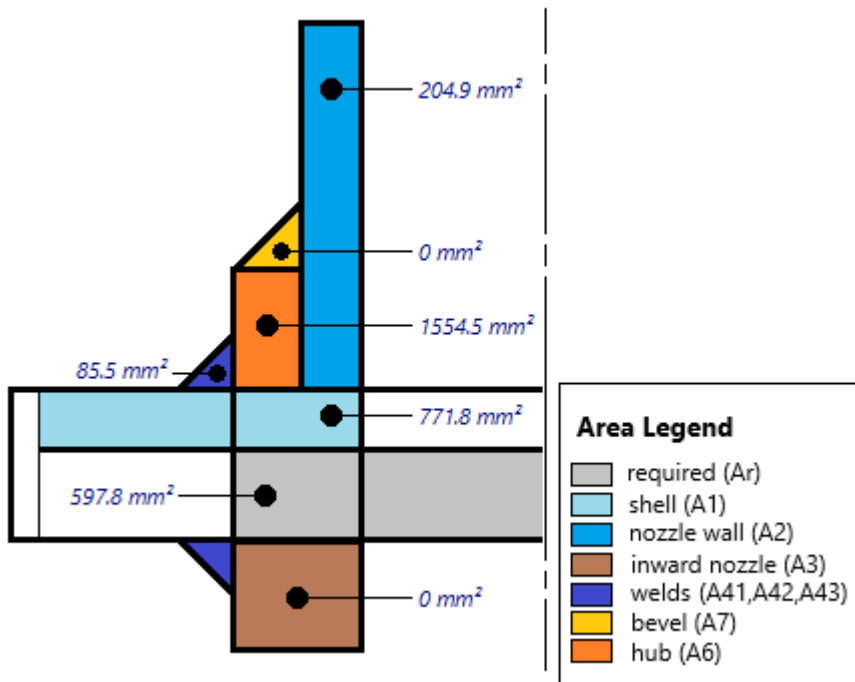
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.1452	mm
Parallel to Vessel Wall, opening length	d	155.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	597.773	246.478	NA
Area in Shell	A1	771.823	859.712	NA
Area in Nozzle Wall	A2	204.918	229.528	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.509	85.509	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1554.504	1554.504	NA
TOTAL AREA AVAILABLE	Atot	2616.753	2729.251	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (155.0726 \cdot 3.5435 \cdot 1.0 + 2 \cdot 47.0 \cdot 3.5435 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 597.773 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 155.073(1.0 \cdot 9.0 - 1.0 \cdot 3.544) - 2 \cdot 47.0$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 3.5435) \cdot (1 - 0.855)$$

$$= 771.823 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(6.6 - 1.28)0.8551$$

$$= 204.918 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$



$$= 85.509 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(22.5, 250.0, 100.0)) \cdot (50.0 - 9.6) \cdot 0.855 \\ &= 1554.504 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.2757 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 4.2757 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.601$, $t_r = 1.276$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.193$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.231$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.248$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.80/5.10 = 0.353$

Weld Size Calculations, Description: H2-6"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 9.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (597.773 - 771.8228 + 2 \cdot 47.0 \cdot 0.8551 \cdot \\ &\quad (1.0 \cdot 9.0 - 3.5435)) \cdot 138) \\ &= 36501.66 \text{ N} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot fr_2) \cdot S_v \\ &= (1759.4218 + 85.5088 - 0.0 \cdot 0.86) \cdot 138 \end{aligned}$$



$$= 254574.16 \text{ N}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (1759.4218 + 0.0 + 85.5088 + (723.4044)) \cdot 138 \\ &= 354393.66 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (1759.4218 + 0.0 + 85.5088 + 0.0 + (723.4044)) \cdot 138 \\ &= 354393.66 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118 \\ &= 152820. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 80.8369) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 118 \\ &= 985826. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 168.275 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 138 \\ &= 242911. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (152820 + 985827) = 1138647 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (152820 + 0 + 242911 + 0) = 395732 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (152820 + 242911 + 0) = 395732 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1138646 N	, must exceed W = 36501 N	or W1 = 254574 N
Path 2-2 = 395731 N	, must exceed W = 36501 N	or W2 = 354393 N
Path 3-3 = 395731 N	, must exceed W = 36501 N	or W3 = 354393 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 22.7951 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 284.7951 mm

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Input, Nozzle Desc: H1-6"

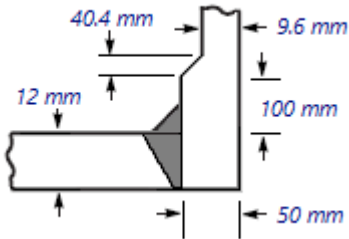
From: 140

Pressure for Reinforcement Calculations	P	1.8001	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	333.35	mm
Design Length of Section	L	3569.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		13677.20	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.4088	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H1-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 169.6748 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 2.2308 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.8 \cdot 84.1375 / (138 \cdot 1.0 + 0.4 \cdot 1.8)$
 $= 1.0918 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5132 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.399 / 0.5773$
 $= 69.9728 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

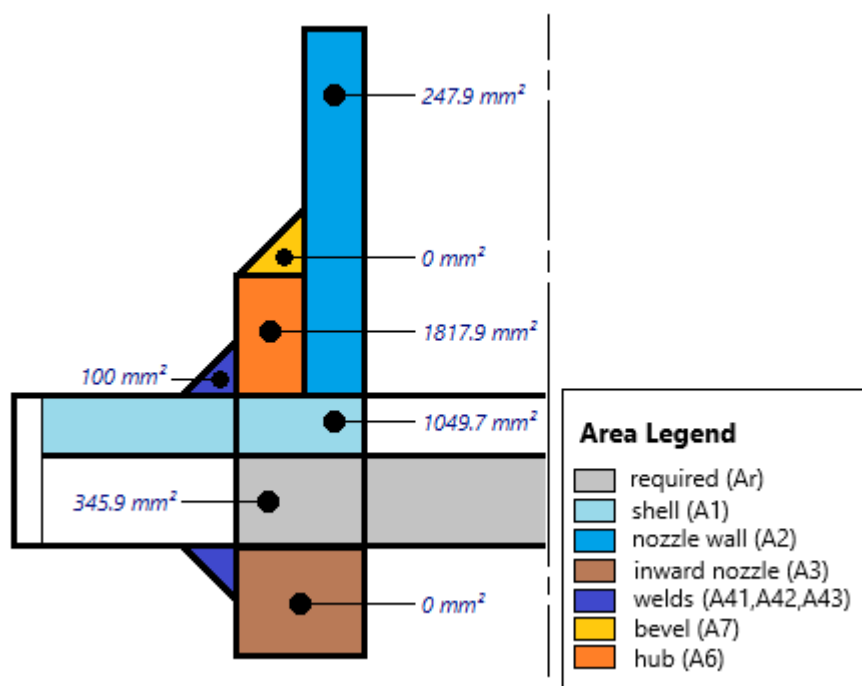
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.1452 mm
Parallel to Vessel Wall, opening length	d	155.0726 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	345.932	232.786	NA
Area in Shell	A1	1049.721	930.082	NA
Area in Nozzle Wall	A2	247.922	273.961	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1817.946	1817.946	NA
TOTAL AREA AVAILABLE	Atot	3215.589	3121.988	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (155.0726 \cdot 2.2308 \cdot 1.0 + 2 \cdot 47.0 \cdot 2.2308 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 345.932 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 155.073(1.0 \cdot 9.0 - 1.0 \cdot 2.231) - 2 \cdot 47.0 \\
 &\quad (1.0 \cdot 9.0 - 1.0 \cdot 2.2308) \cdot (1 - 1.0) \\
 &= 1049.721 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 22.5)(6.6 - 1.09)1.0 \\
 &= 247.922 \text{ mm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 \\
 &= 100.000 \text{ mm}^2
 \end{aligned}$$



Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(22.5, 150.0, 100.0)) \cdot (50.0 - 9.6) \cdot 1.0$
 $= 1817.946 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.0918 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 5.2308 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 5.2308 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 9.2200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[9.22, \max(5.2308, 4.5)]$
 $= 5.2308 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(4.0918, 5.2308)$
 $= 5.2308 \text{ mm}$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 9.601$, $t_r = 1.092$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.165$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.231$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.248$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.80/5.10 = 0.353$

Weld Size Calculations, Description: H1-6"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) S_v)$
 $= \max(0, (345.9325 - 1049.7208 + 2 \cdot 47.0 \cdot 1.0 \cdot$



$$(1.0 \cdot 9.0 - 2.2308) \cdot 138 \\ = \max(0, -9311.48) \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can}) / .707)^2 \cdot f_r2 \cdot S_v \\ = (2065.8684 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ = 298859.03 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v \\ = (2065.8684 + 0.0 + 100.0 + (846.0)) \cdot 138 \\ = 415594.97 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\ = (2065.8684 + 0.0 + 100.0 + 0.0 + (846.0)) \cdot 138 \\ = 415594.97 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ = (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 138 \\ = 178719. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ = (3.1416 \cdot 80.8369) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138 \\ = 1152916. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ = (3.1416/2.0) \cdot 168.275 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 138 \\ = 242916. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (178719 + 1152916) = 1331634 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (178719 + 0 + 242916 + 0) = 421635 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (178719 + 242916 + 0) = 421635 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1331634 N	, must exceed W = 0 N	or W1 = 298859 N
Path 2-2 = 421634 N	, must exceed W = 0 N	or W2 = 415594 N
Path 3-3 = 421634 N	, must exceed W = 0 N	or W3 = 415594 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 22.7951 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 184.7951 mm

Ammonia Scrubber
PV Elite 28 U2 Licensee: SL User
FileName : 123-D_Ope_W0 Overfill_Rev 02
Nozzle Calcs.: H1-6"



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HEXAGON

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Input, Nozzle Desc: N2-3"

From: 140

Pressure for Reinforcement Calculations	P	1.8001	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	333.35	mm
Design Length of Section	L	3569.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		16227.20	mm
-----------------------------------	--	----------	----

User Entered Minimum Design Metal Temperature		-28.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		N/A	

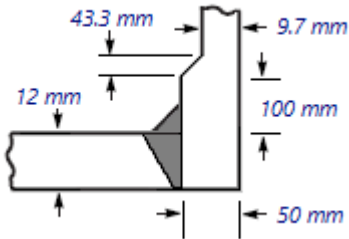
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	43.3325	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.734 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 169.6748 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 2.2308 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 37.72 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 0.4958 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4399 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 30.981 / 0.5773$
 $= 53.6605 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.8618	mm
Parallel to Vessel Wall, opening length	d	75.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N2-3".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4959	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.2308	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.2308	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(5.2308, 4.5)]$

= 5.2308 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.4959, 5.2308)
 = 5.2308 mm

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	39.3,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	305.7 MPa	Passed
Occasional	:	4.6,	Allowable	:	183.5 MPa	Passed
Shear	:	28.3,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.735, tr = 0.496, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.074, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 2.231, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.248, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.80/10.20 = 0.176

Weld Size Calculations, Description: N2-3"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 18.0455 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.0455 mm

Input Echo, WRC297 Item 1, Description: N2-3" :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm
Shell External Corrosion Allowance	0.0000 mm



Shell Diameter		333.350	mm
Shell Thickness		12.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		69.431	mm
Nozzle Thickness		50.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	1.80	MPa
Radial Load	P	-5997.68	N
Circumferential Shear	Vc	3998.47	N
Longitudinal Shear	Vl	3998.47	N
Circumferential Moment	Mc	896046.44	N-mm
Longitudinal Moment	Ml	896046.44	N-mm
Torsional Moment	Mt	0.128E+07	N-mm
Include Axial Pressure Thrust		Yes	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dn/Tn (3.389) is < 10
 check the Limitations of WRC bulletin 297.*

Note: The radial load includes the effect of the pressure thrust force (Pressure times Area). This may lead to overly conservative results in some cases.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 169.431 + 2 \cdot 1.65 \cdot \sqrt{(174.175(12.0 - 3.0))}$
 $= 300.086 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	348.350	mm
Nozzle Outside Diameter	(d) =	169.431	mm
Vessel Thickness used	(T) =	9.000	mm
Nozzle Thickness used	(t) =	47.000	mm
T / t	=	0.191	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.026$$

Nr/P	=	0.058
Mr/P	=	0.106
M0/P	=	0.032
N0/P	=	0.060
MrD/Mc	=	0.266
NrDL/Mc	=	0.093
M0d/Mc	=	0.080



NØDL/Mc = 0.084
MrD/Ml = 0.082
NrDL/Ml = 0.066
MØD/Ml = 0.024
NØDL/Ml = 0.074

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	32	-32	32	-32
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	33	34	33	34
Outplane Stress Summary	62	16	90	7

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	109	-109	109	-109
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-32	32	32	-32
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	99	-55	173	-111

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	100	72	173	119

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	32	-32	32	-32
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	23	19	96	-31



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	109	-109	109	-109
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-104	104	104	-104
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	34	33	34
Inplane Stress Summary	42	33	263	-163

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	42	33	263	163

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	6	8	16	18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	1	-2	6	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0

Shear due to Torsion	0	0	0	0
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Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	6	11	16	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	6	8	16	19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	0	0	9	-2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6	8	17	22

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	34	33	34	33	34	33	34
Circ. Pl (SUS)	6	6	14	14	4	4	16	16
Circ. Q (SUS)	23	-23	41	-41	5	-5	213	-213
Long. Pm (SUS)	16	16	16	16	16	16	16	16



Long. P1 (SUS)	6	6	14	14	5	5	15	15
Long. Q (SUS)	77	-77	141	-141	1	-1	63	-63
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	33.0	34.0	33.0	34.0	33.0	34.0	33.0	34.0
Pm+P1 (SUS)	39.1	40.1	47.1	48.1	37.1	38.1	49.1	50.1
Pm+P1+Q (Total)	99.4	72.4	171.0	118.1	42.2	33.3	262.1	163.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.00	138.00	Passed
Pm+P1 (SUS)	50.05	207.00	Passed
Pm+P1+Q (TOTAL)	262.10	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	3	1	3	1	3	1	3
Circ. P1 (SUS)	6	6	14	14	5	5	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. P1 (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	2	-2	2	-2	-1	1	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0
Pm+P1 (SUS)	8.0	10.0	15.0	17.0	7.0	9.0	16.0	18.0
Pm+P1+Q (Total)	7.0	12.0	15.0	18.0	8.0	8.0	16.0	22.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+P1 (SUS)	18.00	207.00	Passed
Pm+P1+Q (TOTAL)	22.00	414.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: N3-3"

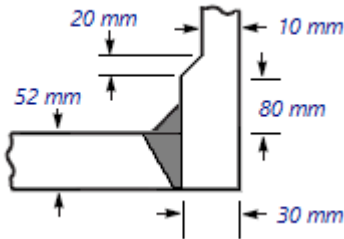
From: 160

Pressure for Reinforcement Calculations	P	1.8001	MPa
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-105	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Outside Diameter of Bolted Blind Flange	D	535.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-28.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.7346	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.8 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 1.8)$
 $= 0.5403 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4595 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 27.0$, less than the hub height, use (e-2)
 $= 67.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

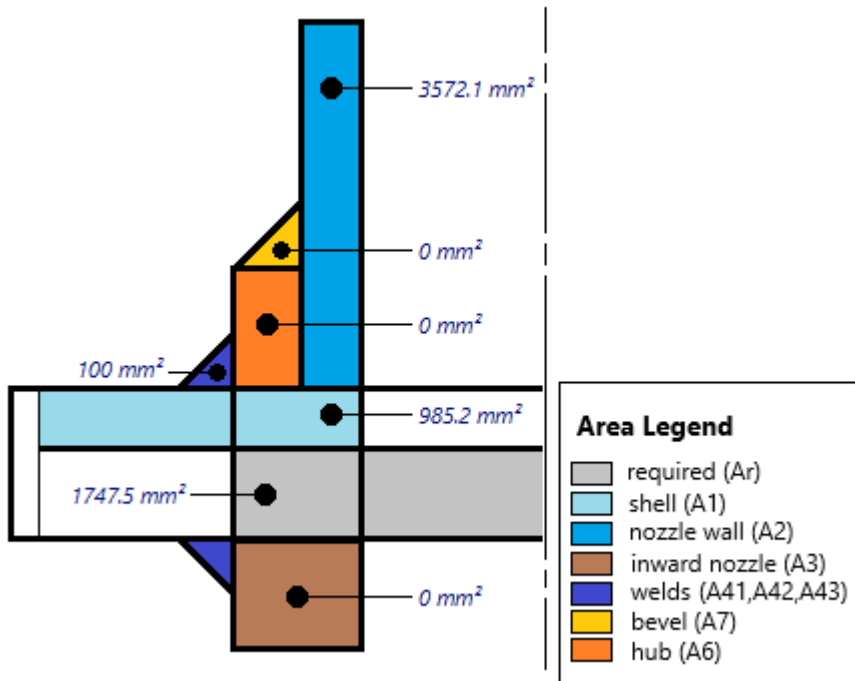
Parallel to Vessel Wall (Diameter Limit)	Dl	234.2000 mm
Parallel to Vessel Wall	Rn+tn+t	117.1000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	67.5000 mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1747.504	1748.484	NA
Area in Shell	A1	985.211	981.586	NA
Area in Nozzle Wall	A2	3572.053	3582.963	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	4657.264	4664.549	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 \cdot d \cdot t + t \cdot \tan(1 - fr_1)$ per UG-39(b)(1)
 $= 0.5 \cdot 82.2 \cdot 42.5183 + 42.5183 \cdot 27.0(1 - 1.0)$
 $= 1747.504 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E_1 \cdot t - F \cdot tr) - 2 \cdot \tan(E_1 \cdot t - F \cdot tr) \cdot (1 - fr_1)$
 $= 152.0(1.0 \cdot 49.0 - 1.0 \cdot 42.518) - 2 \cdot 27.0$
 $(1.0 \cdot 49.0 - 1.0 \cdot 42.5183) \cdot (1 - 1.0)$
 $= 985.211 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot \tan p)(\tan - \tan)fr_2$
 $= (2 \cdot 67.5)(27.0 - 0.54)1.0$
 $= 3572.053 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - \tan/0.707)^2 \cdot fr_2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$



There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5403 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 45.5183 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 45.5183 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[8.02, max(45.5183, 4.5)]
 = 8.0200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.5403, 8.02)
 = 8.0200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.54, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.077, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 30.0, tr = 0.54, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.02, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.80/5.10 = 0.353

Weld Size Calculations, Description: N3-3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
 = max(0, (1747.5039 - 985.2113 + 2 * 27.0 * 1.0 *
 (1.0 * 49.0 - 42.5183))138)
 = 153481.83 N

For hub type nozzles, A2 includes the area of the hub.



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (3572.0527 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 506691.03 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$$

$$= (3572.0527 + 0.0 + 100.0 + (2646.0)) \cdot 138$$

$$= 871801.31 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$$

$$= (3572.0527 + 0.0 + 100.0 + 0.0) \cdot 138$$

$$= 871801.31 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 136.2 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 144653. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 54.6) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 447349. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 136.2 \cdot (9.7346 - 0.0) \cdot 0.74 \cdot 138$$

$$= 212660. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (144653 + 447349) = 592002 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (144653 + 0 + 212660 + 0) = 357313 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (144653 + 212660 + 0) = 357313 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 592001 N	, must exceed W = 153481 N	or W1 = 506691 N
Path 2-2 = 357313 N	, must exceed W = 153481 N	or W2 = 871801 N
Path 3-3 = 357313 N	, must exceed W = 153481 N	or W3 = 871801 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Cut Length for this Nozzle is, Drop + Ho + H + T : 302.0508 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: Wind/Earthquake Deflection Exceeded the Specified Allowable!**

Please review all reports carefully!

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